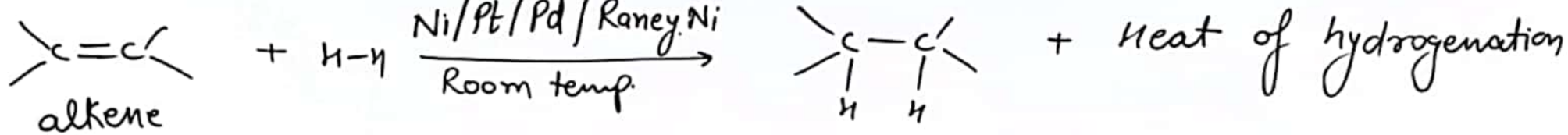
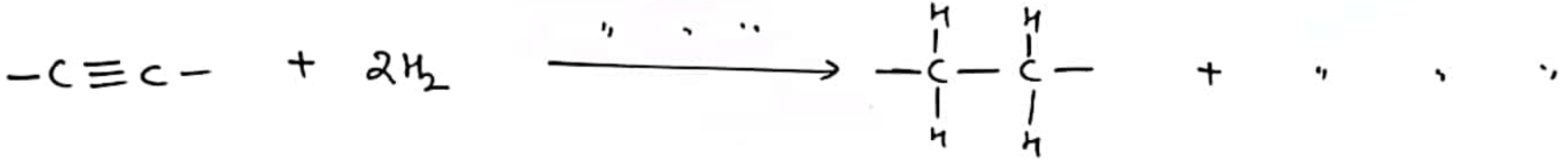
HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

① By Hydrogenation of alkenes and alkynes:-

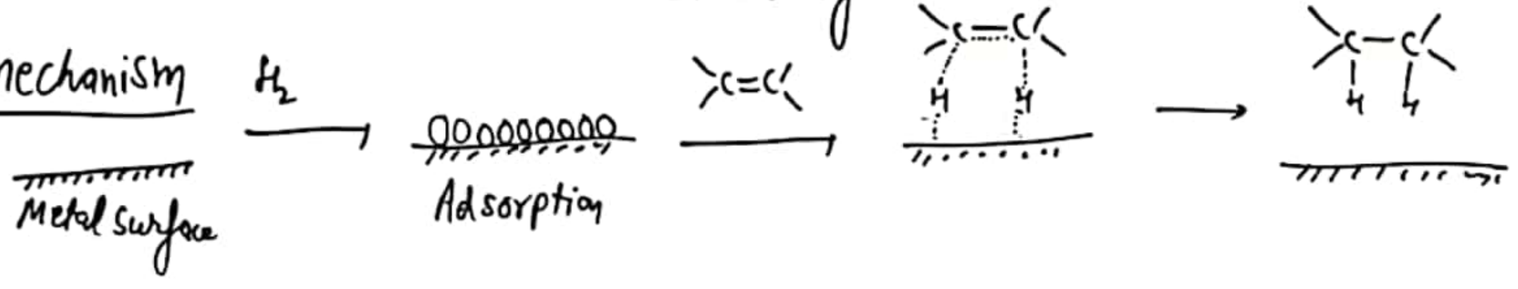




⇒ Heat of hydrogenation ∝ no. of π bonds in reactant.

⇒ ∴ ∴ ∴ ∴ ∝ $\frac{1}{\text{Stability}}$ (When no. π bonds are same in reactants)

⇒ Mechanism

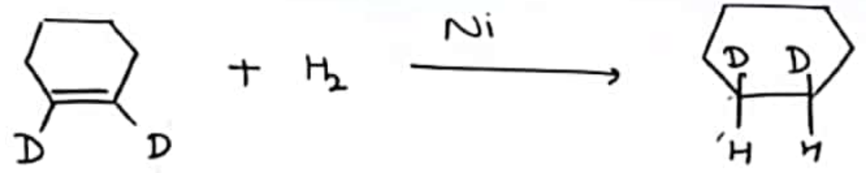


HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

① By Hydrogenation of alkenes and alkynes:-

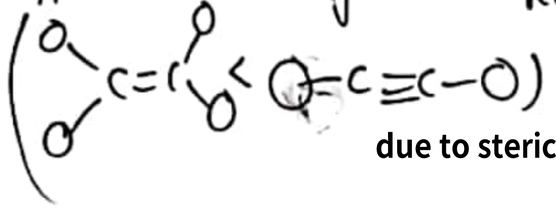
⇒ Syn addition reaction

eg 

Cis

⇒ Reactivity for hydrogenation $\alpha \frac{1}{\text{crowding / steric hindrance of carbon-carbon multiple bond}}$

eg Alkene < Alkyne Reactivity

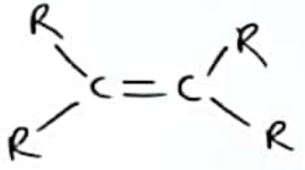
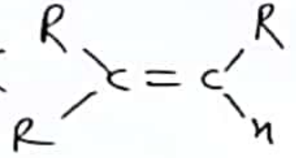


due to steric hindarance in alkene

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

① By Hydrogenation of alkenes and alkynes:-

eg  <  < $R-CH=CH-R$ < $R-CH=CH_2$ < $CH_2=CH_2$

⇒  $\xrightarrow[\text{Room temp}]{H_2 + Ni}$ No Rxn

aromatic
highly stable $\xrightarrow{H_2 + Ni\ \Delta}$ 

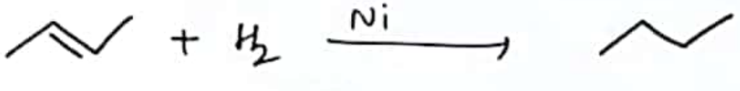
⇒ $R-C\equiv C-R + H_2 \xrightarrow{Ni} \underset{\text{Alkane}}{R-CH_2-CH_2-R}$

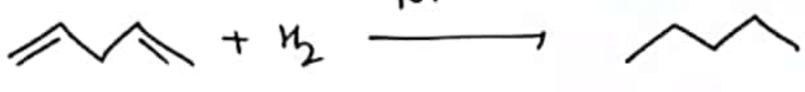
⇒ In presence of metal catalyst Heterogenous hydrogenation take place. but in presence of Wilkinson's catalyst $[Rh(PPh_3)_3Cl]$ Homogeneous hydrogenation.

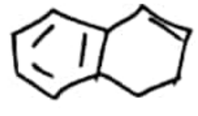
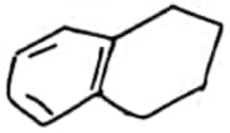
HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

① By Hydrogenation of alkenes and alkynes:-

eg  + H_2 $\xrightarrow{Ni}$

eg  + H_2 $\xrightarrow{Ni}$

eg  + H_2 $\xrightarrow[\text{no temp.?}\\ \text{assume R.T.}]{Ni}$ 

eg  + H_2 $\xrightarrow{\text{''}}$ 

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

(2) By Wurtz Reaction of alkyl halides

$$R{-}X + 2Na + X{-}R \xrightarrow[-2NaX]{\text{dry ether}} R{-}R$$

R–R: Alkane

can't take moist ether as moisture/water would react with Na

mechanism

$$2Na \longrightarrow 2Na^{+} + 2e^{-}$$

$$R\overset{\delta+}{-}X^{\delta-} \xrightarrow{2e^{-}} R^{\ominus} \quad X^{\ominus}$$

$R^{\ominus}$: good Nucleophile

$$R\overset{\delta+}{-}X^{\delta-} + R^{\ominus} \xrightarrow{(S_N2)} R{-}R + X^{\ominus}$$

Nucleophile ⇒ lone pair containing e^- rich species,
⇒ Attack on e^- deficient position of reactant

S_N2 = Bimolecular Nucleophilic Substitution reaction

HYDROCARBONS ⇒ I-ALKANE

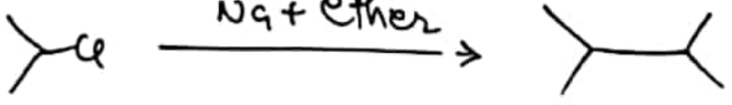
Methods of formation:-

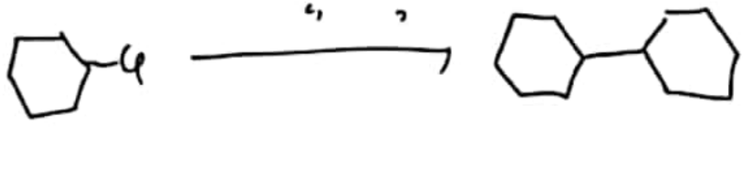
(2) By Wurtz Reaction of alkyl halides

Eg 1 $CH_3-Cl + Na \xrightarrow{\text{ether}} CH_3-CH_3$

Eg 2 $C_2H_5-Cl + Na \xrightarrow{"} C_2H_5-C_2H_5$

Eg 3 $(CH_3-Cl + C_2H_5-Cl)_{mix} \xrightarrow{Na + \text{ether}} CH_3-C_2H_5 + CH_3-CH_3 + C_2H_5-C_2H_5$

Eg 4 

Eg 5 

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

(2) By Wurtz Reaction of alkyl halides

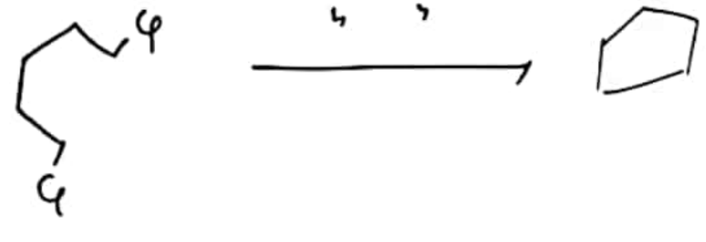
eg 6 $Ph-CH_2-Cl \xrightarrow{Na + ether} Ph-CH_2-CH_2-Ph$

Intramolecular wurtz reaction:-

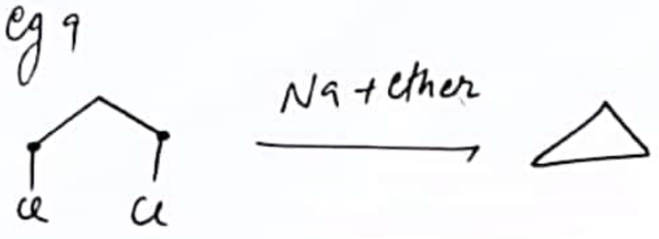
eg 7



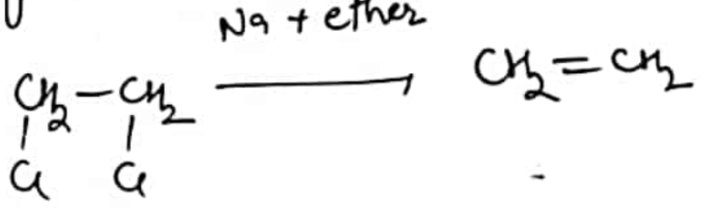
eg 8



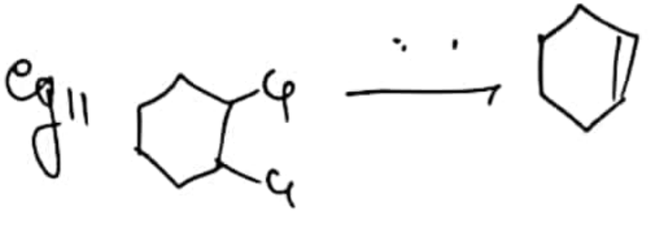
eg 9



eg 10



eg 11



HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

(2) By Wurtz Reaction of alkyl halides

⇒ Wurtz-Fittig Rxn

$$R-X + Ar-X \xrightarrow{Na + Ether} R-Ar$$

$$CH_3-Cl + Ph-Cl \xrightarrow{"} CH_3-Ph \text{ (Toluene)}$$

⇒ Fitting Rxn

$$Ar-X + Ar-X \xrightarrow{Na + ether} Ar-Ar$$

$$2Ph-Cl \xrightarrow{"} Ph-Ph \text{ (Biphenyl}$$

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

(3) By Kolbe's electrolysis of carboxylic acid salt:-

$$R\text{-}COOK_{aq} \xrightarrow{\text{electrolysis}} R\text{-}R + \underset{\text{anode}}{CO_2\uparrow} + \underset{\text{cathode}}{H_2\uparrow} + KOH$$

mechanism (free radical mech)

$$R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O^{\ominus}K^{\oplus} \xrightarrow{aq.} R\text{-}\overset{O}{\overset{\|}{C}}\text{-}O^{\ominus} + K^{\oplus}$$

⇒ on anode (oxidation)

$$R\text{-}\overset{O}{\overset{\|}{C}}\text{-}\ddot{O}:^{\ominus} \longrightarrow R\text{-}\overset{O}{\overset{\|}{C}}\text{-}\ddot{O}: + e^-$$

$$R\text{-}\overset{O}{\overset{\|}{C}}\text{-}\ddot{O}: \longrightarrow R^{\bullet} + CO_2$$

$$R^{\bullet} + R^{\bullet} \longrightarrow R\text{-}R$$

on cathode (Reduction)

$$H_2O + e^- \longrightarrow \frac{1}{2}H_2 + OH^{\ominus}$$

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

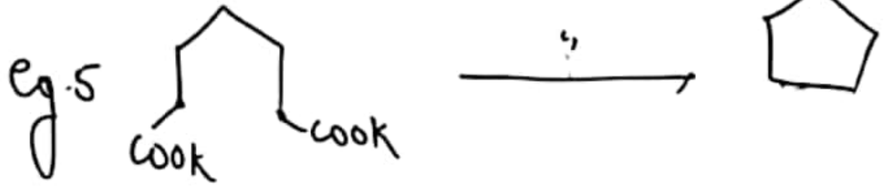
(3) By Kolbe's electrolysis of carboxylic acid salt:-

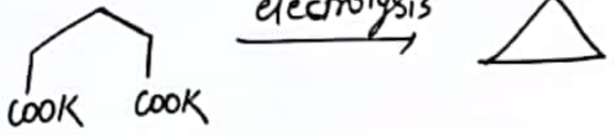
eg 1 $CH_3-COOK \xrightarrow{\text{electrolysis}} CH_3-CH_3$

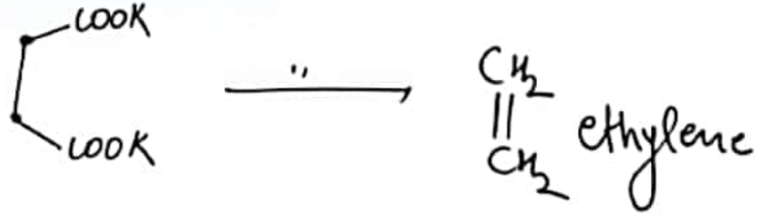
eg 2 $C_2H_5-COOK \xrightarrow{"} C_2H_5-C_2H_5$

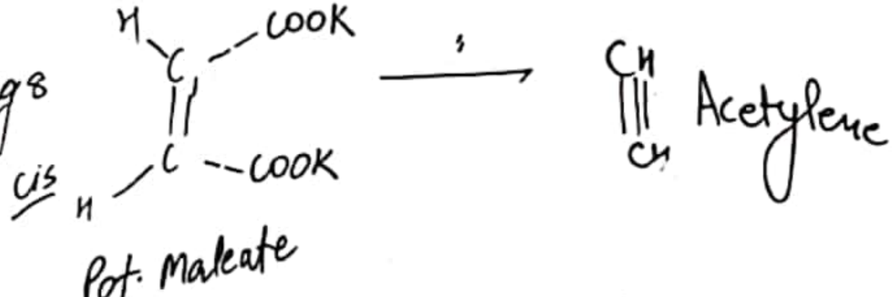
eg 3 $CH_3-COOK + C_2H_5-COOK \xrightarrow{"} CH_3-CH_3 + C_2H_5-C_2H_5 + CH_3-C_2H_5$

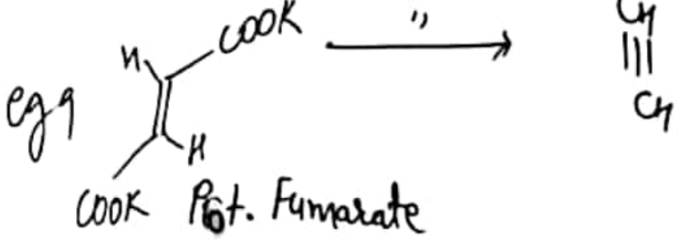
eg 4 $Ph-CH_2-COONa \xrightarrow{"} Ph-CH_2-CH_2-Ph$

eg 5 

eg 6 

eg 7 

eg 8 

eg 9 

HYDROCARBONS ⇒ I-ALKANE

Methods of formation:-

(4) By Decarboxylation of carboxylic acid:-

$$\left(R\text{-}COOH / R\text{-}COONa\right)_{aq} \xrightarrow[\text{Soda lime}]{(NaOH + CaO)} R\text{-}H + Na_2CO_3$$

Mechanism (intermediate = Carbanion)

$$R\text{-}\overset{O}{\overset{\|}{C}}\text{-}OH + \overset{\ominus}{O}H \longrightarrow R\text{-}\overset{O}{\overset{\|}{C}}\text{-}\overset{\ominus}{O} + H_2O$$

RDS = Rate Determining Step

RDS

$$R^{\ominus} + CO_2$$

$$R^{\ominus} + H_2O \longrightarrow R\text{-}H + \overset{\ominus}{O}H$$

Base

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

⇒ Rate of decarboxylation ∝ stability of anion intermediate

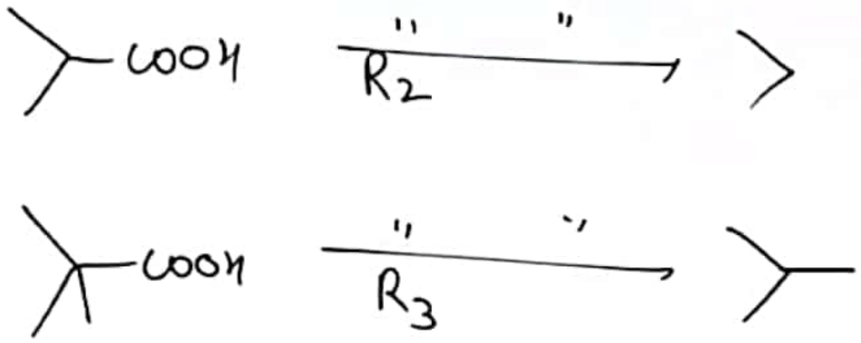
$1^{\circ} > 2^{\circ} > 3^{\circ}$

HYDROCARBONS ⇒ I-ALKANE

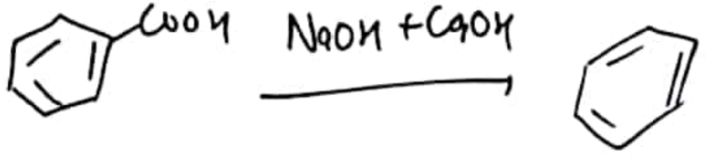
Methods of formation:-

(4) By Decarboxylation of carboxylic acid:-

eg1 $CH_3-COOH \xrightarrow[R_1]{NaOH + CaO,\ \Delta} CH_4$



Reactivity of decarboxylation = $R_1 > R_2 > R_3$

eg2 

HYDROCARBONS ⇒ I-ALKANE

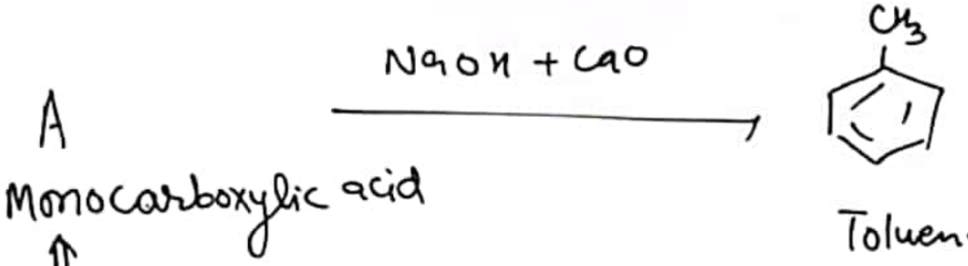
Methods of formation:-

(4) By Decarboxylation of carboxylic acid:-

Eg 3



Q



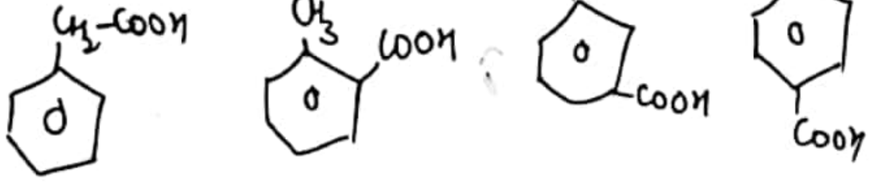
A

Monocarboxylic acid

⇑

Possible structures?

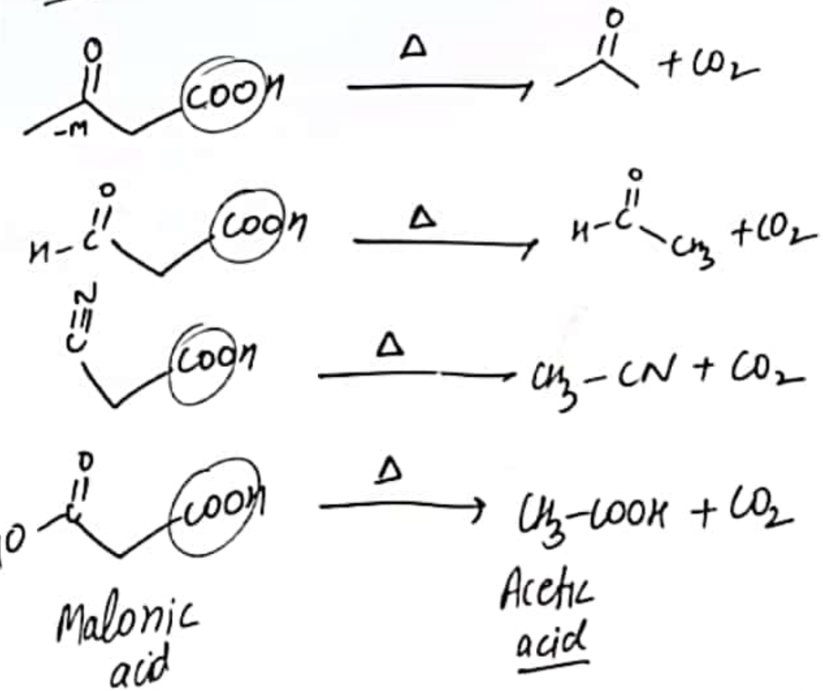
Sol:- A



Toluene

if the carboxylic acid has a strong -M group at its β position, then it undergoes decarboxylation simply on heating; soda lime is not required

Note



Malonic acid

Acetic acid

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION:-

(5) By Reduction of alkyl halide:-

(a) with Metal + Acid:-

water can also be used instead of acid

$$R-X \xrightarrow[1^\circ]{\text{Metal + Acid}} R-H + \text{Metal halide}$$

$Zn + HCl$ / $Zn-Cu + CH_3COOH$ /

$Zn-Cu + C_2H_5OH$ / $Zn-Cu + H_2O\ \Delta$

$Na-Hg + HCl$ etc.

(b) with $LiAlH_4$:-

$$R-X \xrightarrow[1^\circ,2^\circ]{LiAlH_4} R-H + LiX$$

$$R-X \xrightarrow[3^\circ]{LiAlH_4,\ (-HX)} \text{Alkene}$$

(c) with $NaBH_4$

$$R-X \xrightarrow[2^\circ,3^\circ]{NaBH_4} R-H$$

$$R-X \xrightarrow[1^\circ]{NaBH_4} R-H$$ ✗

(d) with Ph_3SnH (Triphenyl tin hydride)

$$R-X \xrightarrow[1^\circ,2^\circ,3^\circ]{Ph_3SnH} R-H$$

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION :-

⑥ Reduction of various functional group with HI + Red-P

↑ very Stoong reducing agent

$R-X + 2H\overset{-1}{I} \xrightarrow{\text{Red-P}} R-H + H-X + \overset{0}{I_2}$

$R-OH + 2HI \xrightarrow{\Delta} R-H + H_2O + I_2$

$R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{HI + \text{Red P}} R-CH_3 + H_2O + I_2$

$R-CH=O \xrightarrow{\Delta\ \Delta} R-CH_3 + \text{〃} + \text{〃}$

$R-\overset{O}{\overset{\|}{C}}-R \xrightarrow{\Delta\ \Delta} R-CH_2-R + \text{〃} + \text{〃}$

$R-\overset{O}{\overset{\|}{C}}-O-R' \xrightarrow{\Delta\ \Delta} R-CH_3 + R'-H + H_2O + I_2$

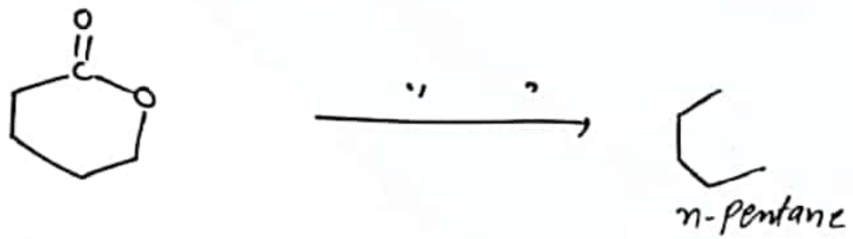
$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow{\Delta\ \Delta} R-CH_3 + HCl + \text{〃} + \text{〃}$

HYDROCARBONS ⇒ (I) ALKANE

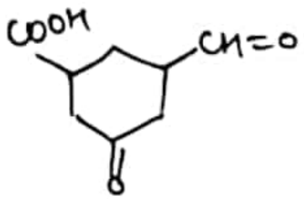
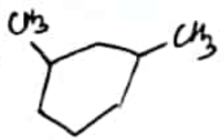
METHODS OF FORMATION :-

⑥ Reduction of various functional group with HI + Red-P

eg $CH_3-\overset{O}{\overset{\|}{C}}-O-C_2H_5 \xrightarrow{HI + Red\text{-}P} CH_3-CH_3$

eg  →

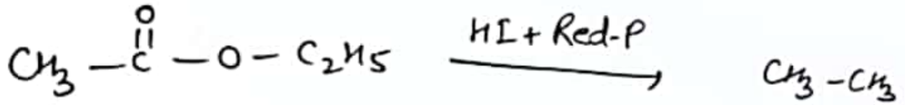
n-Pentane

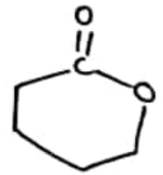
eg  → 

HYDROCARBONS ⇒ (I) ALKANE

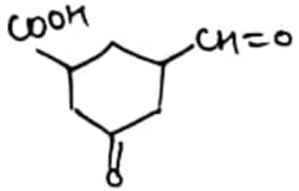
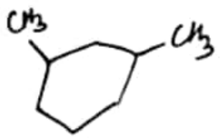
METHODS OF FORMATION:-

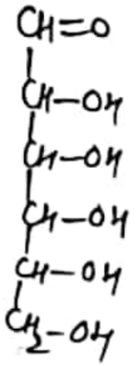
⑥ Reduction of various functional group with HI + Red-P

eg  $\xrightarrow{HI + Red\text{-}P}$ CH_3-CH_3

eg  $\xrightarrow{\quad " \quad "\quad}$ 

n-Pentane

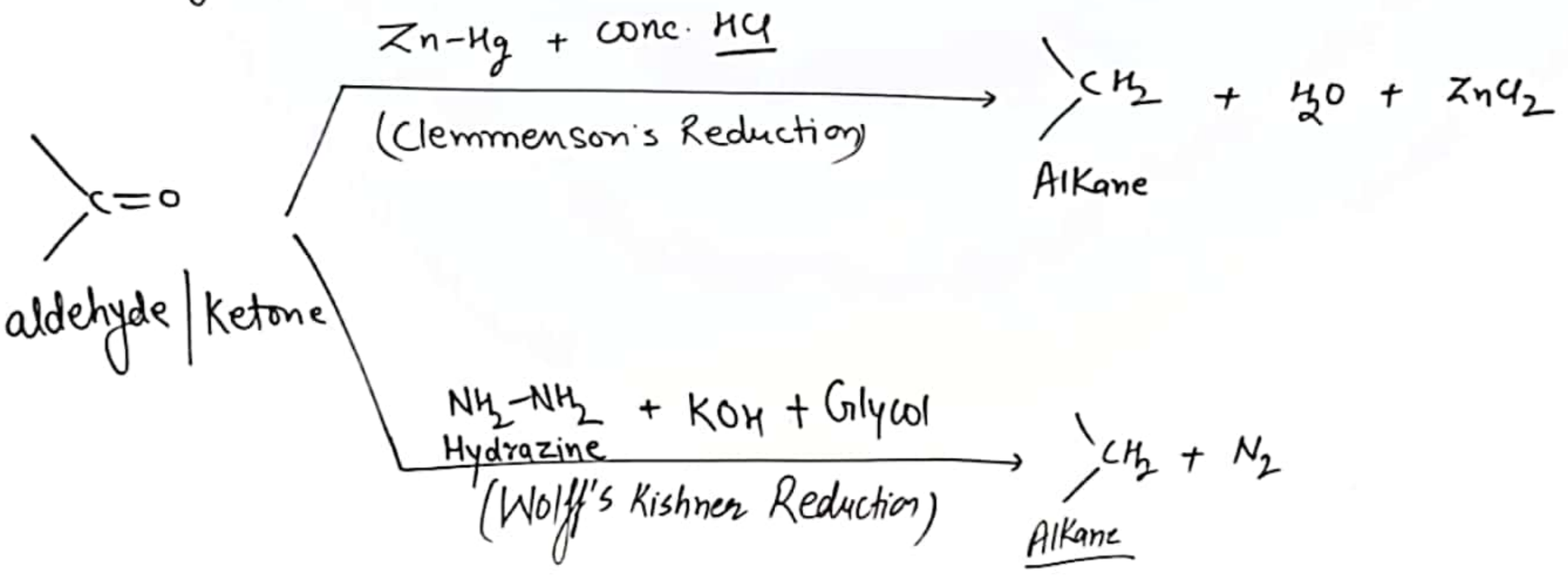
eg  $\xrightarrow{\quad " \quad "\quad}$ 

eg  $\xrightarrow{\quad " \quad "\quad}$ n-Hexane

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION:-

(7) By Clemmenson's Reduction and Wolff-Kishner Reduction of aldehyde & Ketone



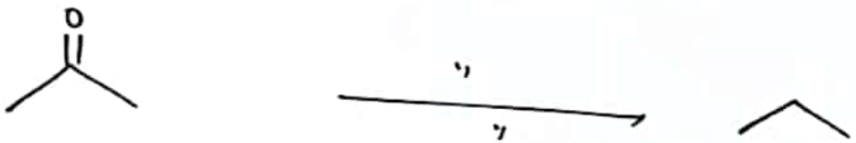
CH2-OH
|
CH2-OH
glycol

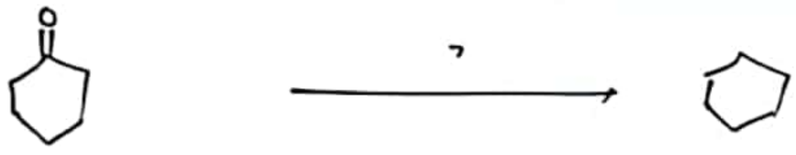
HYDROCARBONS ⇒ (I) ALKANE

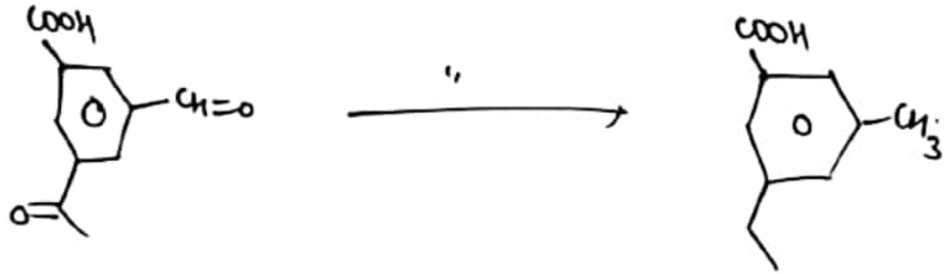
METHODS OF FORMATION :-

(1) By Clemmenson's Reduction and Wolff-Kishner Reduction of aldehyde & Ketone

Eg 1 $CH_3-CH=O \xrightarrow[\text{or } N_2H_4+KOH]{Zn-Hg+HCl} CH_3-CH_3$

Eg 2 

Eg 3 

Eg 4 

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION :-

(8) By Grignard Reagent (RMgX)

⇒ Introduction of Grignard Reagent (An Organometallic compound)

$R^{\delta-}-Mg^{\delta+}X$

- Alkyl (1°, 2°, 3°)
- $-CH_2=CH_2$, $-C\equiv C-R$
- $-Ph$ etc.

X = I, Br, Cl

$X \neq F$

Metal-carbon bond containing compounds.

⇒ RMgX Strong Base as well as Strong Nucleophile

⇒ Formation of RMgX

$$R-X + Mg \xrightarrow[\text{solvent}]{\text{ether}} R-MgX$$

$X \neq F$

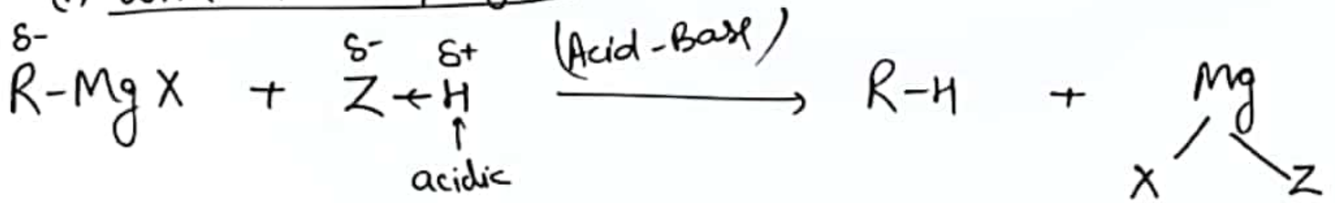
HYDROCARBONS ⇒ (I) ALKANE

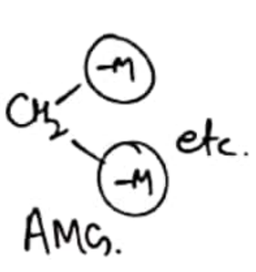
METHODS OF FORMATION :-

(8) By Grignard Reagent ($RMgX$)

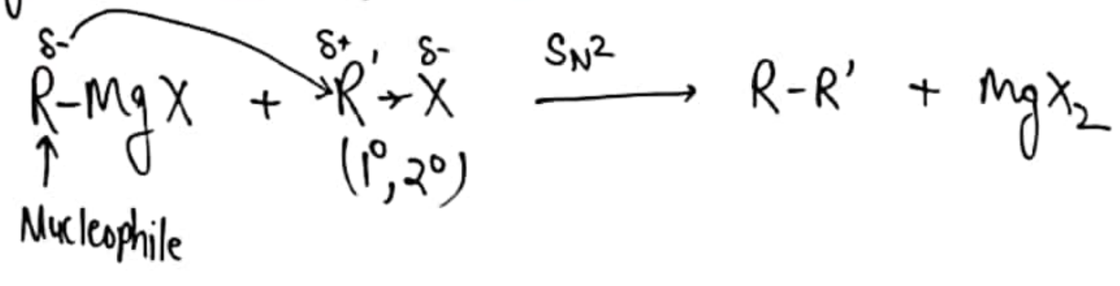
⇒ Formation of Alkane

(i) with acidic hydrogen :-



Where Z-H ⇒ $-O-H$, $-S-H$, $-\underset{|}{N}-H$, $X-H$, $-C\equiv C-H$ (sp),  , 

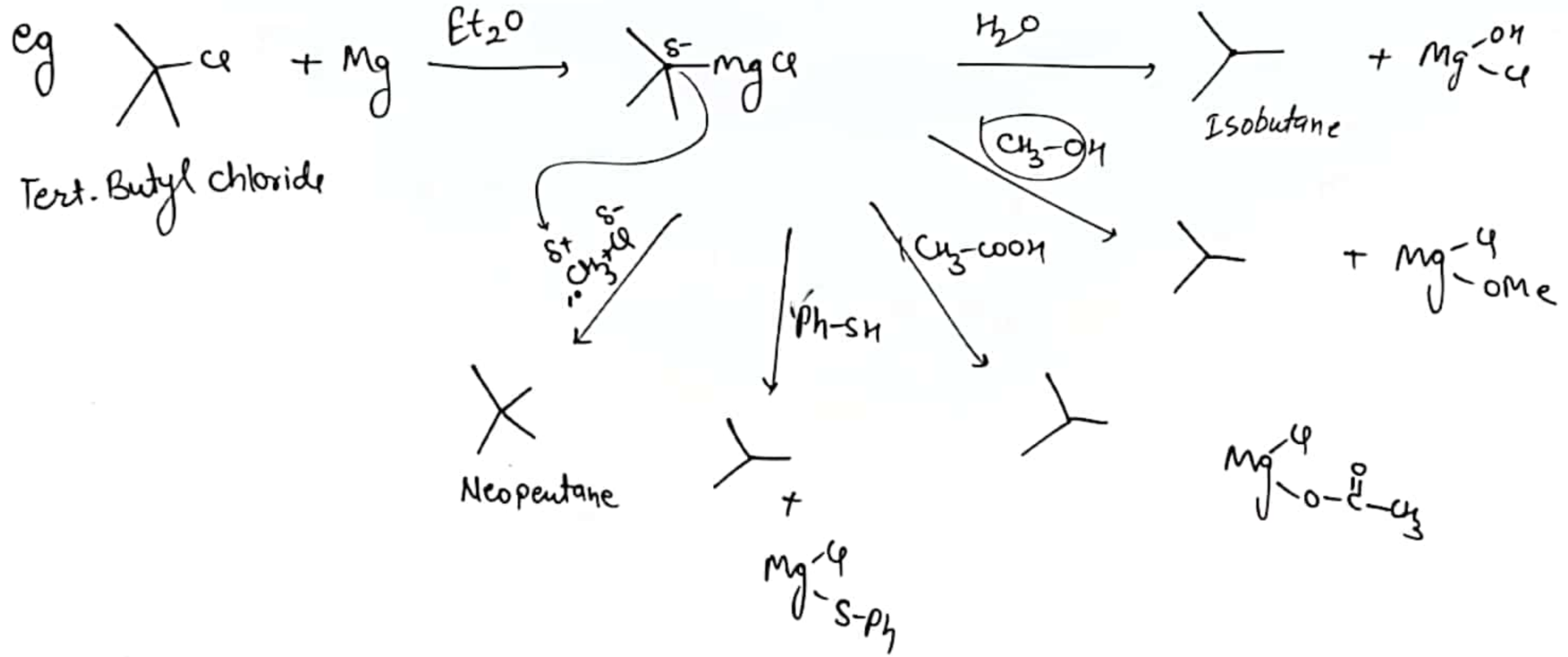
(ii) By reaction of RMgx with R-X :-



HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION:-

(8) By Grignard Reagent (RMgX)



HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION:-

(8) By Grignard Reagent ($RMgX$)

eg $CH_2=CH-Cl$ (Vinyl Chloride) $+ Mg \xrightarrow{\text{THF}} CH_2=CH-MgCl \xrightarrow{H_2O} CH_2=CH_2 + Mg(Cl)(OH)$

eg $R-C\equiv C-CH_2-Cl + Mg \xrightarrow{\text{ether}} R-C\equiv C-\overset{\delta-}{CH_2}-MgCl \xrightarrow{Ph-OH} R-C\equiv C-CH_3$

$R-C\equiv C-\overset{\delta-}{CH_2}-MgCl + CH_3-\overset{\delta+}{CH_2}-\overset{\delta-}{Cl} \longrightarrow R-C\equiv C-CH_2-\underset{CH_3}{CH_2}$ (CH_3-CH_2-Cl: 1°)

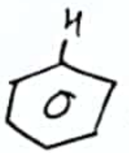
$R-C\equiv C-\overset{\delta-}{CH_2}-MgCl + Cl \longrightarrow R-C\equiv C-CH_2-\underset{CH_3}{\overset{CH_3}{CH}}$

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION:-

(8) By Grignard Reagent (R Mg X)

eg $Ph-Br + Mg \xrightarrow{ether} \overset{\delta-}{Ph}MgBr \xrightarrow{CH_3-C\equiv C-\overset{\delta+}{H}} Ph-H +$



Benzene

+

$CH_3-C\equiv C-MgBr$

New G.R.

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION :-

(9) By Corey-House Synthesis :-

$$\overset{\delta-}{R_2}CuLi + \overset{\delta+}{R'}\rightarrow\overset{\delta-}{X} \longrightarrow R-R' + LiX + RCu$$

saturated alkyl (1°, 2°, 3°)

Lithium dialkyl cuprate (Gilmann's Reagent)

$R'\rightarrow X$: (1°, 2°) should not be tertiary

Formation of Gilmann's reagent:-

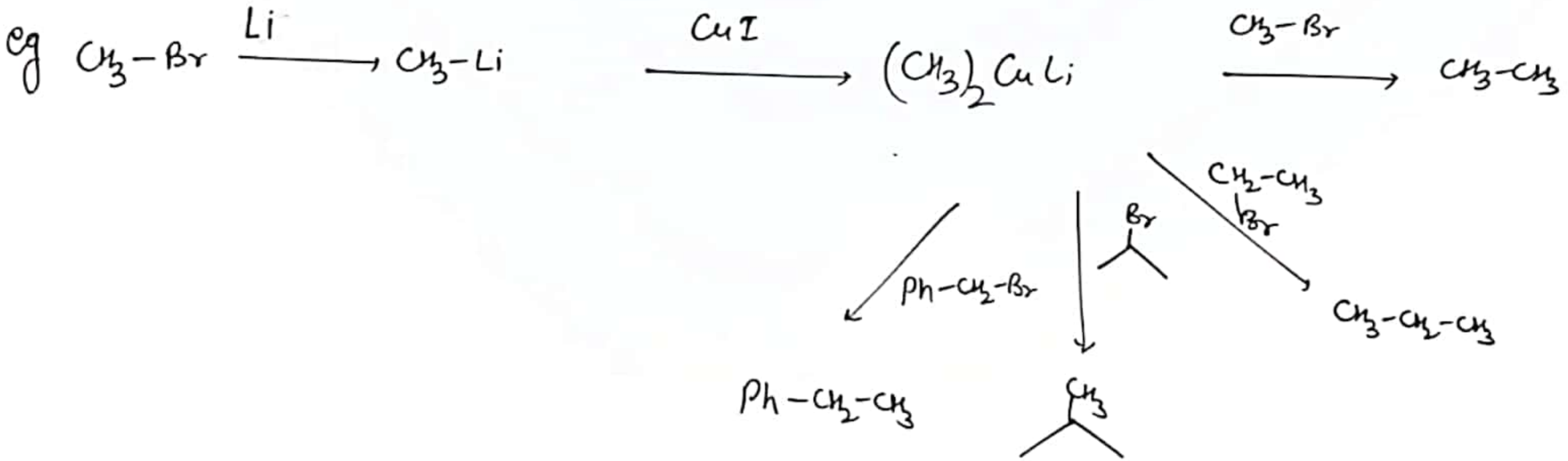
$$R-X \underset{(1^\circ,2^\circ,3^\circ)}{} + 2Li \longrightarrow RLi + LiX$$

$$2RLi + CuI \longrightarrow \underset{\text{Gilmann's Reagent}}{R_2CuLi} + LiI$$

HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION :-

⑨ By Corey - House Synthesis :-



HYDROCARBONS ⇒ (I) ALKANE

METHODS OF FORMATION :-

(10) By Reaction of alkyl halide with Zn dust :-

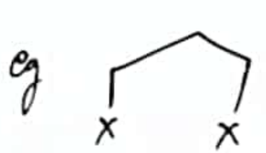
$$R-X + Zn + X-R \xrightarrow[-ZnX_2]{\Delta} R-R$$

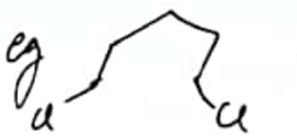
alkyl halide

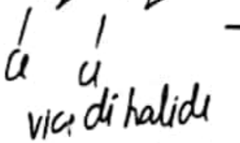
eg $CH_3-Cl \xrightarrow{Zn,\Delta} CH_3-CH_3 + ZnCl_2$

eg $CH_3-CH_2-Cl \xrightarrow{Zn\ \Delta}$  + ''

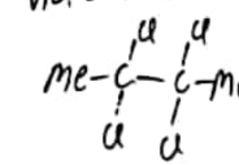
eg $Ph-CH_2-Cl \xrightarrow{Zn\ \Delta} Ph-CH_2-CH_2-Ph$ + ''

eg  $\xrightarrow{Zn,\ \Delta}$ 

eg  $\xrightarrow{\ ''\ }$ 

eg  $\xrightarrow{\ ;\ }$ $CH_2=CH_2$ ethylene

vic. di halide

eg  $\xrightarrow{\ ''\ }$ $H-C\equiv C-H$ acetylene

HYDROCARBONS ⇒ (I) ALKANE

↳ Paraffin's

PHYSICAL PROPERTIES OF ALKANES :-

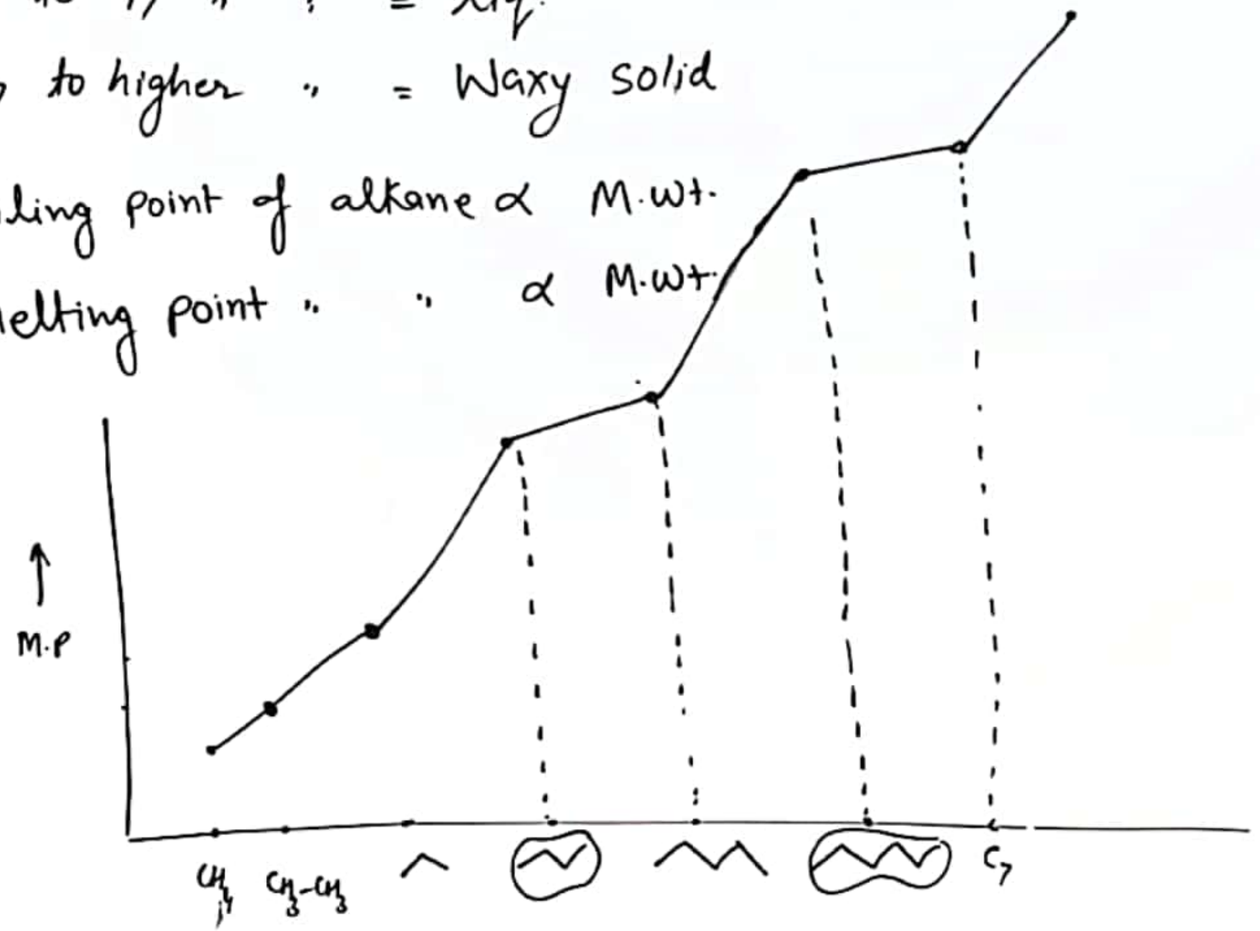
⇒ C_1 to C_4 alkane = Gas

⇒ C_5 to C_{17} " " = liq.

⇒ C_{17} to higher " = Waxy solid

⇒ Boiling point of alkane ∝ M.wt.

⇒ Melting point " " ∝ M.wt.



even no. of C containing alkanes have symmetrical structure; thus their MP increases sharply

like dissolves like

⇒ Solubility of Alkane ∝ $\frac{1}{M.wt}$

- insoluble in water
- Soluble in organic non polar Solvent

⇒ Density of alkane ∝ M.wt.

lighter than water

approx. 0.8 or 0.9 g/ml
hence, float in water

HYDROCARBONS ⇒ (I) · ALKANE

CHEMICAL REACTIONS OF ALKANES :-

$\overset{2.5}{C} - \overset{2.1}{C}$ $\qquad$ $\overset{2.5}{C} - \overset{2.1}{H}$ $\Big\}$ Free Radical Substitution

↑ Non Polar $\qquad$ ↑ Strong bond

HYDROCARBONS ⇒ (I) - ALKANE

CHEMICAL REACTIONS OF ALKANES :-

① Halogenation :-

$$R-H + X_2 \xrightarrow[\text{or high temp.}]{h\nu \,/\, U.V \,/\, \text{Sun light}} R-X + HX$$

+ Peroxide

rex. of sp3 C-H bond

alkyl halide

$$R-H + SO_2Cl_2 \xrightarrow[S_FR]{h\nu \,/\, \Delta} R-Cl + SO_2 + HCl$$

Sulfuryl chloride

⇒ Reactivity of X_2 ⇒ $F_2 > Cl_2 > Br_2 > I_2$

highly exothermic — useful. — Reversible ⟷

CFC
CF_2Cl_2
$CFCl_3$
CF_3Cl

⇒ Iodination of alkane is reversible reactive because HI (Product) is strong reducing agent, it can reduce alkyl halid back to alkane. But iodination possible in presence of oxidising agent HIO_3, HNO_3 & H_2O.

$$R-H + I_2 \rightleftharpoons R-I + HI \quad ; \quad 5HI + HIO_3 \longrightarrow 3I_2 + 3H_2O$$

best suited
since it regenerates iodine
which is again used for iodination
though others will also produce
iodine, but in lesser amount

HYDROCARBONS ⇒ (I) · ALKANE

CHEMICAL REACTIONS OF ALKANES :-

① Halogenation :-

⇒ Mechanism of halogenation :- (Free radical Substitution Rx^n SFR)

- Chain Reaction :-

$R-H + .X_2$

ⓐ Chain Initiation Step.

$X-X \xrightarrow[\text{Homolysis or Homolytic diss.}]{h\nu / \Delta} 2X^{\bullet}$ (Endothermic)

highly reactivity

ⓑ Chain propagation Step.

$R-H + X^{\bullet} \xrightarrow{RDS} R^{\bullet} + H-X$ (F → Exothermic; Cl, Br, I → Endo)

$R-H + X^{\bullet} \xrightarrow[Eq = high]{X} R-X + H^{\bullet}$

$X-X + R^{\bullet} \longrightarrow R-X + X^{\bullet}$ (Exothermic)

ⓒ Chain Termination step.

$X^{\bullet} + X^{\bullet} \longrightarrow X-X$

$R^{\bullet} + X^{\bullet} \longrightarrow R-X$

$R^{\bullet} + R^{\bullet} \longrightarrow R-R$

} Exothermic.

⇒ Reactivity ∝ Stability of R° (free radical)

$(3^{\circ}\,C-H > 2^{\circ}\,C-H > 1^{\circ}\,C-H)$

HYDROCARBONS ⇒ (I) · ALKANE

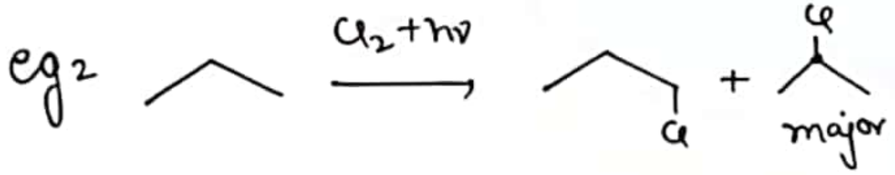
CHEMICAL REACTIONS OF ALKANES :-

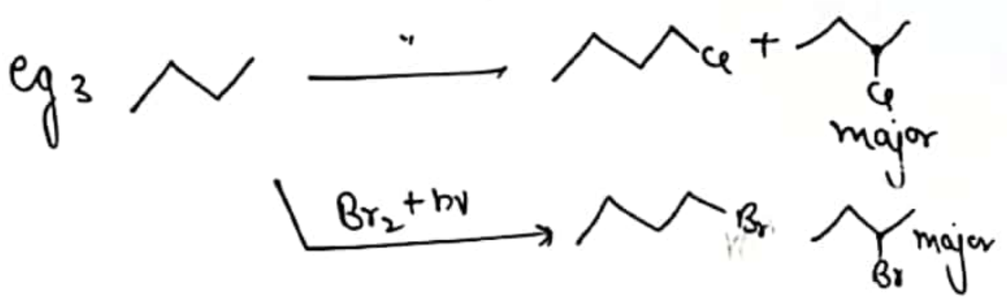
① Halogenation :-

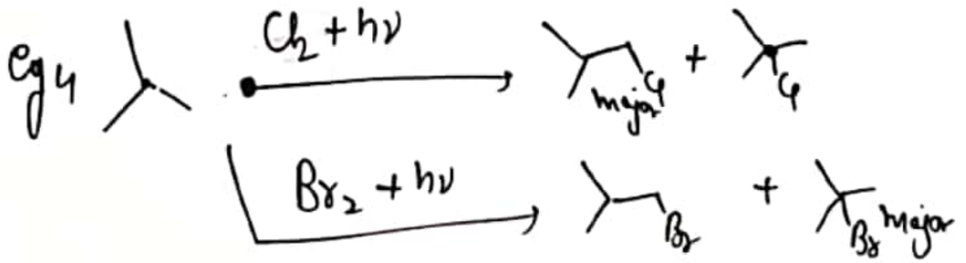
eg 1 $CH_4 \xrightarrow{Cl_2+h\nu} CH_3-Cl \xrightarrow{Cl_2+h\nu} CH_2Cl_2 \xrightarrow{Cl_2} CHCl_3 \longrightarrow CCl_4$

monochloro product

Chloroform

eg 2 

eg 3 

eg 4 

⇒ Yield of Products depends upon following factors.

① Stability of $R^{\cdot}$ ⇒ $3^{\circ} > 2^{\circ} > 1^{\circ}$

② reactivity of C-H Bond ⇒ $3^{\circ}C-H > 2^{\circ}C-H > 1^{\circ}C-H$

③ Probability factor :- Availability of same types of H. atoms.

④ Selectivity and Reactivity of $X^{\cdot}$ ⇒

$F^{\cdot} > Cl^{\cdot} > Br^{\cdot} > I^{\cdot}$

$I^{\cdot} > Br^{\cdot} > Cl^{\cdot} > F^{\cdot}$

HYDROCARBONS ⇒ (I) · ALKANE

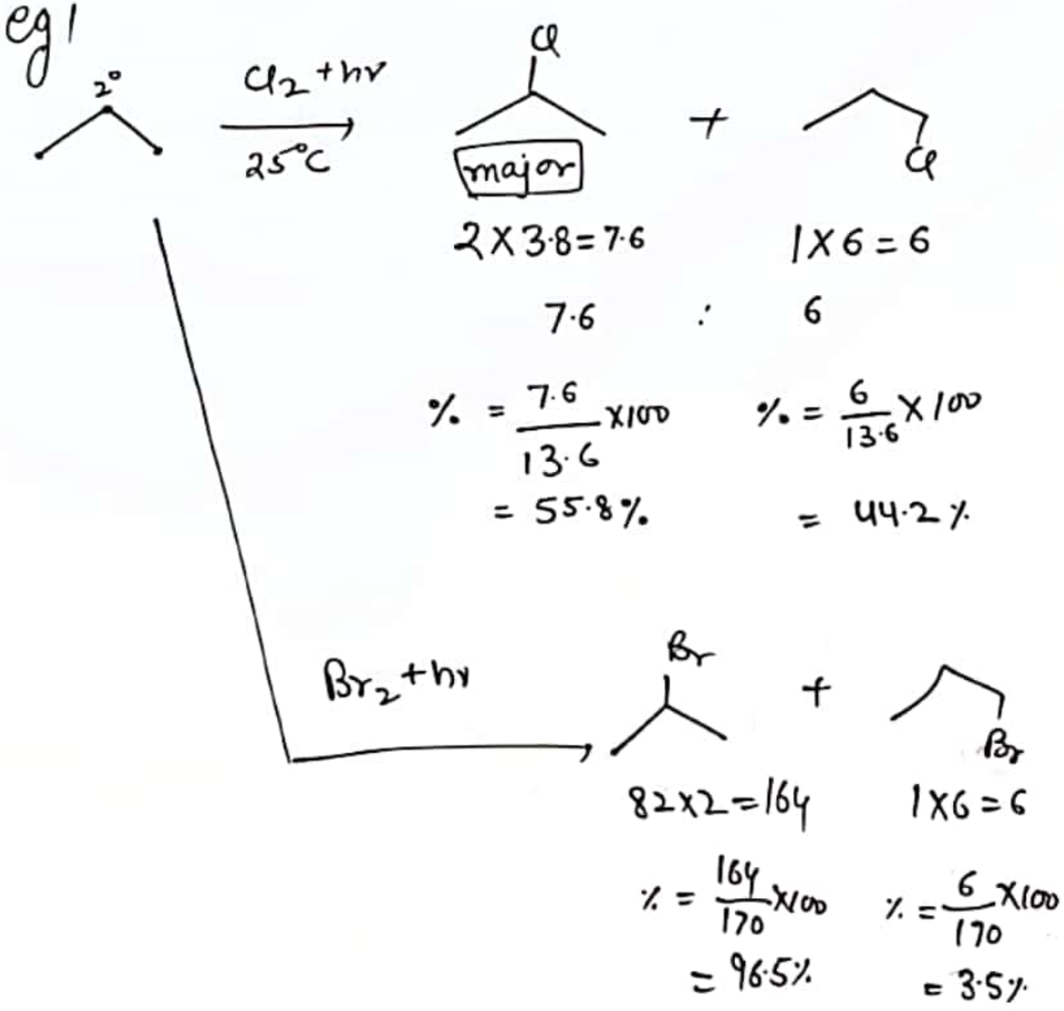
CHEMICAL REACTIONS OF ALKANES :-

① Halogenation :-

⇒ Relative rate for replacement of one hydrogen atom in alkane

	$1^\circ H$	:	$2^\circ H$	:	$3^\circ H$
F ⇒	1	:	1	:	1
Cl (25°C) ⇒	1	:	3·8	:	5
Br (125°C) ⇒	1	:	82	:	1600

eg 1



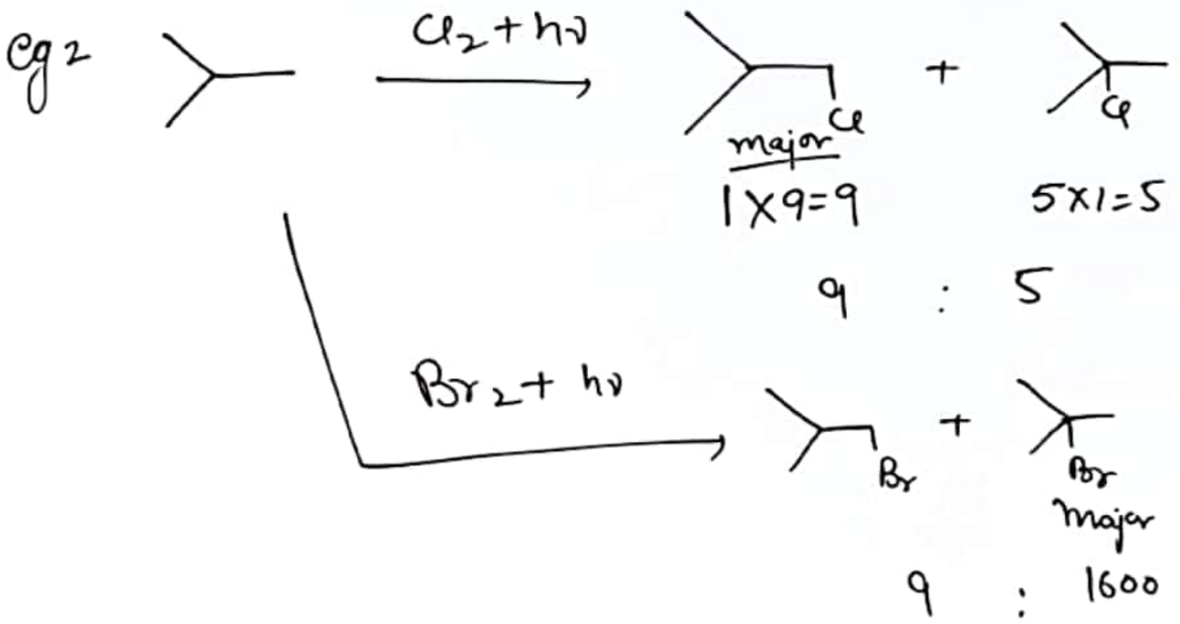
HYDROCARBONS ⇒ (I) · ALKANE

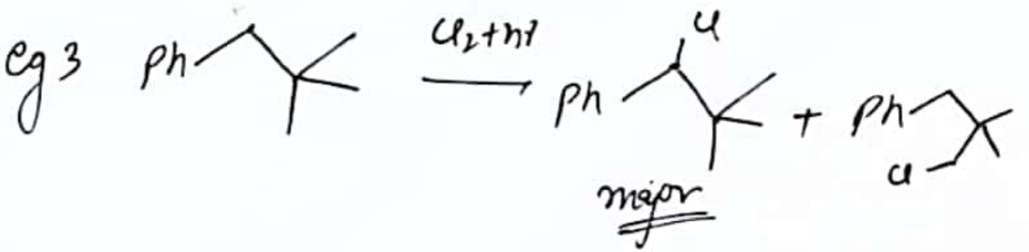
3·8 X 2 = 7·6 1 X 9 = 9

7·6 : 9

CHEMICAL REACTIONS OF ALKANES :-

① Halogenation :-





HYDROCARBONS ⇒ (I) · ALKANE

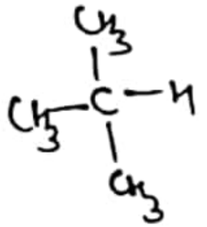
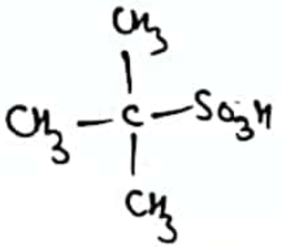
CHEMICAL REACTIONS OF ALKANES :-

② Sulphonation :-

$$R-H \xrightarrow[(S_FR)]{\text{conc.}\ H_2SO_4\ (\text{or oleum}),\ \Delta} R-SO_3H + H_2O$$

n. alkane (min. 6 carbon containing) → Sulphonic acid

⇒ Reactivity ⇒ $3^\circ\ C-H > 2^\circ\ C-H > 1^\circ\ C-H$

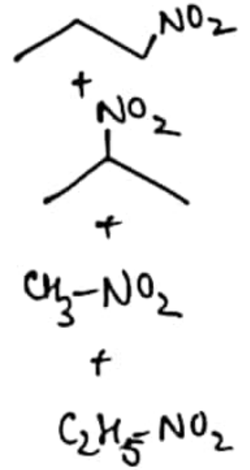
⇒  $\xrightarrow{\text{Conc.}\ H_2SO_4\ \Delta}$ 

agar bracnched ho and tertiary carbon ho, then easily sulphonation ho jaata hai (6 carbon not required)

③ Nitration of alkanes :-

$$CH_4 + \text{conc.}\ HNO_3 \xrightarrow[(S_FR)]{\Delta} CH_3-NO_2 + H_2O$$

$$CH_3-CH_3 + \text{" "} \xrightarrow{\Delta} CH_3-CH_2-NO_2 + CH_3-NO_2$$

$$CH_3-CH_2-CH_3 \xrightarrow{\text{Conc.}\ HNO_3,\ \Delta} \text{  }$$

$CH_3-NO_2 + C_2H_5-NO_2$

the C-C bond also undergoes nitration

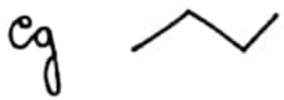
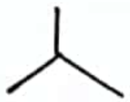
HYDROCARBONS ⇒ (I) - ALKANE

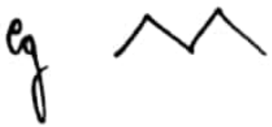
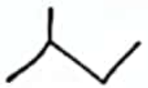
CHEMICAL REACTIONS OF ALKANES :-

(4) Isomerisation / Reformation :-

n-Alkane (lesser Stable) $\xrightarrow{AlCl_3 + HCl,\ \Delta}$ Branched alkane isomer (More Stable)

can also use Br

eg  $\xrightarrow{AlCl_3 + HCl,\ \Delta}$ 

eg  $\xrightarrow{\text{,,}}$ 

both these reactions are used in petroleum industry

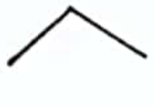
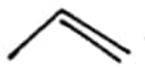
isoalkanes are generally good petrols
isooctane is the best petrol having an octane no. of 100

(5) Pyrolysis of alkane / Cracking

higher alkane $\xrightarrow[\text{(in absence of } O_2)]{\Delta}$ lower alkane + lower alkene + H_2

alkyne are not formed

used to break down larger alkanes into petrol

eg  $\xrightarrow{\Delta}$ $CH_4 + CH_2=CH_2$  $+ H_2$

eg $C_4H_{10} \xrightarrow{\Delta}$ $C_4H_8 + H_2$

$C_3H_6 + CH_4$

$C_2H_6 + CH_3{-}CH_3$

alkane-alkene pairs are formed

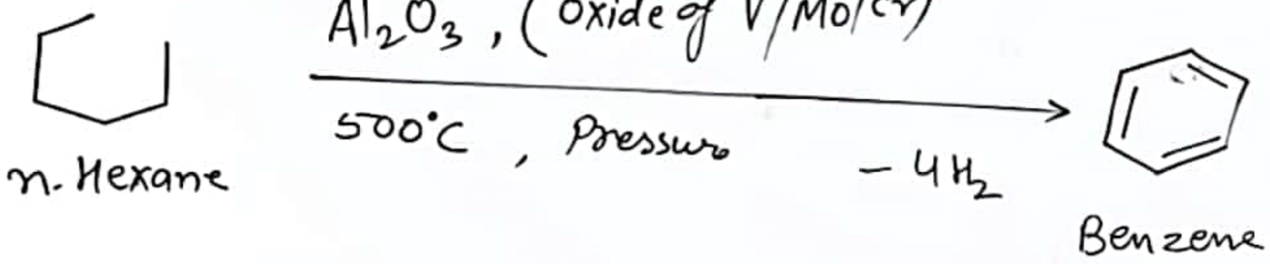
eg $C_5H_{12} \xrightarrow{\Delta}$ $C_5H_{10} + H_2 + C_4H_8 + CH_4 + C_3H_6 + C_2H_6$

$C_2H_6 + C_3H_8$

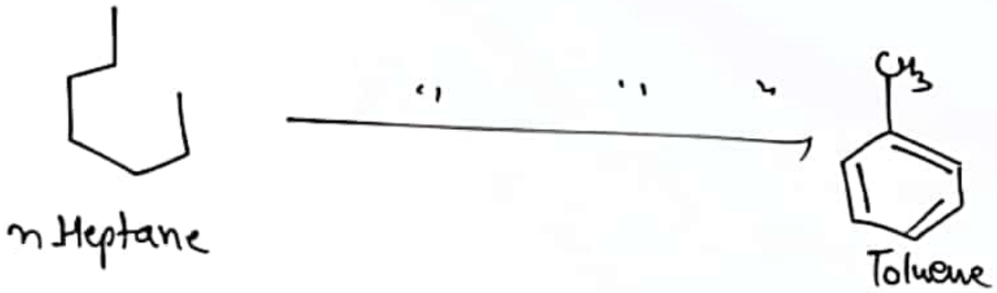
HYDROCARBONS ⇒ (I) · ALKANE

CHEMICAL REACTIONS OF ALKANES :-

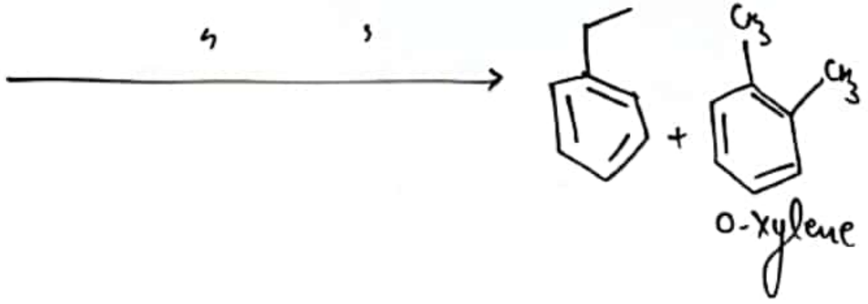
⑥ Aromatisation :-



n-Hexane → Benzene



n-Heptane → Toluene



n-octane → + o-xylene

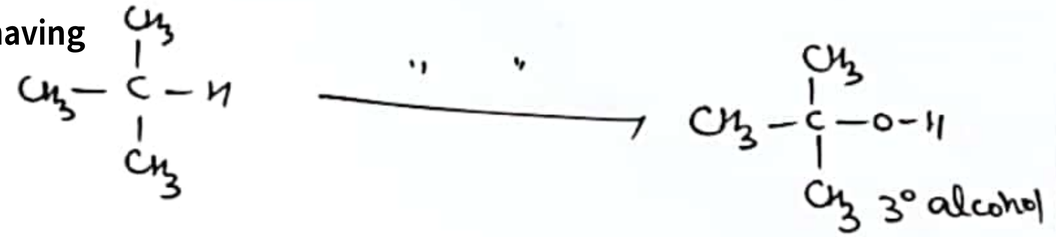
HYDROCARBONS ⇒ (I) · ALKANE

CHEMICAL REACTIONS OF ALKANES :-

⑦ Oxidation :-

$$\text{n.alkane} \xrightarrow{KMnO_4\ \Delta} \text{No oxidation}$$

isoalkanes having tertiary H



⇒ Controlled . oxidation :-

$$CH_4 + O_2 \xrightarrow{Cu,\ 300^\circ C,\ \text{high pressure}} CH_3{-}OH$$

$$CH_4 + O_2 \xrightarrow{Mo_2O_3,\ \Delta} CH_2{=}O$$

$$CH_4 + H_2O \xrightarrow{Ni\ \Delta} CO + 3H_2$$

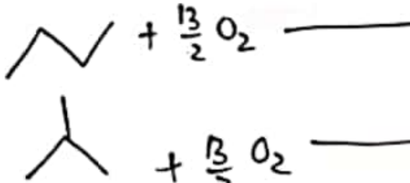
⇒ Combustion

$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)H_2O + \text{Heat of combustion.}$$

Heat of combustion $\propto$ no. of atom in reactan.

∴ ∴ ∴ $\propto \frac{1}{\text{Stability}}$ (for isomers)

> Branched isomers > Unbranched isomer (Stability)

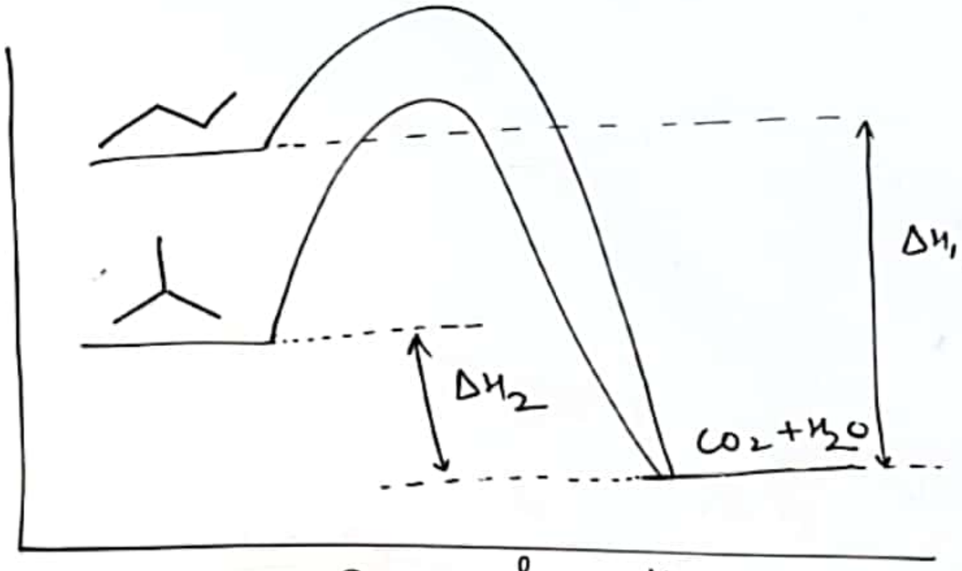
eg  $+ \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O \quad \Delta H_1$

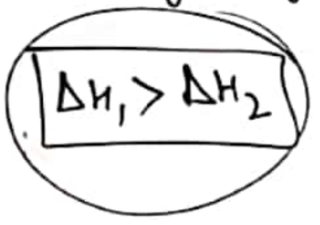
 $+ \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O \quad \Delta H_2$

more stable

HYDROCARBONS ⇒ (I) · ALKANE

CHEMICAL REACTIONS OF ALKANES :-





comparing magnitude (not sign)

ΔH = -ve exothermic

ΔH = +ve Endothermic

⇒ **Combustion**

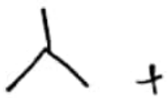
$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)H_2O + \text{Heat of combustion}$$

Heat of combustion $\propto$ no. of atom in reactan.

∴ ∴ $\propto \frac{1}{\text{Stability}}$ (for isomers)

Branched isomers > Unbranched isomer (Stability)

eg  $+ \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O \quad \Delta H_1$

 $+ \frac{13}{2}O_2 \longrightarrow 4CO_2 + 5H_2O \quad \Delta H_2$

more stable

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

Olefins

METHODS OF FORMATION :-

① By Grignard Reagent : (For C=C, C≡C)

eg. $CH_2=\overset{\delta-}{C}H-MgX \xrightarrow[\text{(}-O-H,\ -\overset{|}{N}-H,\ -S-H,\ -C\equiv C-H,\ X-H\ \text{AMG etc.)}]{Z-\overset{\delta+}{H}\ \text{acidic}} CH_2=CH_2 + Mg\begin{matrix}X\\Z\end{matrix}$

eg $R-C\equiv\overset{\delta-}{C}-MgBr \xrightarrow{H_2O} R-C\equiv C-H + Mg\begin{matrix}Br\\OH\end{matrix}$

(2.) By Kolbe's electrolysis :- (for C=C, C≡C)

CH_2-COOK (Pot. Succinate) $\xrightarrow[(S_{FR})]{\text{electrolysis}}$ $CH_2=CH_2$ (ethylene) $+ CO_2 + H_2$

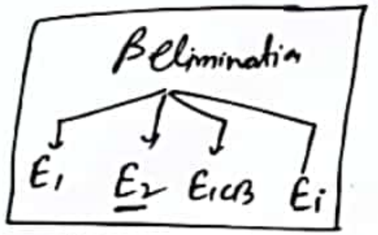
eg 2 (Trans) Pot. Fumarate $\xrightarrow{\text{electrolysis}}$ $CH\equiv CH$

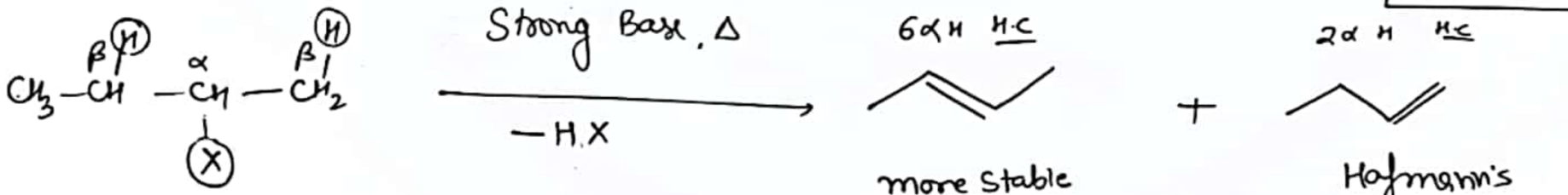
eg 3 (cis) Pot. Maleate. $\xrightarrow{\text{"}}$ $CH\equiv CH$

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(3) By Dehydrohalogenation in alkyl halide (for C=C, C≡C)





more Stable
(Saytzeff's Product)
(generally major)

Hofmann's product

⇒ This elimination is an example of β-elimination
↳ (elimination of 2 groups from adjacent position)

⇒ This is E_2 elimination.

⇒ Elemination reactions are endotheric reaction.

⇒ E_2 elimination is favourable on Strong Base (like alc. KOH, Δ or R-OH + KOH, Δ or $RO^{\ominus}$, Δ

$NaNH_2$, Δ)

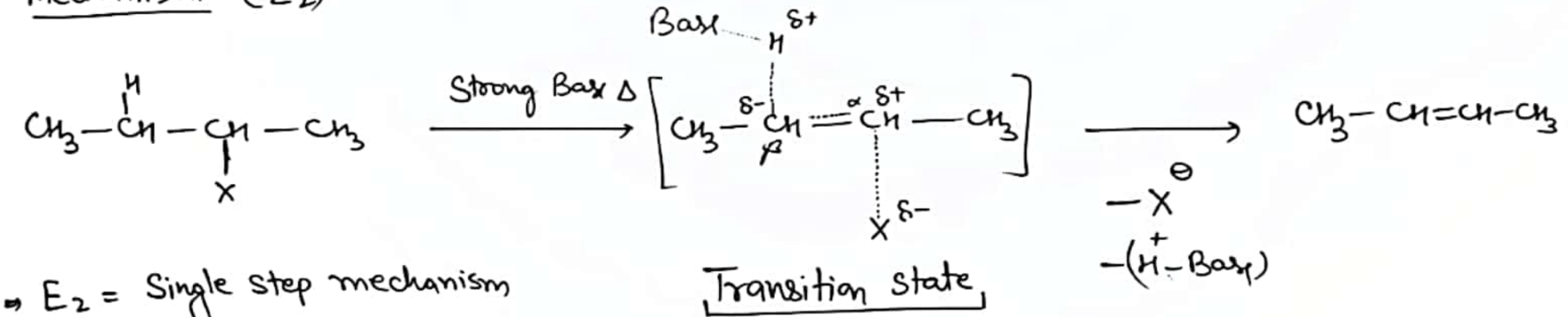
stronger base than alc. KOH

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(3) By Dehydrohalogenation in alkyl halide (for C=C, C≡C)

Mechanism (E_2)



Transition state

⇒ E_2 = Single step mechanism

⇒ No Rearrangement

⇒ Anti elimination reaction (elimination of 2 groups from oppsite plane)

⇒ Generally Saytzeff's product form as major product (Saytzeff's product = more no. of alkyl group containing carbon-carbon multiple bond)

⇒ Reactivity of R-X ⇒ R-I > R-Br > R-Cl > R-F

⇒ " " " " ⇒ 3° R-X > 2° R-H > 1° R-X

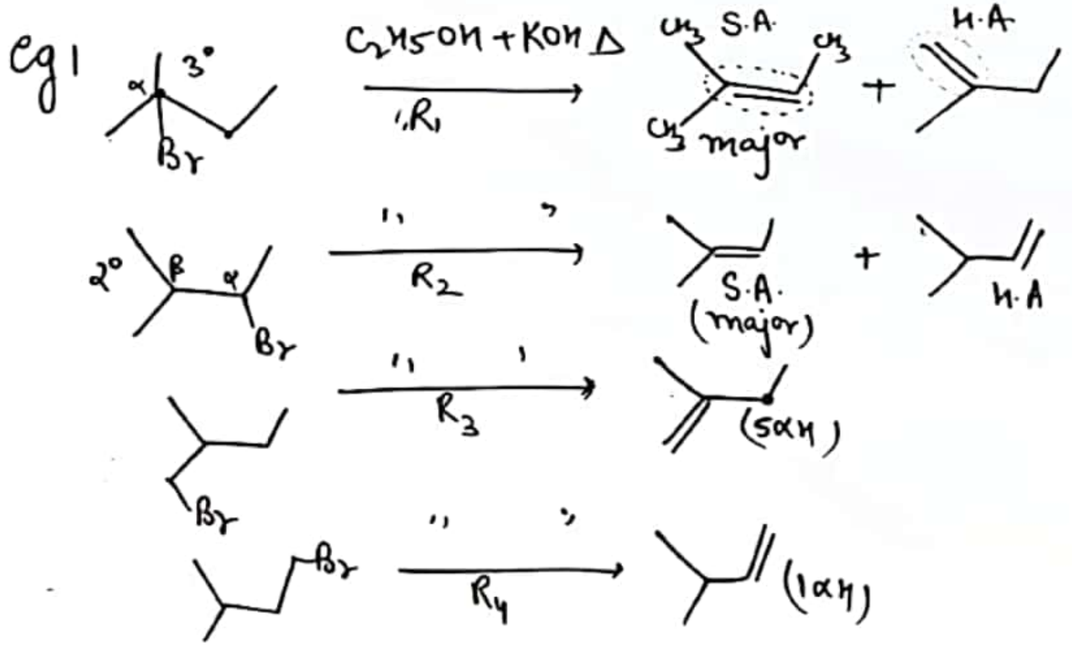
HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(3) By Dehydrohalogenation in alkyl halide (for C=C, C≡C)

⇒ Rate of E_2 elimination $\propto [R\text{-}X]^1 [Base]^1$

eg 1



reactivity of E_2 $\propto$ stability of alkene

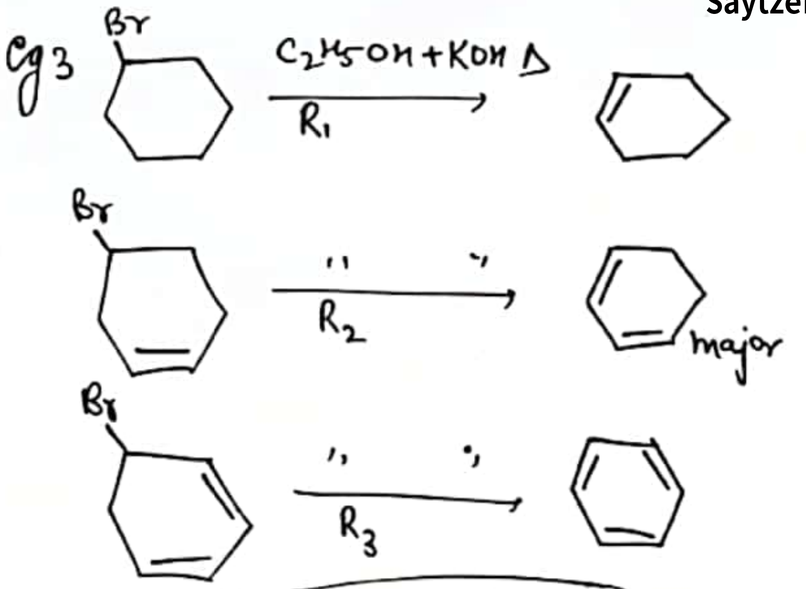
↳ $(R_1 > R_2 > R_3 > R_4)$

eg 2



Saytzeff product, yet minor

eg 3

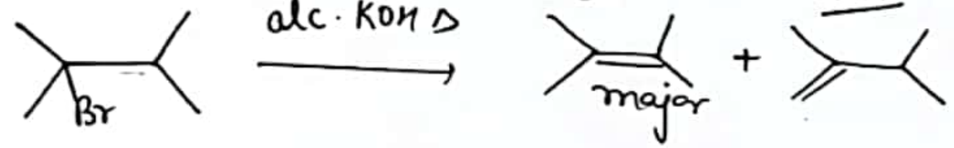


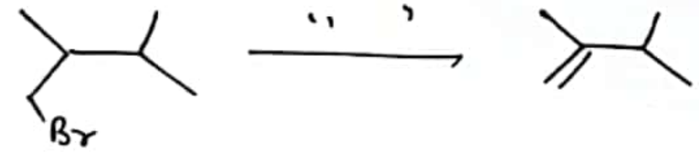
Reactivity = $R_3 > R_2 > R_1$

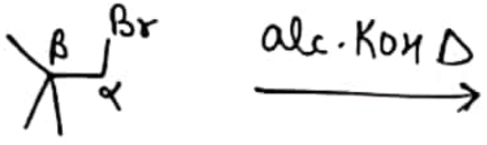
HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

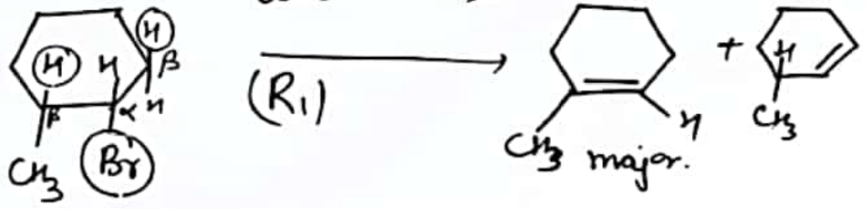
(3) By Dehydrohalogenation in alkyl halide (for C=C, C≡C)

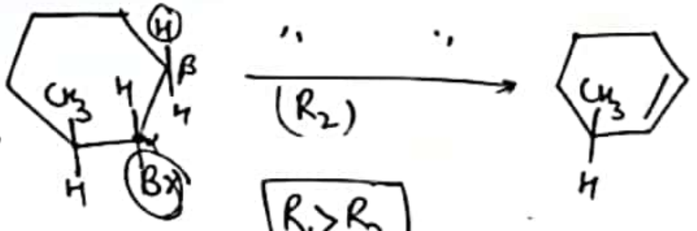
eg 

eg 

eg 

eg

Cis 

Trans 

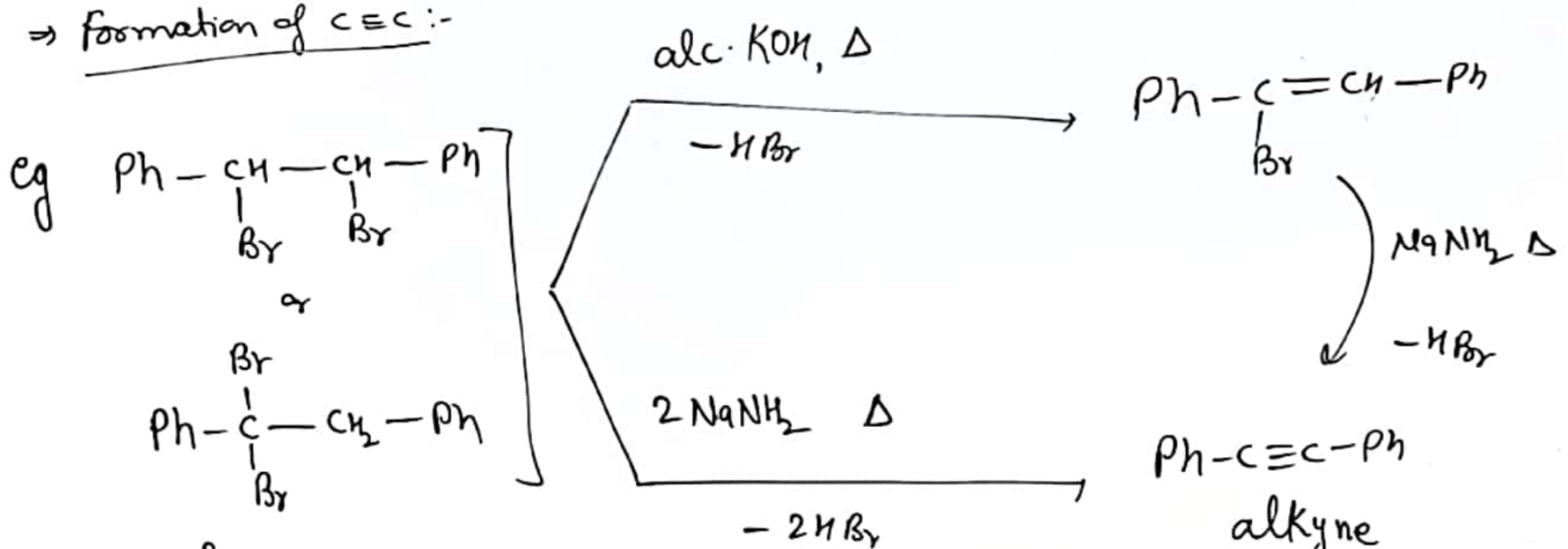
$R_1 > R_2$

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(3) By Dehydrohalogenation in alkyl halide (for C=C, C≡C)

⇒ Formation of C≡C :-

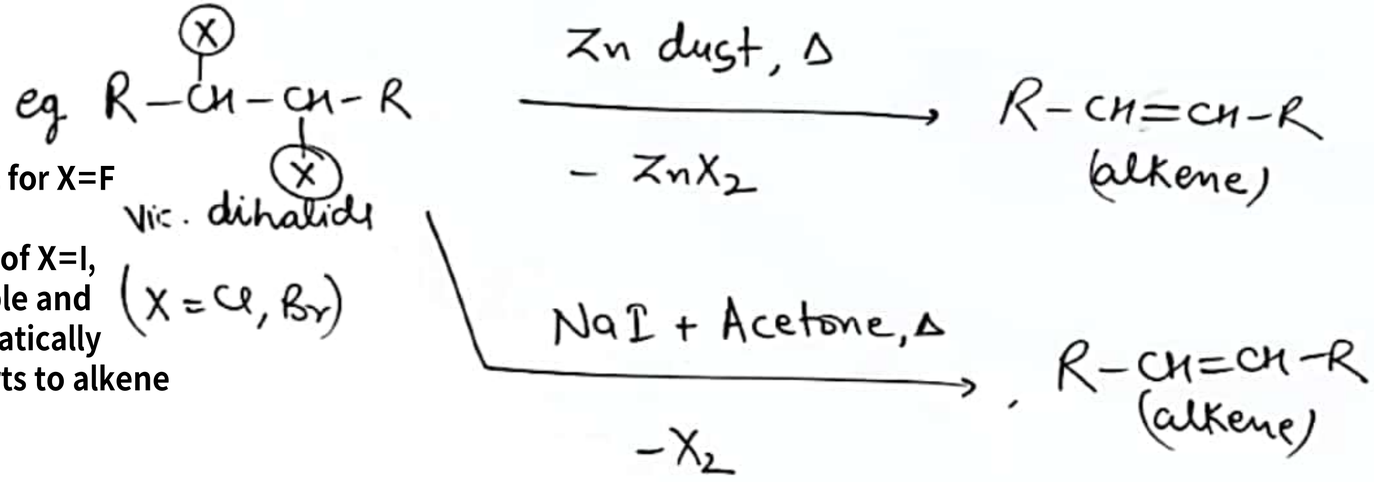




HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

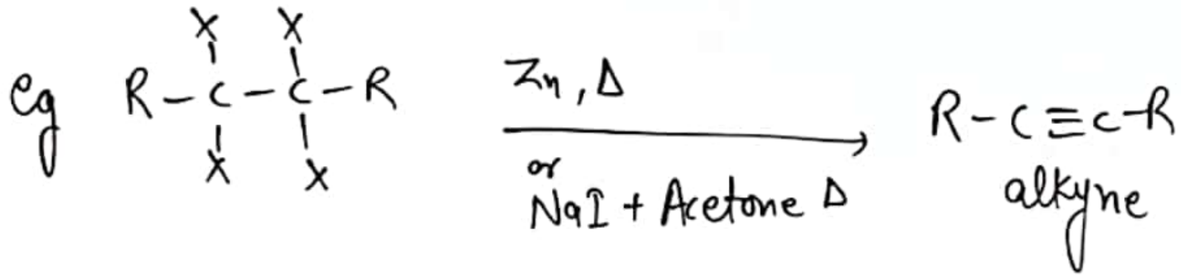
METHODS OF FORMATION :-

(4) By Dehalogenation of Vicinal dihalides :- (for C=C, C≡C)



no rex. for X=F

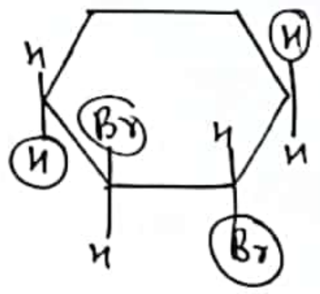
incase of X=I, unstable and automatically converts to alkene



HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(4) By Dehalogenation of Vicinal dihalides :- (for C=C , C≡C)

eg 

alc. KOH Δ
(dehydrohalogenation)



Zn , Δ
(Dehalogenation)



eg 

Zn , Δ



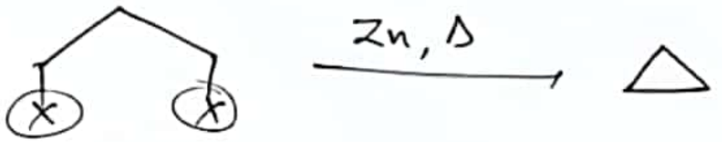
HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

Other dehalogenation reaction with Zn.

① $R-\underset{\cdot\cdot}{X} + Zn + \underset{\cdot\cdot}{X}-R \xrightarrow{\Delta} R-R$

②



③



④ $R-CH_2-CH_2-R \xrightarrow[-2ZnX_2]{\Delta} R-CH=CH-R$

⑤ $R-C\equiv C-R \xrightarrow[-AgX\downarrow]{\Delta} R-C\equiv C-R$

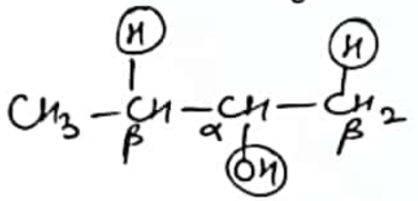
X = Cl, Br, I

for all 5 reactions

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(5) By Dehydration in alcohols :- (only for formation of C=C)



$$\xrightarrow[-H_2O]{\text{Dehydrating reagents}} CH_3-CH=CH-CH_3 + CH_3-CH_2-CH=CH_2$$

Saytzeff's alkene — Hafmann's alkene

Dehydrating agents:-

① Conc. H_2SO_4, Δ
② Conc. H_3PO_4, Δ

⇒ Mainly E_1 elimination
⇒ Carbocation = intermediate
⇒ Rearrangement possible
⇒ Major product = Saytzeff's product

③ Al_2O_3, Δ
④ P_2O_5, Δ

⇒ E_2
⇒ No rearrangement
⇒ Saytzeff product = major

(5.) ThO_2 Δ
(6.) WO_2, Δ

⇒ E_2
⇒ No rearrangement
⇒ Hafmann's product = major

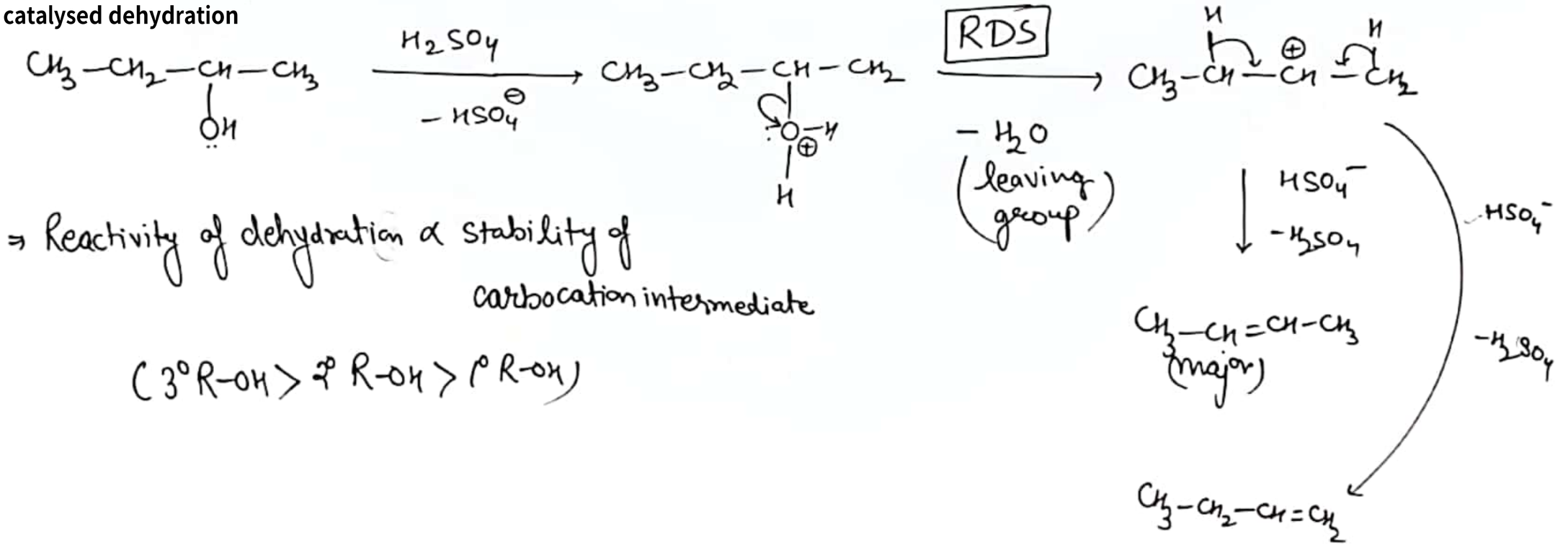
HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(5) By Dehydration in alcohols :- (only for formation of C=C)

Mechanism (E_1)

acid catalysed dehydration



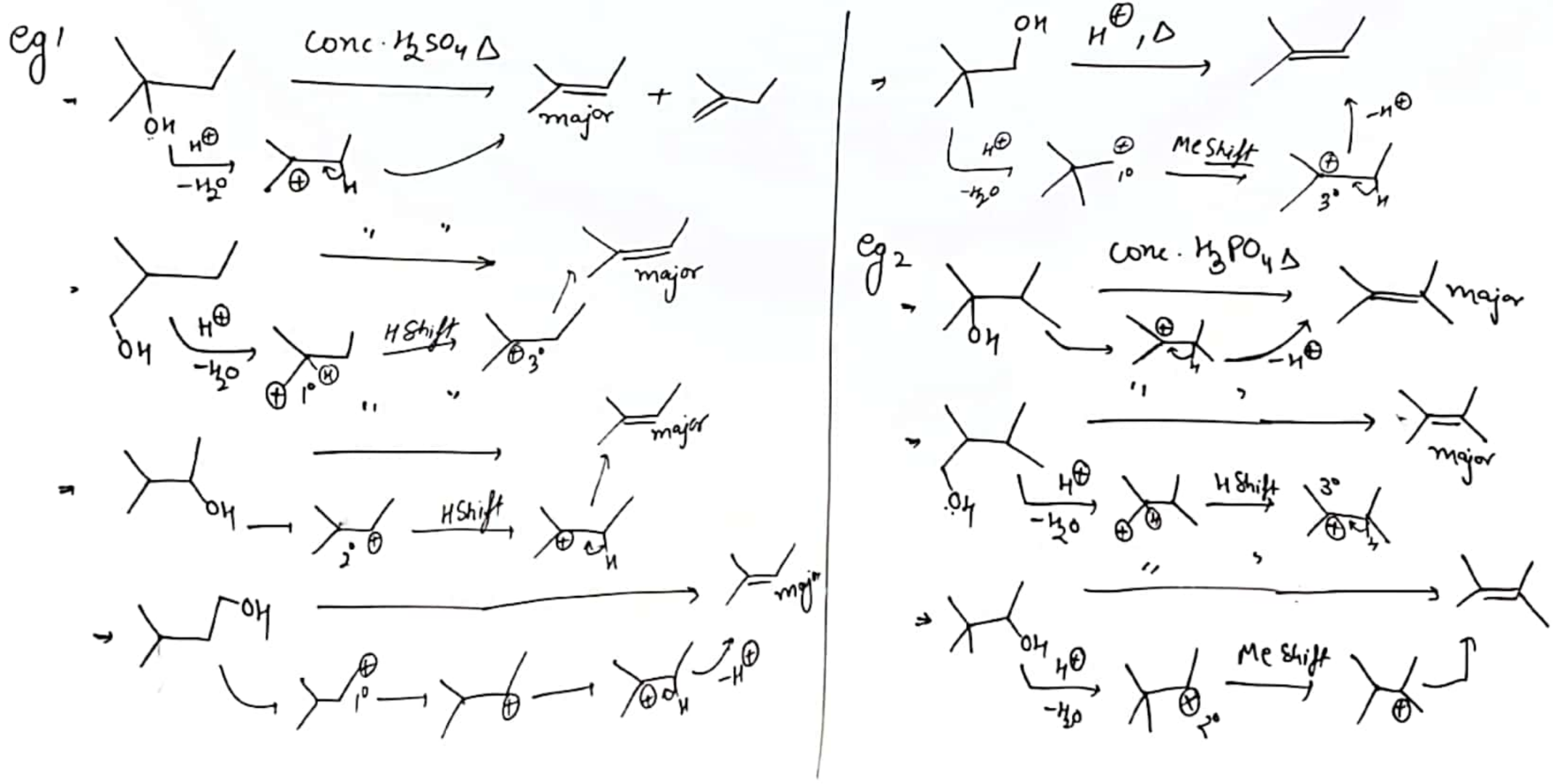
⇒ Reactivity of dehydration ∝ stability of carbocation intermediate

(3° R–OH > 2° R–OH > 1° R–OH)

HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

METHODS OF FORMATION :-

(5) By Dehydration in alcohols :- (only for formation of C=C)

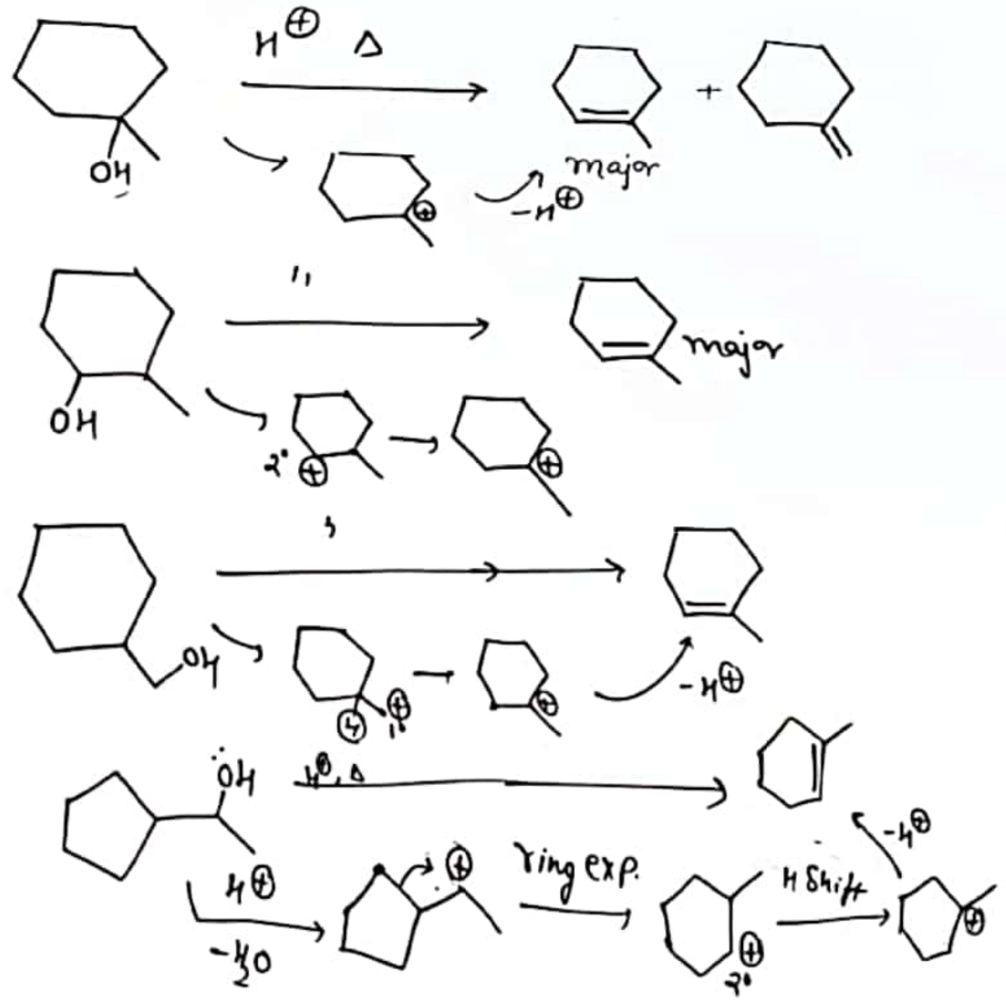


HYDROCARBONS ⇒ (II) ALKENES & ALKYNES

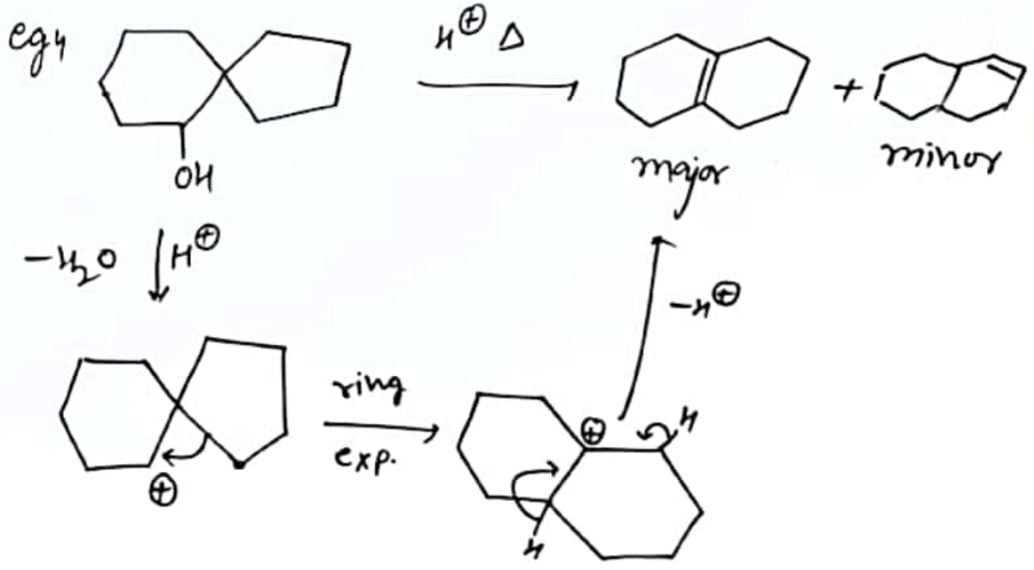
METHODS OF FORMATION :-

(5) By Dehydration in alcohols :- (only for formation of C=C)

eg 3



eg 4

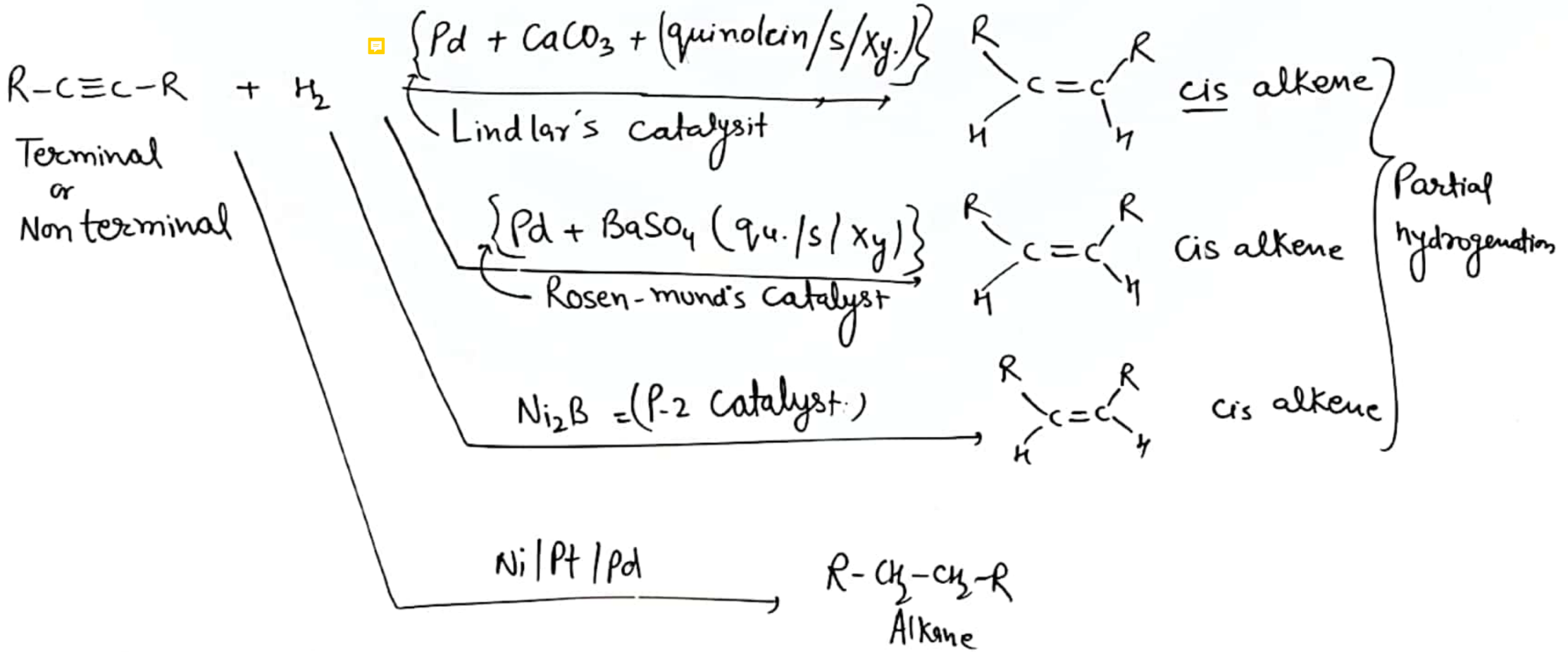


HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

METHODS OF FORMATION :-

(6) By Partial hydrogenation of alkynes:- (For C=C)

poisons (which lower the rate of reaction)

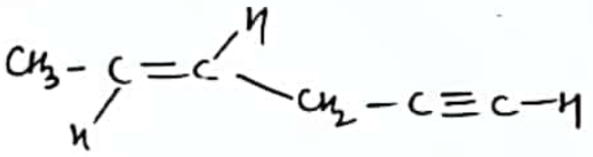


HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

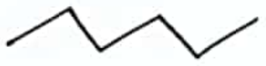
METHODS OF FORMATION :-

(6) By Partial hydrogenation of alkynes:- (For C=C)

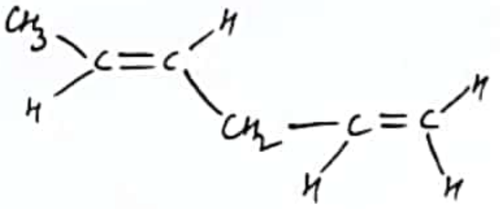
Eg 1



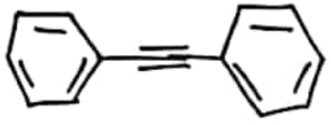
$H_2 + Ni$



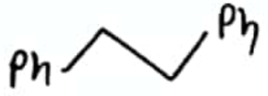
$H_2 + Pd + CaCO_3$



Eg 2



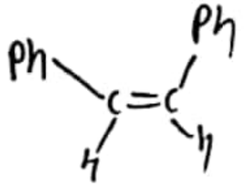
$H_2 + Pd$



1,2-Diphenylethane

$H_2 + Pd$

$BaSO_4 + Qu.$



HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

METHODS OF FORMATION :-

(7) By Birch Reduction of Non terminal alkyne (for ^{n} formation C=C)

$$R-C\equiv C-R \xrightarrow[-M^{\oplus}NH_2^{\ominus}]{(Na/K/Li) + NH_{3(l)}} \underset{\text{Trans alkene}}{\overset{R}{\underset{H}{C=C}}\overset{H}{\underset{R}{}}}$$

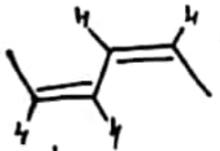
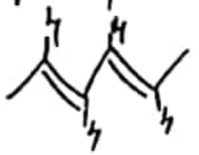
(Birch Reduction)

Non terminal alkyne

H from ammonia gets attached on alkene

$$R-C\overset{sp}{\equiv}C\overset{\delta+}{-}H \xrightarrow[\text{(Acid-Base Rxn)}]{Na + NH_{3(l)}} R-C\equiv C^{\ominus}Na^{\oplus} + \frac{1}{2}H_2$$

Eg $Me-C\equiv C-C\equiv C-Me$

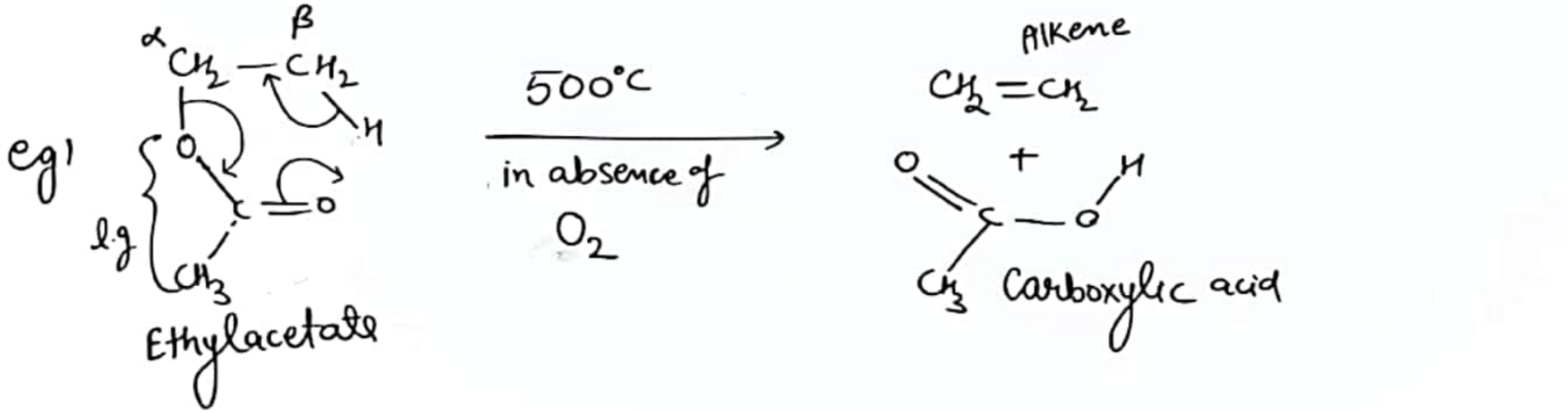
- $H_2 + Pd$ → 
- $H_2 + Pd + CaCO_3$ →  (Cis-Cis)
- $Na + NH_{3(l)}$ →  Trans-Trans.

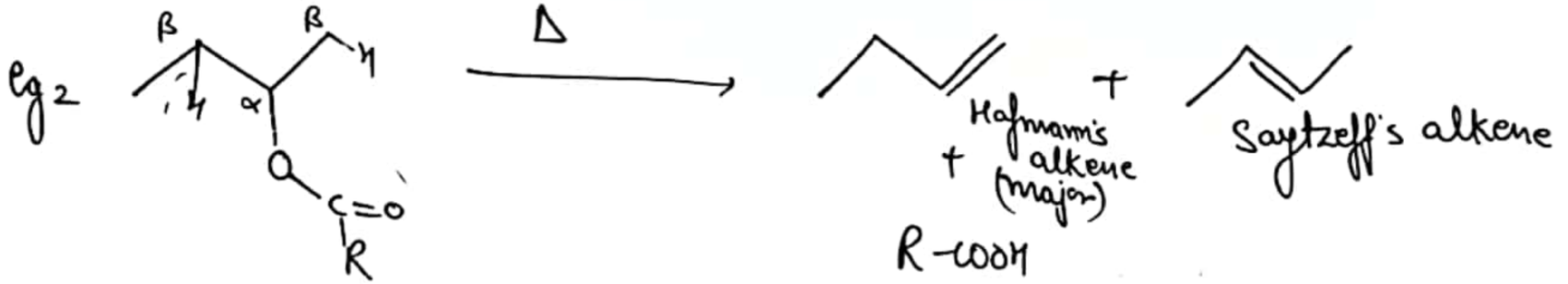
HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

METHODS OF FORMATION :-

⑧ By Pyrolysis of Esters :-

⇒ β-elimination :-





⇒ Hofmann's product = major product.

due to steric factor

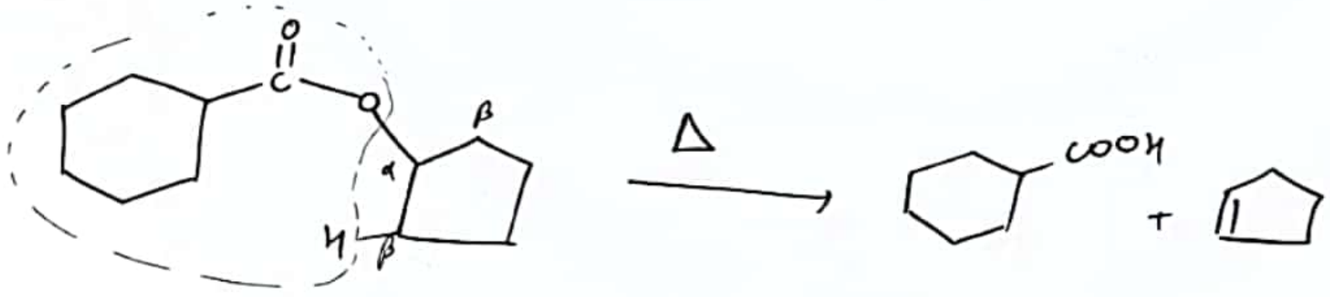
HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

METHODS OF FORMATION :-

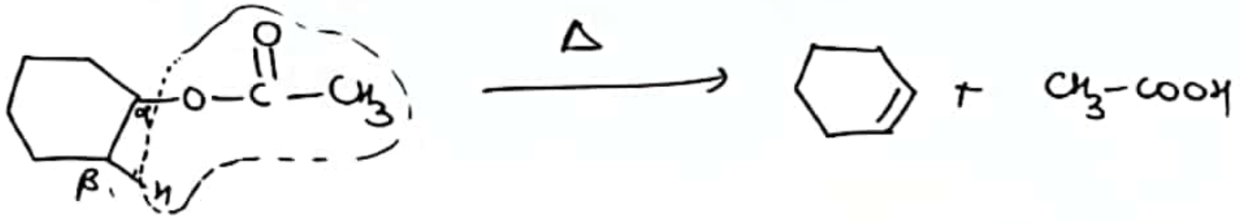
⑧ By Pyrolysis of Esters :-

⇒ β-elimination :-

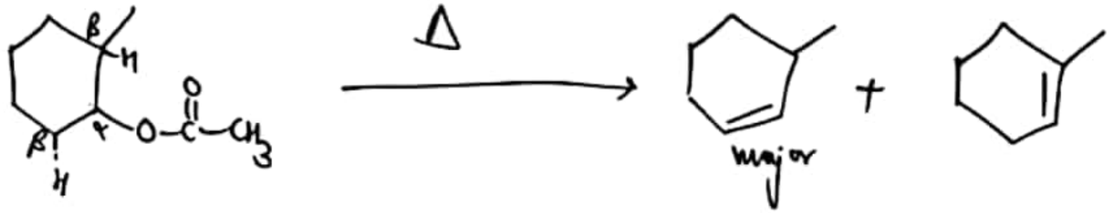
eg3



eg4



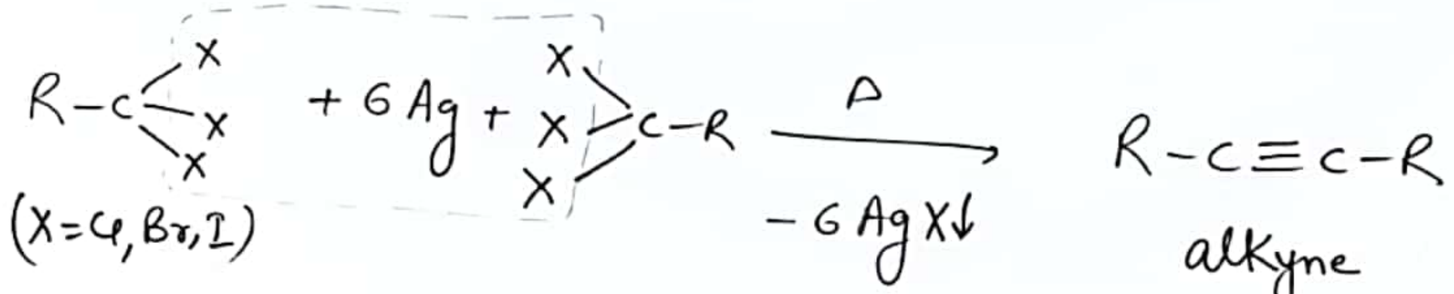
eg5



HYDROCARBONS → (II) ALKENES AND ALKYNES

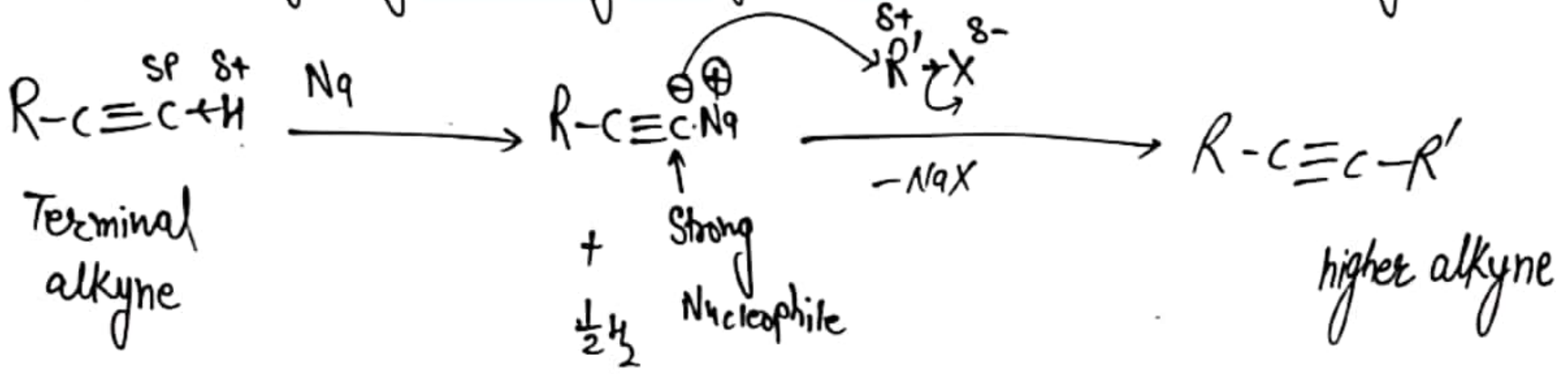
METHODS OF FORMATION :-

(9) By 1,1,1-Trihaloalkane with Ag, Δ :- (for C≡C)



eg $2CHI_3 + 6Ag \xrightarrow{\Delta} CH \equiv CH$

(10) Formation of higher alkyne from lower terminal alkyne (for C≡C)



HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

METHODS OF FORMATION :-

(11) By Hydrolysis of metal carbides :- (for C≡C)

$^{-1}C\equiv C^{-1}$ | $CaC_2 + 2H_2O \longrightarrow H-C\equiv C-H + Ca(OH)_2$
acetylene

$(^{-3}C-C\equiv C^{-1})$ $Mg_2C_3 + 4H_2O \longrightarrow H_3C-C\equiv C-H + 2Mg(OH)_2$
propyne

(C_3^{-4})

Formation of methane

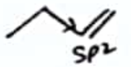
C^{-4} $Be_2C + 4H_2O \longrightarrow CH_4 + 2Be(OH)_2$

C^{-4} $Al_4C_3 + 12H_2O \longrightarrow 3CH_4 + 4Al(OH)_3$

HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

PHYSICAL PROPERTIES OF ALKENES AND ALKYNES

ALKENES



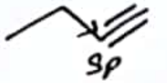
⇒ C_2 to C_4 Alkenes ⇒ Gas

⇒ C_5 to C_{17} Alkenes ⇒ Liquid

⇒ C_{18} and higher Alkenes ⇒ Solid

⇒ Alkenes are insoluble in water but soluble in benzene, CCl_4, $CHCl_3$ etc.

ALKYNES



⇒ $C_2 - C_4$ = Gas

⇒ $C_5 - C_{11}$ → Liquid

⇒ C_{12} – higher ⇒ Solid

⇒ Lower alkynes are partial soluble in water but higher alkynes are insoluble.

⇒ **B.P.** (provided molecular wt. is approx. same)

① B.P. ∝ intermolecular H-bonding

② ∴ ∝ Dipole moment (~ M.Wt)

③ ∴ ∝ M.Wt

④ ∴ ∝ $\frac{1}{\text{Branching}}$ (For isomers)

M.P

① M.P. ∝ M.Wt

② M.P ∝ H-Bonding

③ M.P ∝ Symmetrical Structure

④ M.P ∝ Branching (isomers)

not hard and fast rule generally

Solubility in water

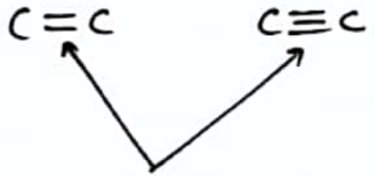
① Solubility ∝ intermolecular H-bonding with water molecules

② ∴ ∴ ∝ $\frac{1}{\text{M.Wt}}$

③ ∴ ∴ ∝ Branching (For isomers)

HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

CHEMICAL REACTIONS OF ALKENES AND ALKYNES:-



loosely bonded πe^- density

⇒ Electrophilic addition reaction given by alkenes and alkynes.

Electrophile ($E^{\oplus}$)

↳ Electron deficient

- Vacant orbital containing
- ⇒ Attacks on higher e^- density containing position.
- ⇒ Examples

ⓐ Charged electrophiles:-

$H^{\oplus}$, $R^{\oplus}$, $X^{\oplus}$, $NO^{\oplus}$, $NO_2^{\oplus}$, $R-\overset{O}{\overset{\|}{C}}{}^{\oplus}$ etc.

ⓑ Neutral electrophiles:-

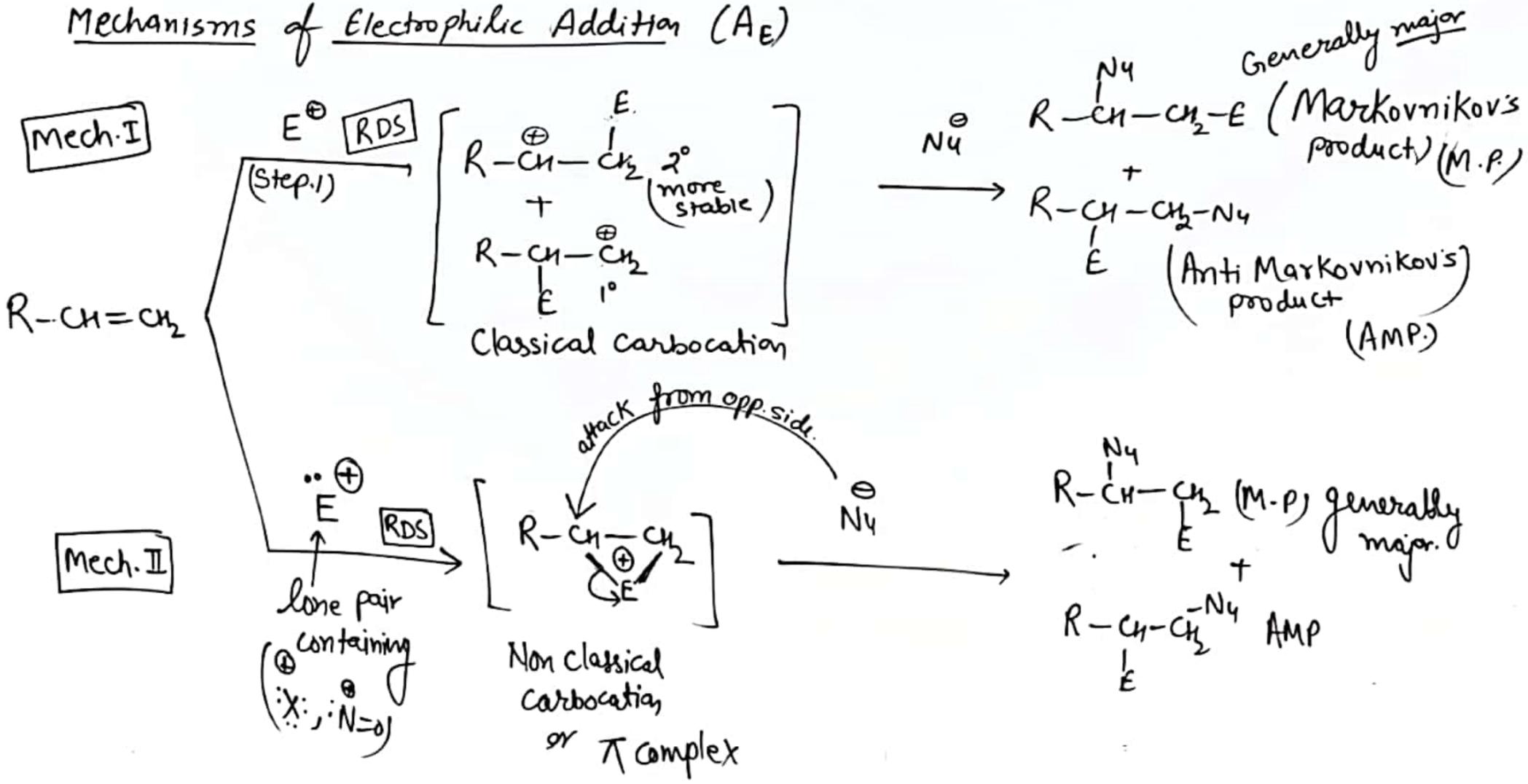
⇒ BX_3, Borane (B_2H_6), $:CH_2$ Carbene

⇒ SO_2, SO_3, CO_2, CCl_4 etc.

HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

CHEMICAL REACTIONS OF ALKENES AND ALKYNES:-

Mechanisms of Electrophilic Addition (A_E)



HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

CHEMICAL REACTIONS OF ALKENES AND ALKYNES :-

⇒ Reactivity of A_E

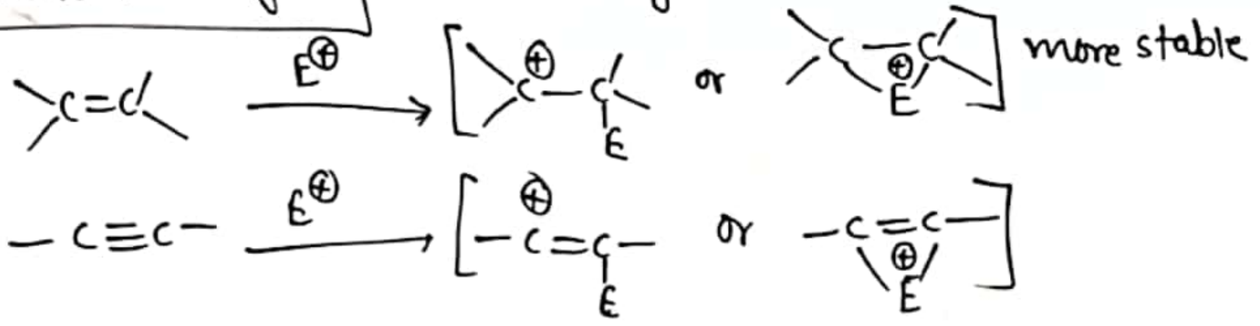
① Reactivity α π e^- density on carbon - carbon multiple bond.

② " " α Stability of intermediate.

③ " " ↑es in presence of EDG. (+M, +H.C, +I) on C=C/C≡C

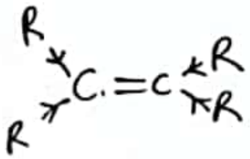
④ " " ↓es " " " EWG (-M, -I ↓) " " "

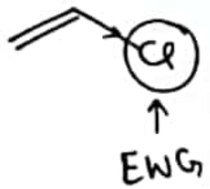
Eg: **Alkene > Alkyne** A_E Reactivity

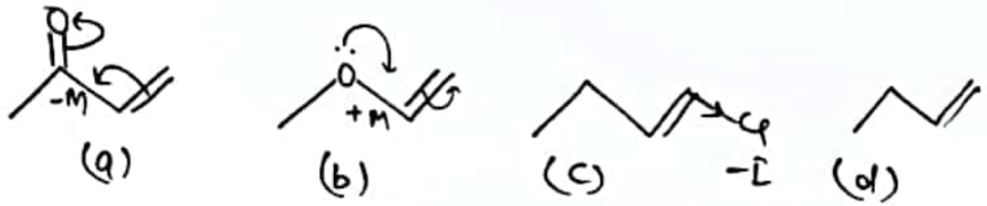


HYDROCARBONS ⇒ (II) ALKENES AND ALKYNES

CHEMICAL REACTIONS OF ALKENES AND ALKYNES :-

Eg 2  > $R_2C=CH-R$ > $R-CH=CH-R$ > $R-CH=CH_2$ > $CH_2=CH_2$ > $H-C\equiv C-H$

Eg 3  < $CH_2=CH_2$

Eg 4 

(a) (b) (c) (d)

Reactivity A_E = b > d > c > a

HYDROCARBONS → (II) ALKENES AND ALKYNES

CHEMICAL REACTIONS OF ALKENES AND ALKYNES :-

Mech. I

① intermediate = classical carbonium ion

② Rearrangement possible

③ Non stereospecific Rxn

④ Example

(i) Addition of HX
(ii) " " " Conc. H_2SO_4
(iii) " " " H_2O
(iv) " " " R-OH etc.

Mech. II

① intermediate = Non classical Carbocation.

② No Rearrangement in intermediate.

③ Stereospecific anti addition.

④ Examples.

(i) Addition of X_2
(ii) " " " HOX ($X_2 + H_2O$)
(iii) " " " X-X' etc

HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

① Addition of Hydrogen (Hydrogenation)

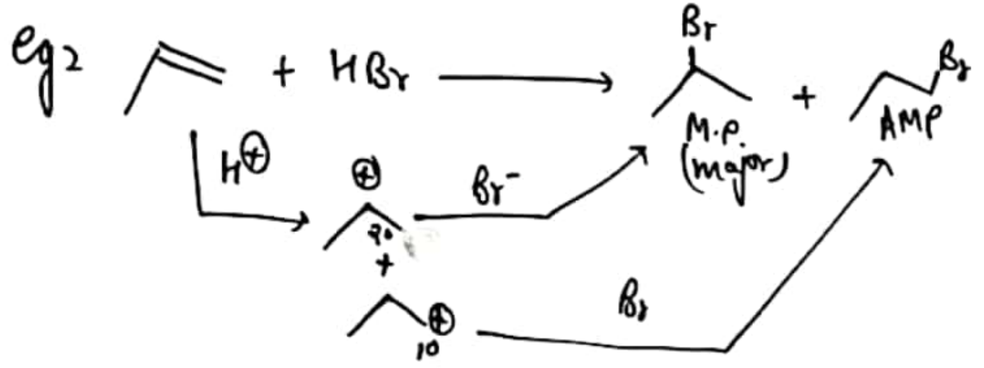
② Addition of HX (Hydrohalogenation)

⇒ According to mech-I
- Classical carbocation
- Rearrangement
- Generally (M.P. Product = major

Eg 1 $CH_2=CH_2 + \underset{\uparrow}{HX} \longrightarrow CH_3-\underset{X}{CH_2}$ (Alkyl halide)

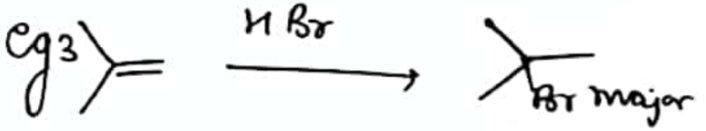
Reactivity = $HI > HBr > HCl > \underset{\uparrow \text{No Rxn.}}{H-F}$

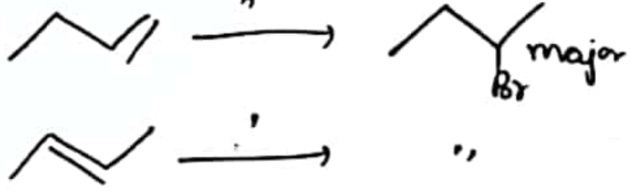
Eg 2

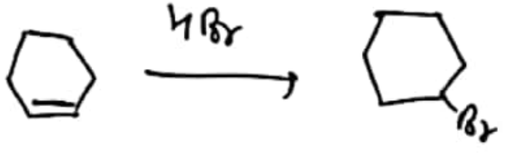


⇒ Markovnikov's Rule :-

On unsymmetrical alkene / alkyne addition of positively charge group take place on carbon of multiple bond, which has greater no. of hydrogen atoms.

Eg 3 

Eg 4 

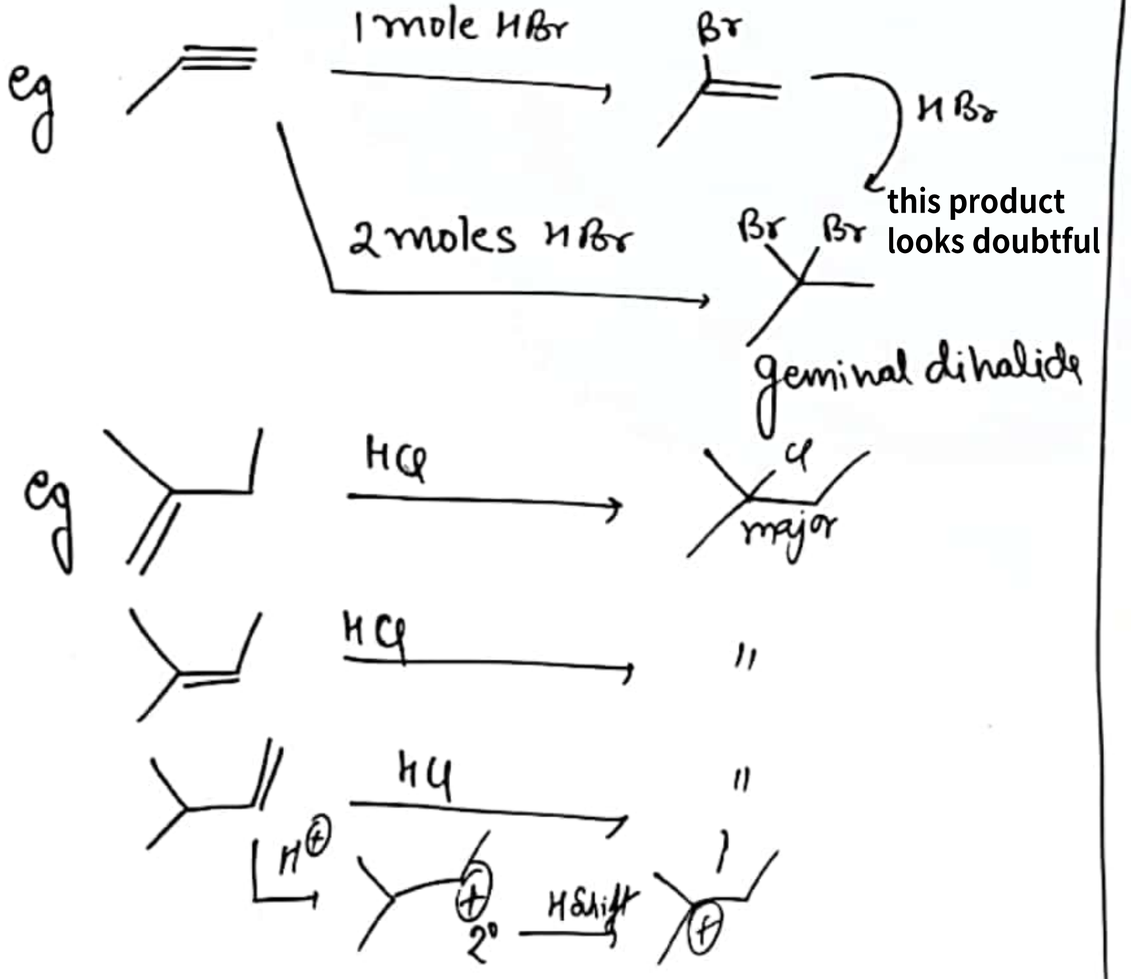
Eg 5 

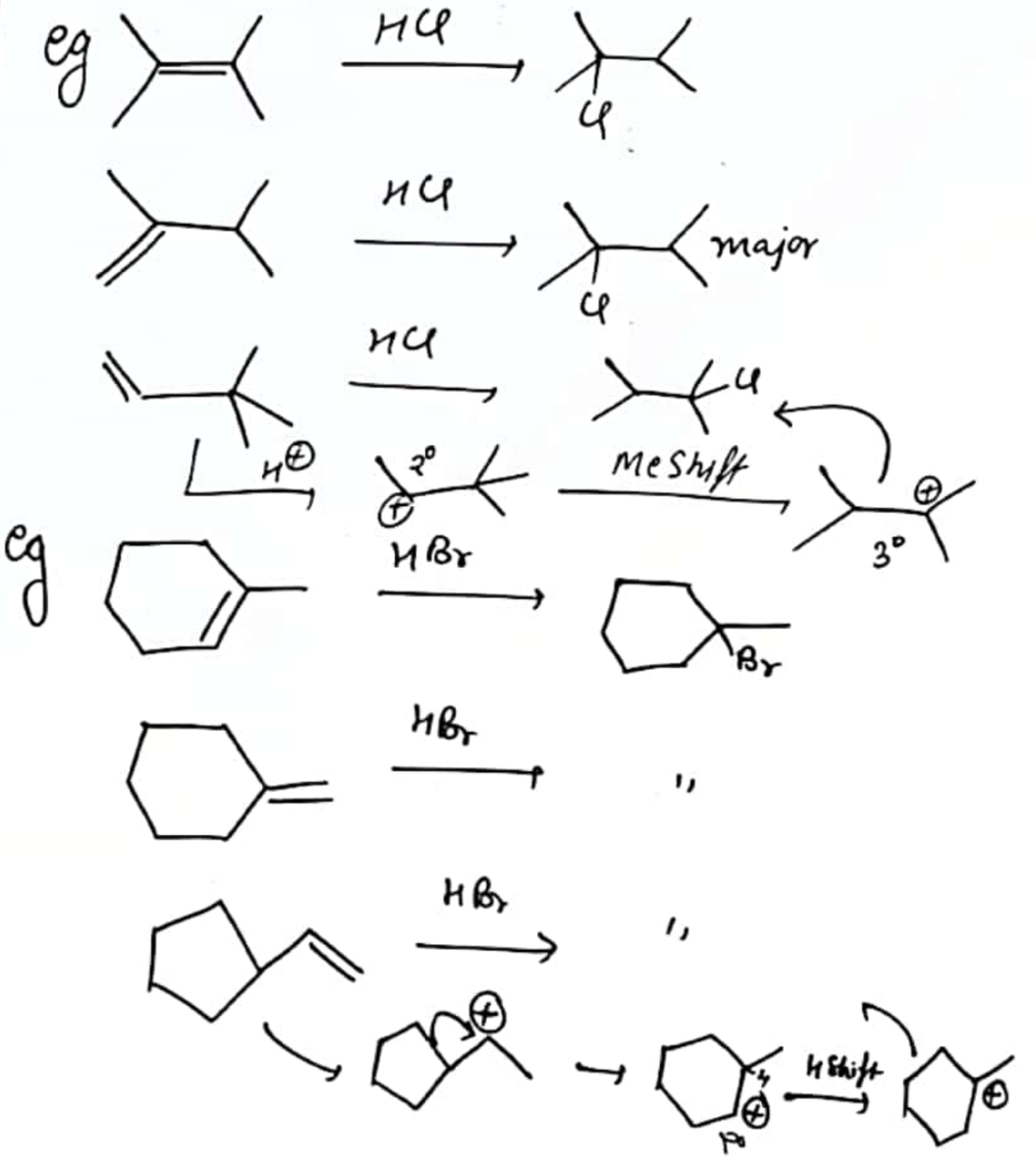
HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

① Addition of Hydrogen (Hydrogenation)

② Addition of HX (Hydrohalogenation)





HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

① Addition of Hydrogen (Hydrogenation)

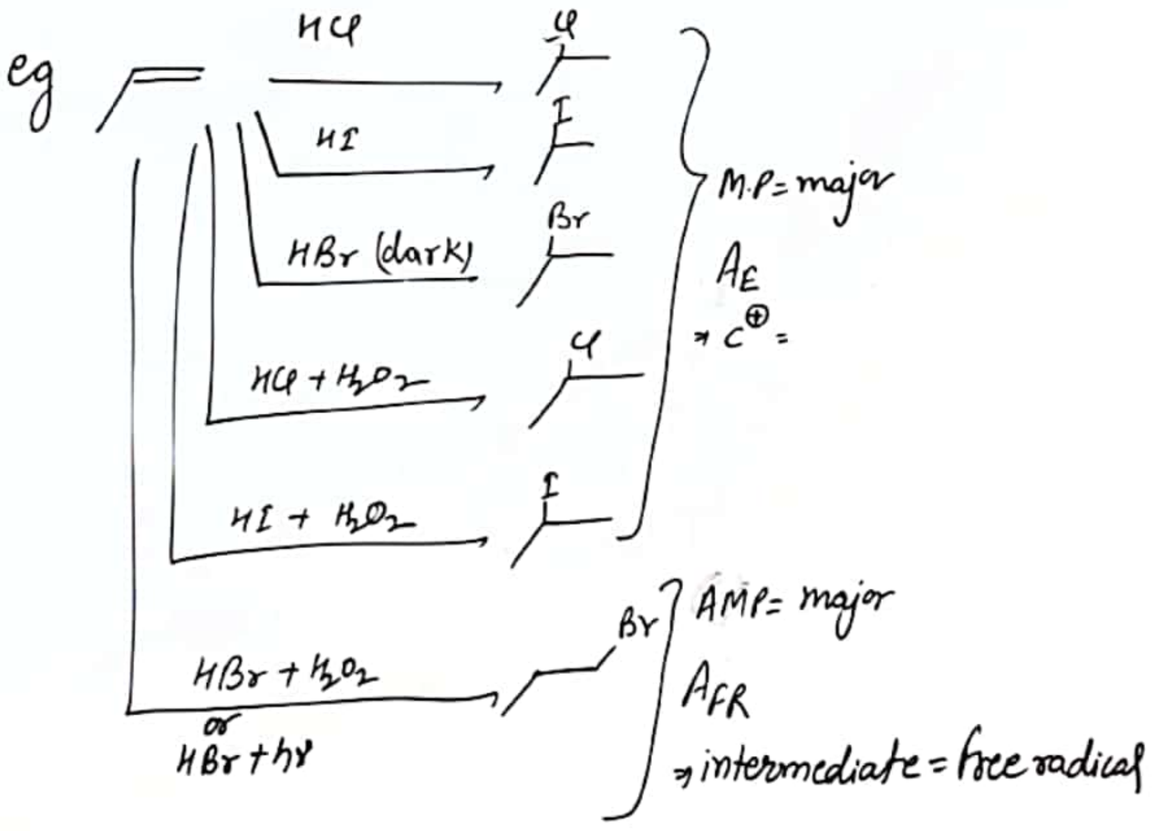
② Addition of HX (Hydrohalogenation)

Anti Markovnikov's addition of HBr :-

On unsymmetrical alkenes or alkyne Anti-markovnikov's addition take place of HBr in presence of Peroxide (H_2O_2, R-O-O-R) / $h\nu$.

- this addition is called "peroxide effect" or "Kharch effect.

⇒ This addition is free radical addition reaction.



HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

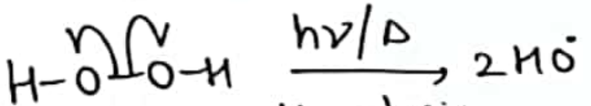
① Addition of Hydrogen (Hydrogenation)

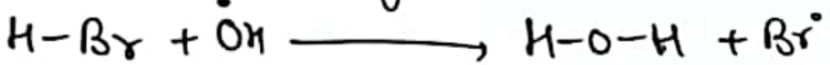
② Addition of HX (Hydrohalogenation)

Mechanism of free radical Addition:-

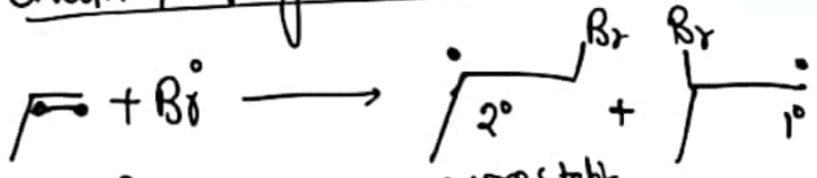
Chain reaction:

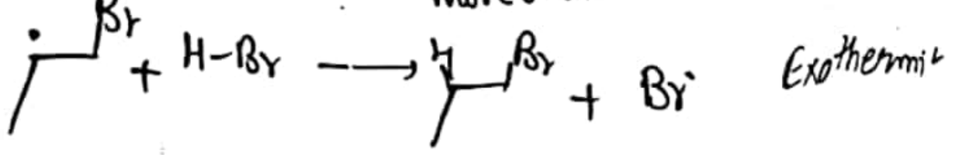
(i) Chain initiation Step:-



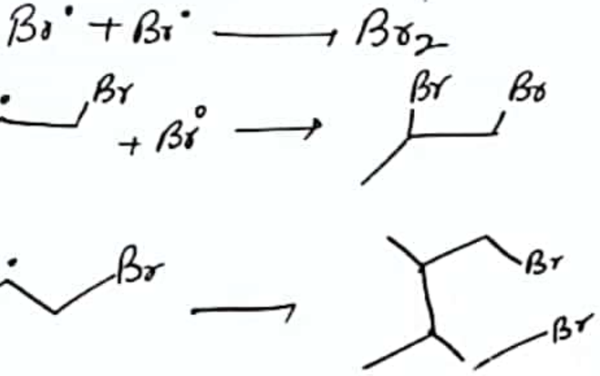


(ii) Chain propagation Step:-





(iii) Chain Termination Step:-



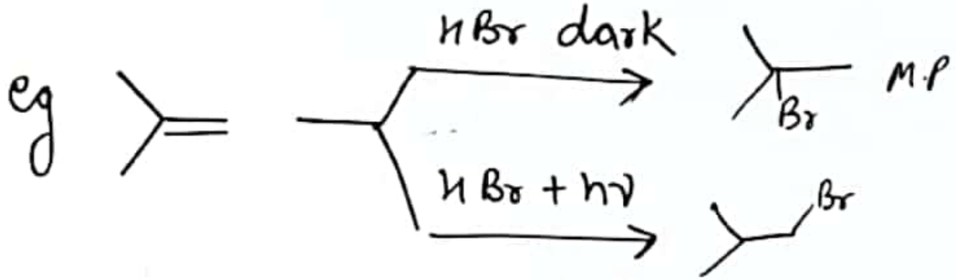
Note Out of four HX only HBr gives A_{FR} in presence of peroxide because both chain propagation steps are exothermic for only HBr.

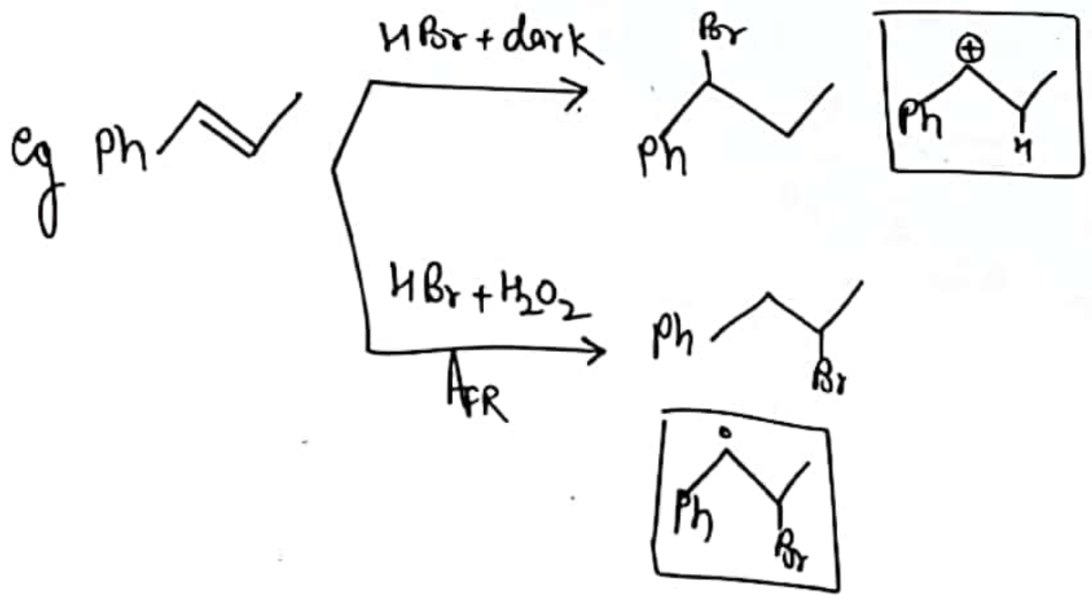
HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

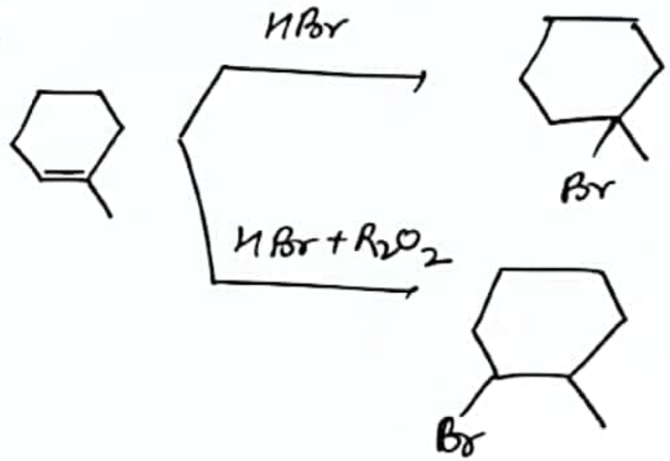
CHEMICAL REACTIONS

① Addition of Hydrogen (Hydrogenation)

② Addition of HX (Hydrohalogenation)

eg 



eg 

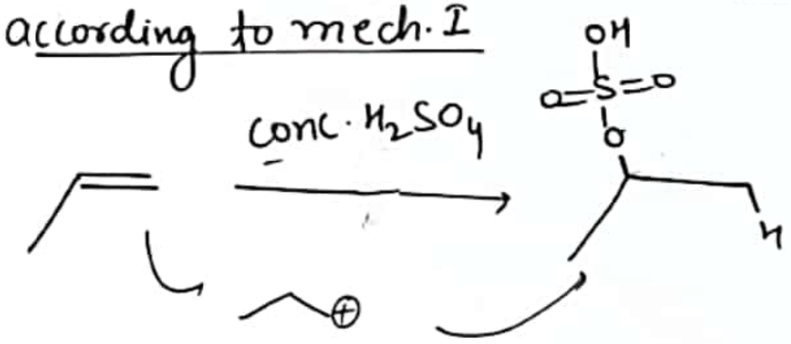
HYDROCARBONS = (II) ALKENE AND ALKYNES

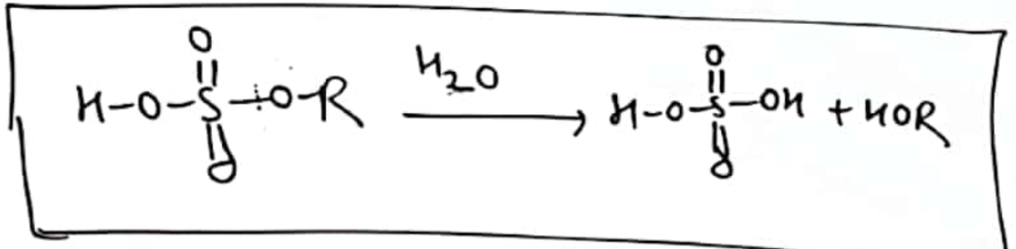
CHEMICAL REACTIONS

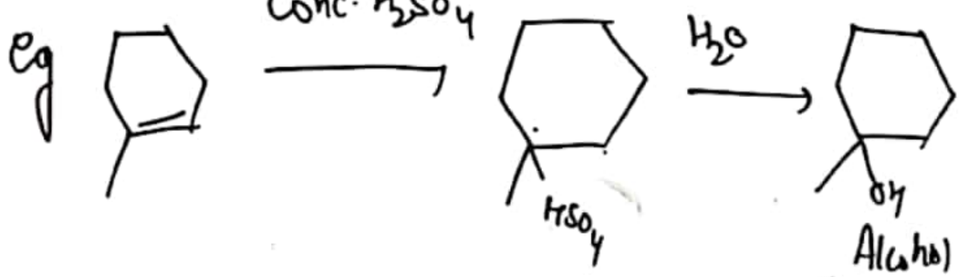
③ Addition of conc. H_2SO_4 :-

$H_2SO_4 \rightleftharpoons H^{\oplus} + HSO_4^{\ominus}$

according to mech. I

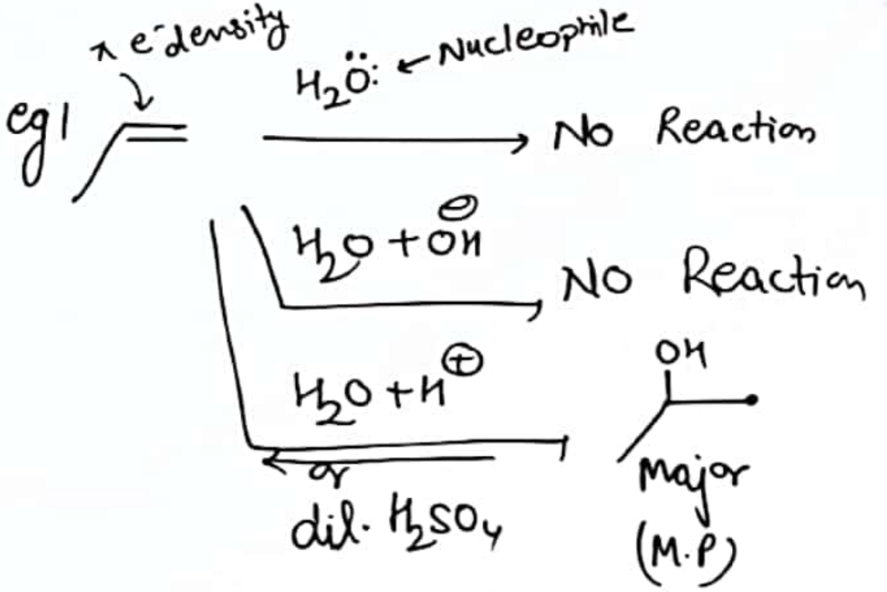


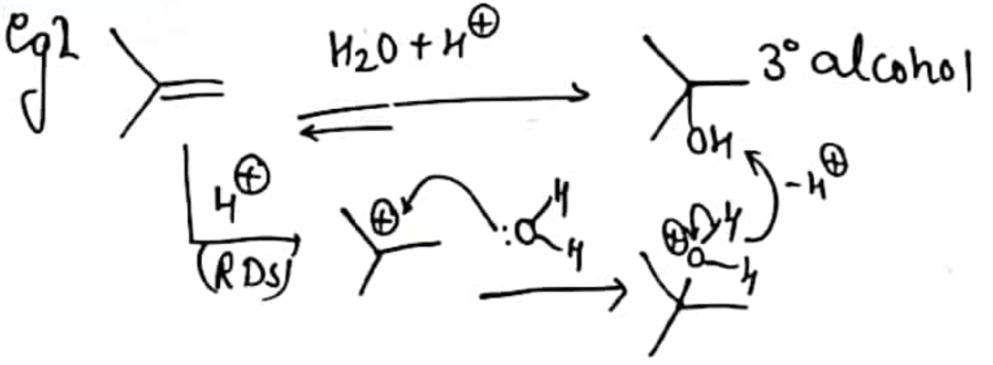




④ Addition of H_2O (Hydration)

according to mechanism - I

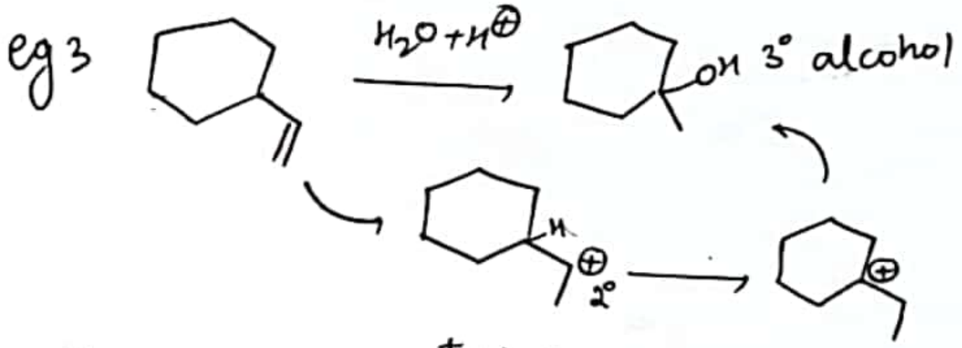




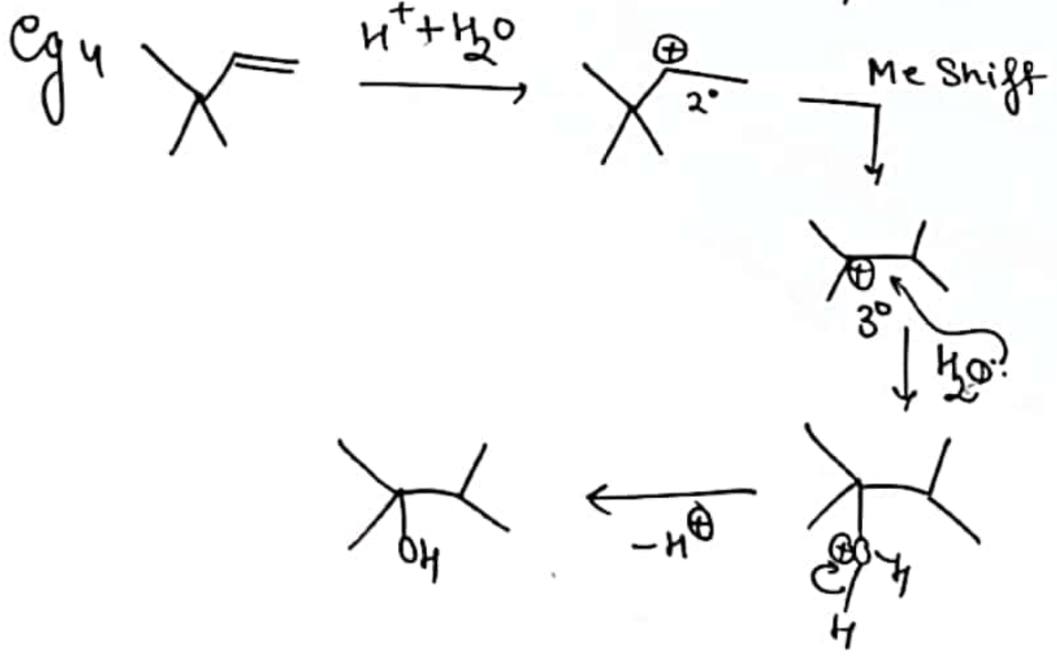
HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

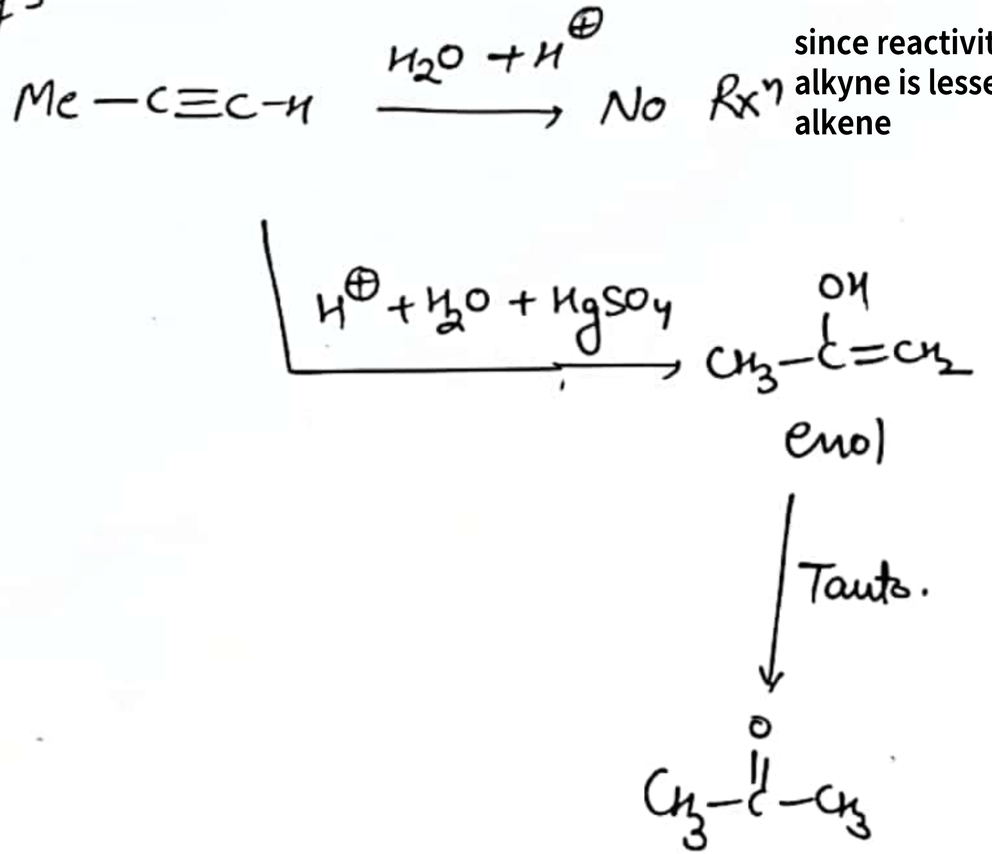
eg 3



eg 4



eg 5



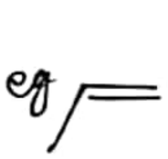
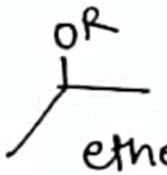
since reactivity of alkyne is lesser than alkene

HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

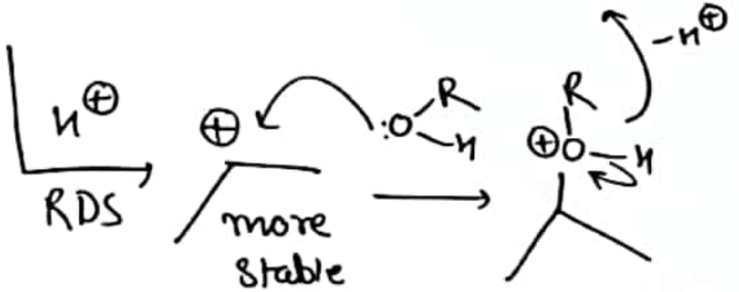
(5) Addition of alcohols:-

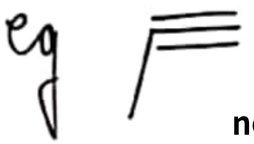
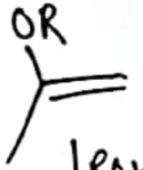
Same as addition H_2O

eg  $\xrightarrow{ROH + H^{\oplus}}$ 

ether

no rex. in basic medium



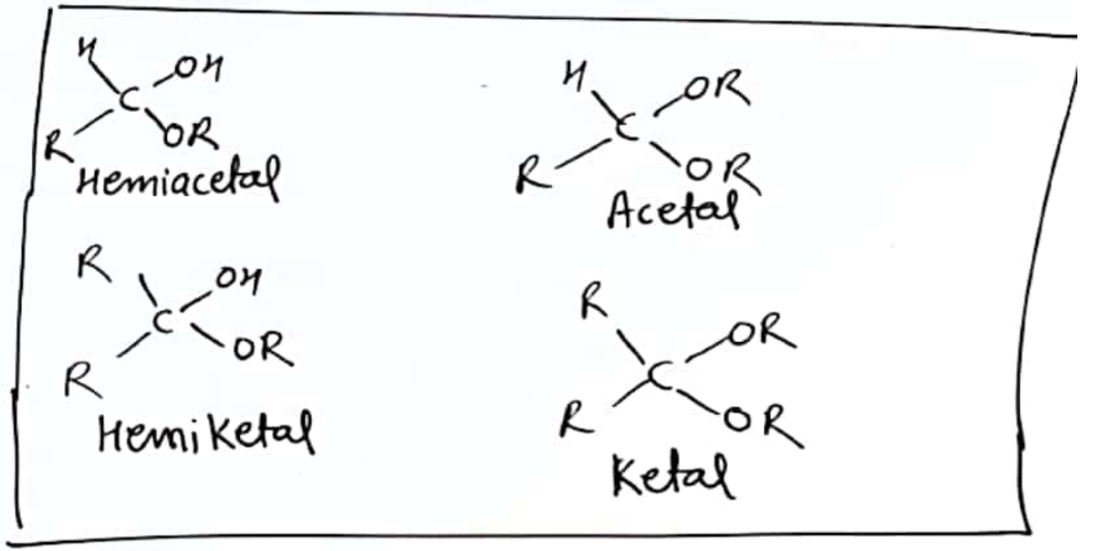
eg  $\xrightarrow{H^{\oplus} + R\text{-}OH + HgSO_4}$ 

$\downarrow ROH + H^{\oplus}$



Ketal

no rex. with base
requires HgSO4 just as in case of addition of water



HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

CHEMICAL REACTIONS

⑥ Addition of halogens (Halogenation)

according to mech. II:
- → Nonclassical carbocation = intermediate
- → No Rearrangement
- → M.P = major

eg 1 $CH_2=CH_2 + X_2 \xrightarrow[\text{Solvent } CCl_4]{} CH_2(X)-CH_2(X)$ vic. dihalide

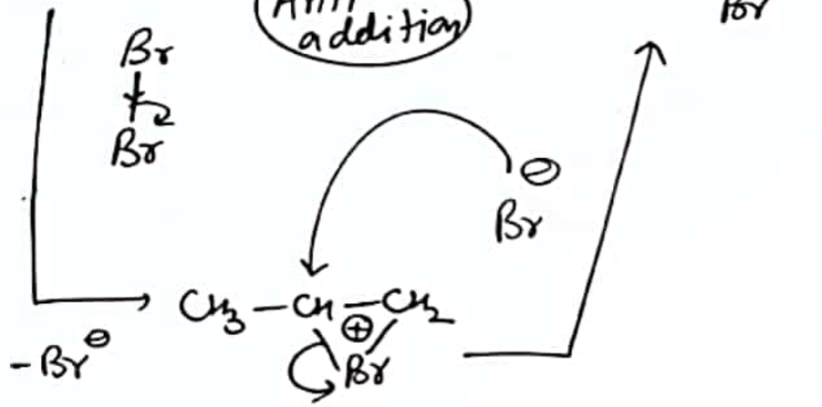
Reactivity ⇒ $F_2 > Cl_2 > Br_2 > I_2$ ($Cl_2 > Br_2$: more useful)

- F_2: highly exothermic
- I_2: reversible

carried out in dark
and/or diluting with inert gas

eg 2

$CH_3-CH=CH_2 \xrightarrow{Br_2}$ (Anti addition) $CH_3-CH(Br)-CH_2(Br)$



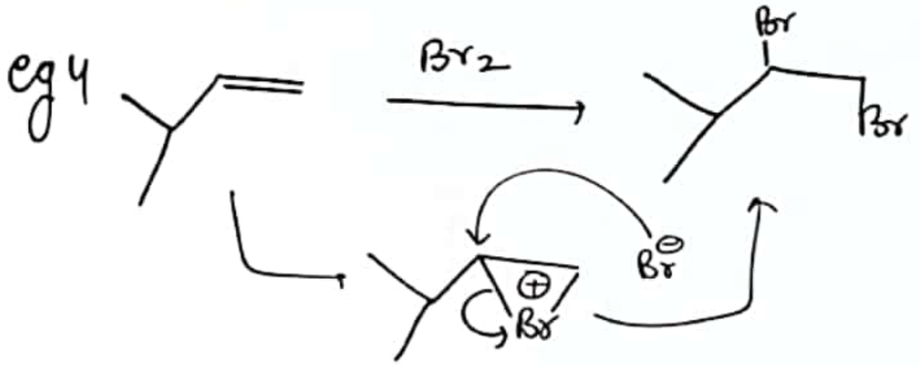
eg 3 $H-C\equiv C-H \xrightarrow{1 \text{ mole } Br_2}$ Trans alkene $\xrightarrow{2 \text{ moles } Br_2} CHBr_2-CHBr_2$

HYDROCARBONS ⇒ (II) ALKENE AND ALKYNES

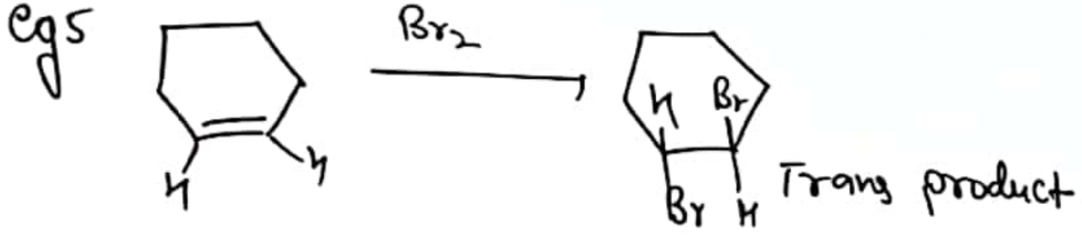
CHEMICAL REACTIONS

⑥ Addition of halogens (Halogenation)

eg 4



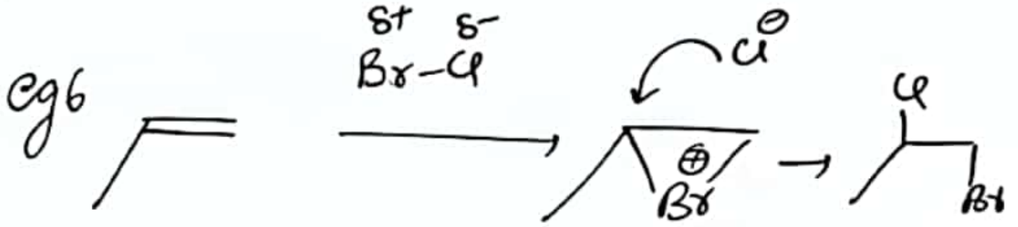
eg 5

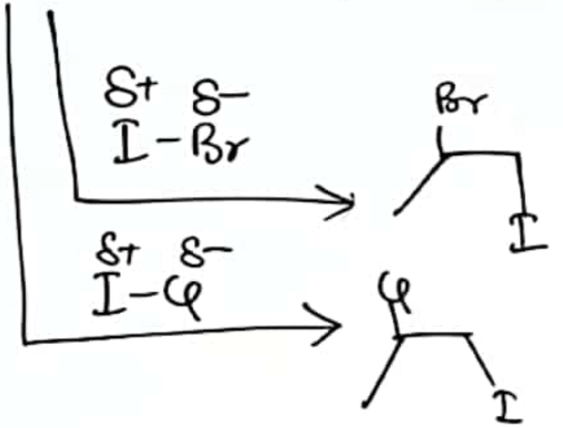


Trans product

Addition of interhalogen compounds:

eg 6





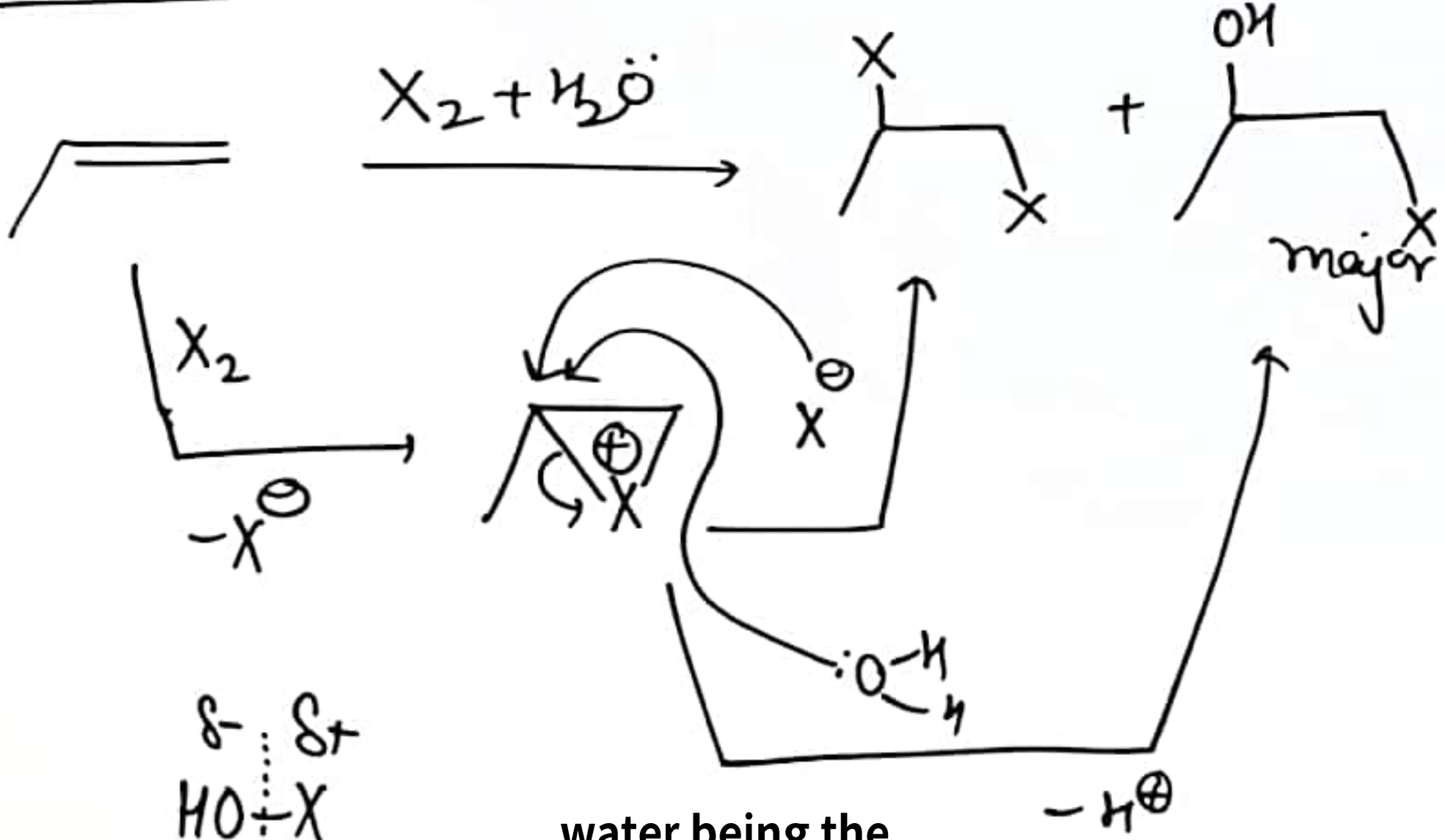
CHEMICAL REACTIONS

⑦ Addition of HOX or $X_2 + H_2O$:-

↑ Hypohalous acid

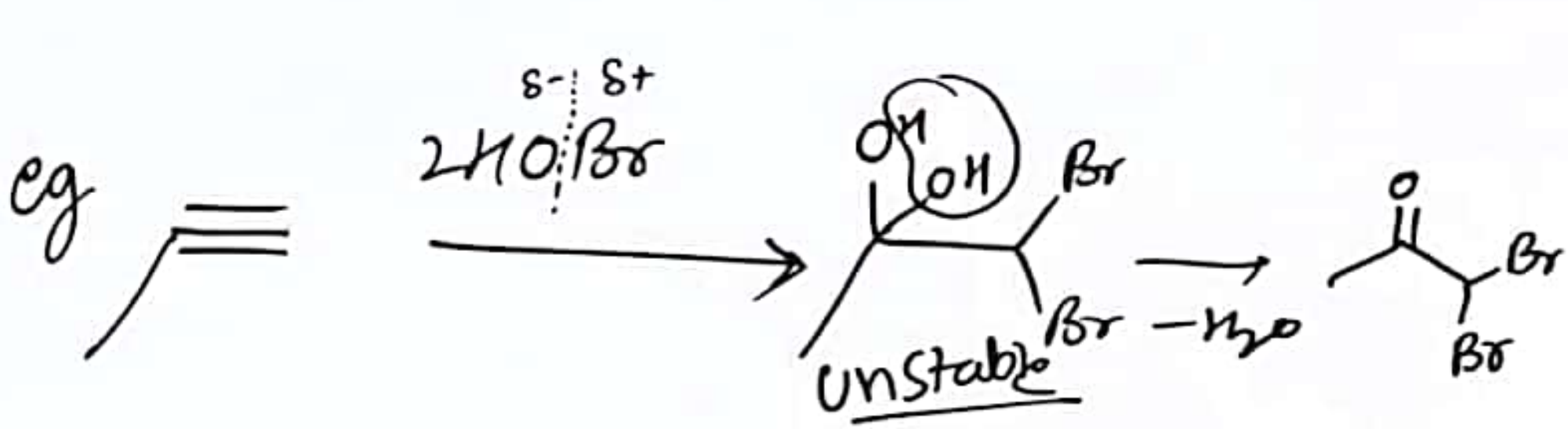
(H-OF) unstable

mech-II

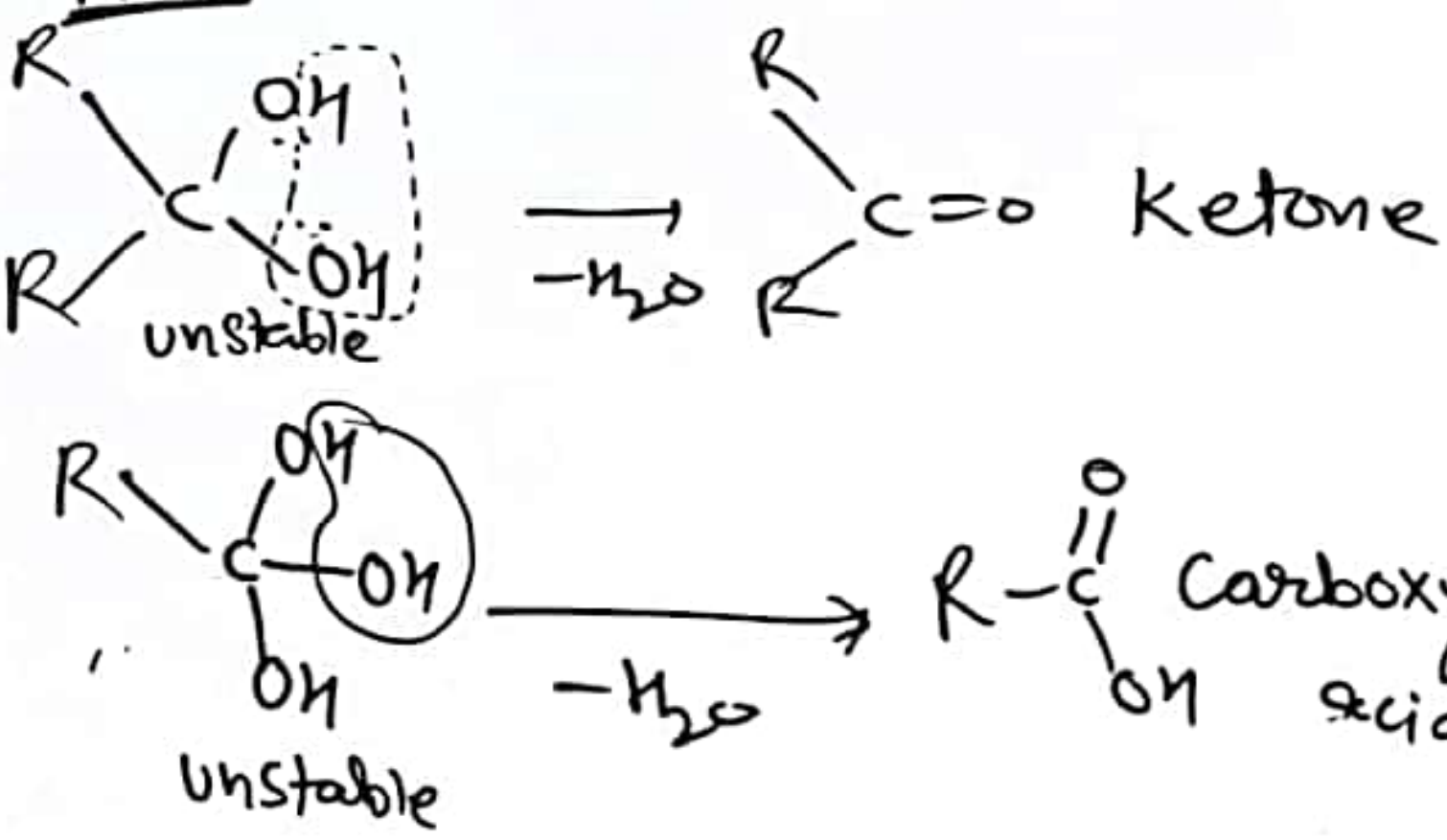




water being the solvent is present in larger amount and hence forms major product



Note



more than one OH groups on the same C atom are unstable due to repulsion b/w the lone pairs of O

HYDROCARBON - ALKENE AND ALKYNE

CHEMICAL REACTIONS

(8) Polymerisation:- Methods of formation of polymer.

Polymer = Macromolecules which are formed by small repeating units (Monomer units)

eg 1 $n CH_2=CH_2$ —Polymerisation→ $[CH_2-CH_2]_n$

Ethylene → Polyethylene (LDPE / HDPE) or polyethene

eg 2 $n CH(CH_3)=CH_2$ —"→ $[CH(CH_3)-CH_2]_n$

Propylene → Polypropylene

eg 3 $n CH(Cl)=CH_2$ —Polymerisation→ $[CH(Cl)-CH_2]_n$

vinyl chloride → Poly vinyl chloride PVC

eg 4 $n CH(Ph)=CH_2$ —"→ $[CH(Ph)-CH_2]_n$

Styrene → Polystyrene / Styron

eg 5 $n CF_2=CF_2$ —"→ $[CF_2-CF_2]_n$

Tetrafluoroethylene → Poly Tetrafluoro ethylene (PTFE) (Teflon)

HYDROCARBON - ALKENE AND ALKYNE

CHEMICAL REACTIONS

(8) Polymerisation:- Methods of formation of polymer.

Polymer = Macromolecules which are formed by small repeating units (Monomer units)

eg6 $n CH_2=CH-CN$ (acrylonitrile) $\xrightarrow{\text{Polymerisation}}$ $[-CH_2-CH(CN)-]_n$ Polyacrylonitrile (PAN)

$CH_2=CH-COOH$
acrylic acid

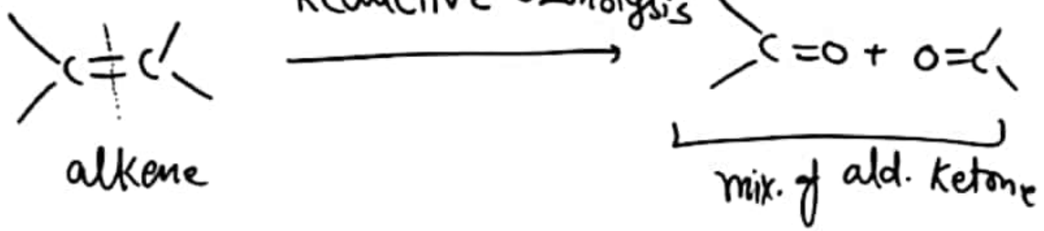
eg7 n $CH_2=CH-O-C(=O)-CH_3$ (Vinyl acetate) $\xrightarrow{\text{pol.}}$ $[-CH_2-CH(O-C(=O)-CH_3)-]_n$ Poly Vinyl acetate (PVA)

HYDROCARBON - ALKENE AND ALKYNE

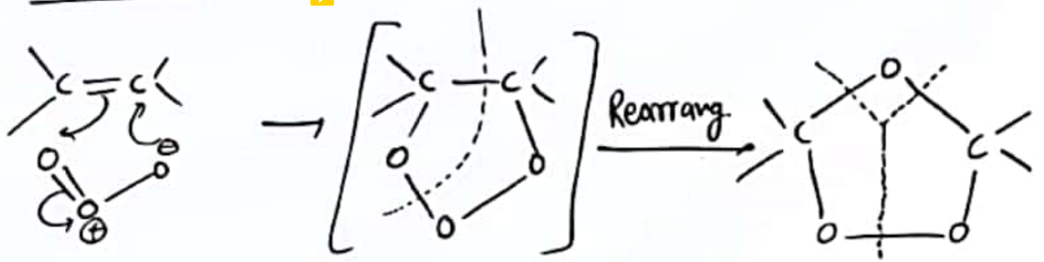
CHEMICAL REACTIONS

(9) Ozonolysis :-

- Reductive ozonolysis
  - (i) O_3
  - (ii) $H_2O + Zn$ / Me_2S
- Oxidative ozonolysis
  - (i) O_3
  - (ii) H_2O / H_2O_2

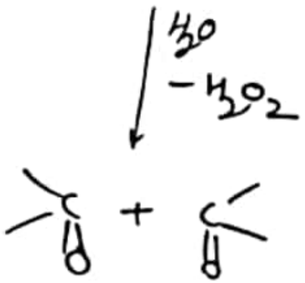


mechanism



monozonide (unstable)

Ozonide



$H_2O_2 + Zn \rightarrow ZnO + H_2O$

$H_2O_2 + Me_2S \rightarrow$ Me–S(=O)–Me + H_2O

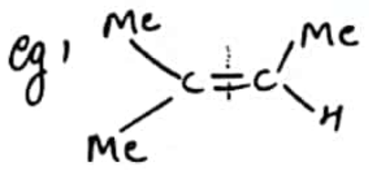
Dimethyl sulphide (DMSO)

H_2O_2 oxidising agent

HYDROCARBON - ALKENE AND ALKYNE

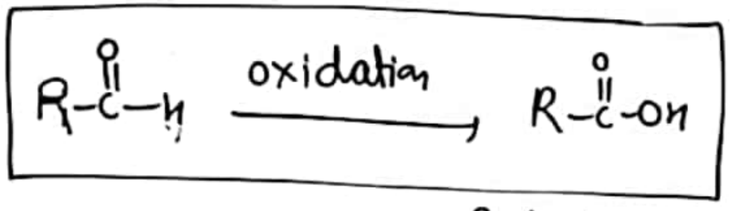
CHEMICAL REACTIONS

(9) Ozonolysis :-

eg1 

(i) O_3 (ii) $H_2O + Zn$ (Red. ozo.) → $Me_2C{=}O + O{=}C(CH_3)H$

(i) O_3 (ii) H_2O (Oxi. ozon) → $>C{=}O + CH_3COOH$



eg2 $R-C\equiv C-R'$

Reductive ozonolysis → $R-C(=O)-C(=O)-R'$ adjacent dione

Oxidative ozo. → $R-COOH + HO-C(=O)-R'$

(10) Oxidation with Strong oxidising agents (SOA).

- ⇒ $KMnO_4 + H^{\oplus}/OH^{\ominus} + \Delta$
- ⇒ $K_2Cr_2O_7 + H^{\oplus} + \Delta$
- ⇒ $CrO_3 + H^{\oplus} + \Delta$

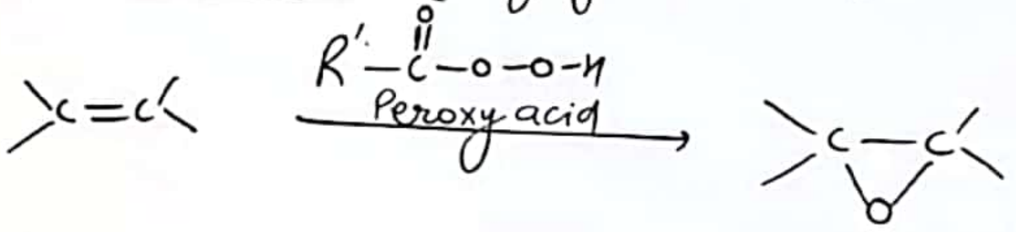
Alkene $\xrightarrow{SOA}$ Products are same as products of oxidative ozonolysis

alkyne $\xrightarrow{SOA}$ Products are same as products of oxidative ozonolysis

HYDROCARBON - ALKENE AND ALKYNE

CHEMICAL REACTIONS

(11) Epoxidation:- (only for alkene) not for alkyne

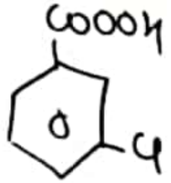


R'-COOOH

① $H-C(=O)-O-OH$ Performic acid (PFA)

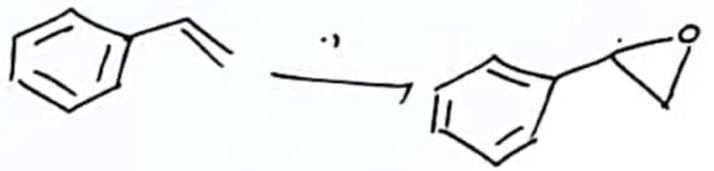
② $CH_3-C(=O)-OOH$ Peracetic acid

③ $Ph-C(=O)-O-OH$ Perbenzoic acid

④  m-Chloroperbenzoic acid (MCPBA)

⑤ $CF_3-COOOH$ etc.

eg 1 

eg 2 

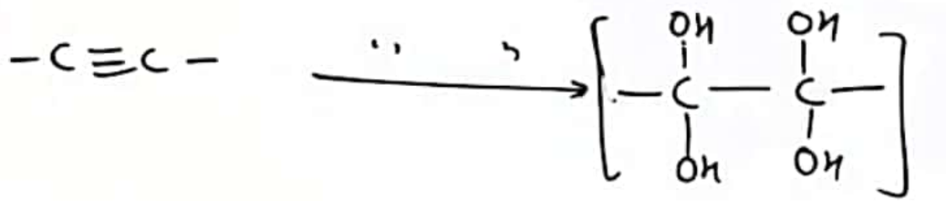
HYDROCARBON - ALKENE AND ALKYNE

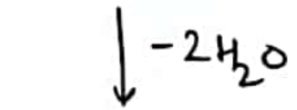
CHEMICAL REACTIONS

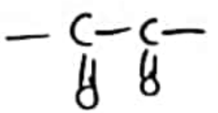
(12) Hydroxylation :-



Vic. diol







dione

Reagents for hydroxylation :-

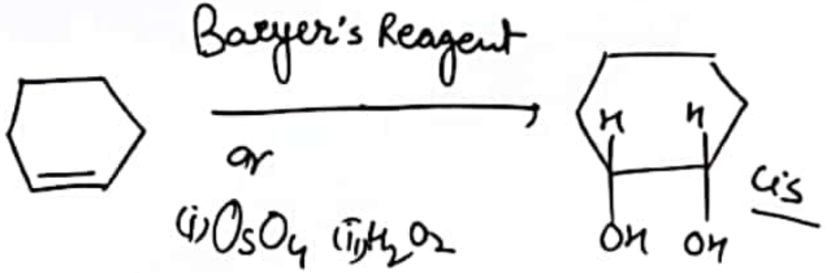
① Cold. dil. $KMnO_4 + \overset{\ominus}{O}H$ Purple
(Baeyer's Reagent)

cold dilute aqueous alkaline KMnO4

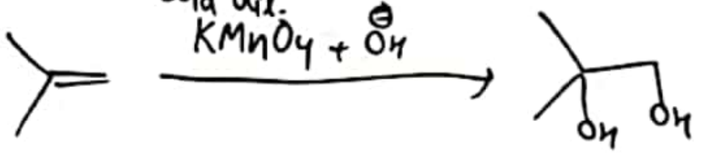
② ⓐ OsO_4 ⓑ H_2O_2 / $NaHSO_3$

syn hydroxylation in both cases

eg 1



eg 2



CHEMICAL REACTIONS

Test of alkenes and alkynes:

(a) Baeyer's Reagent (cold dil $KMnO_4$)

alkene —Baeyer's (Purple)→ Puple colour disappears

alkyne —" "→

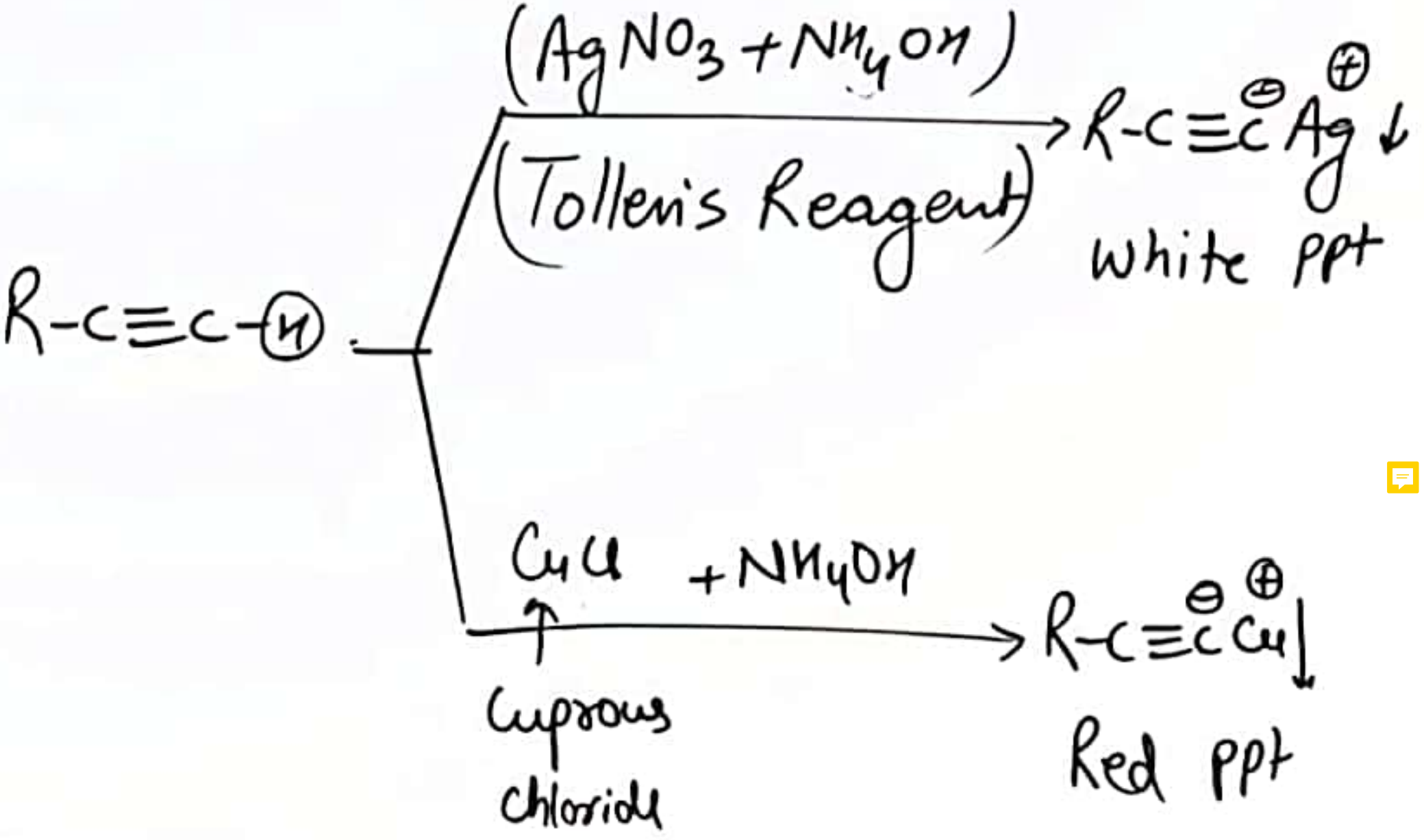
(b) Bromin water test

alkene —$Br_2 + H_2O$ (Red-Brown)→ Colour disappears.

alkyne —" "→ " " "

white colour colloidal ppt. is formed

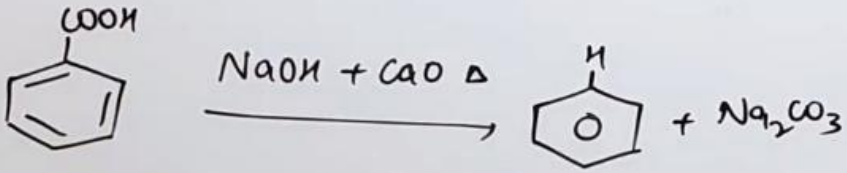
Test of Terminal alkynes.



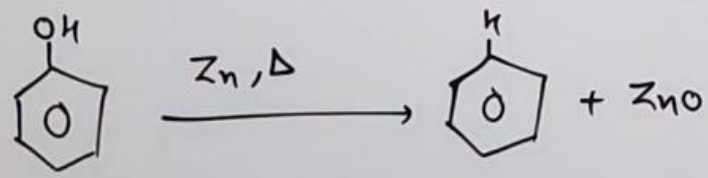
HYDROCARBON - (III) BENZENE

Methods of formation:-

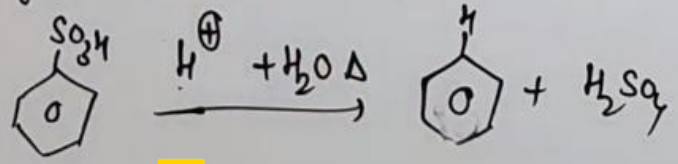
① By decarboxylation of benzoic acid:-



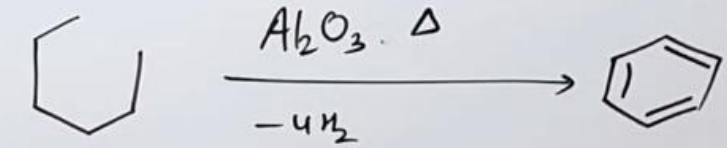
② By Phenol :-



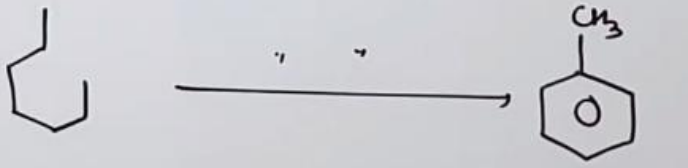
③ By desulphonation of benzene sulphonic acid:



④ By aromatisation of n-Hexane:-



n-Hexane

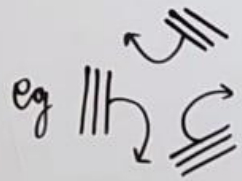
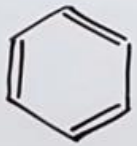


Toluene

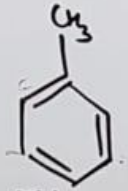
HYDROCARBON - (III) BENZENE

Methods of formation:-

⑤ By tricyclisation of acetylene:-

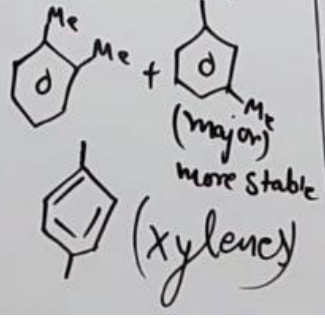
eg  Red hot, Fe-Cu tube, Pressure → 

Benzene

eg $H-C\equiv C-H + CH_3-C\equiv CH \rightarrow$ 

2 : 1

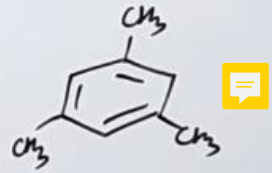
Toluene

eg $H-C\equiv CH + CH_3-C\equiv CH \rightarrow$ 

1 : 2

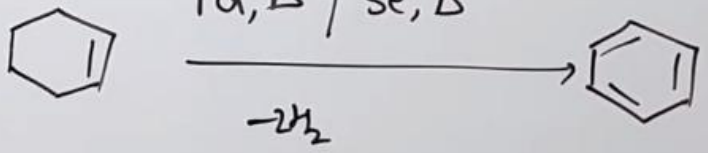
(major) more Stable

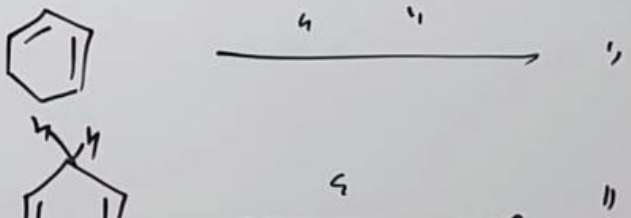
(xylene)

eg $3CH_3-C\equiv CH \xrightarrow[Fe\ Cu\ tube]{Red\ hot}$ 

Mesitylene

⑥ By dehydrogenation of cyclohexene:-

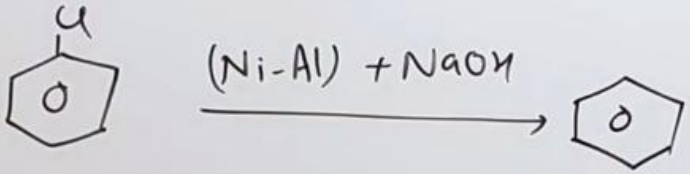




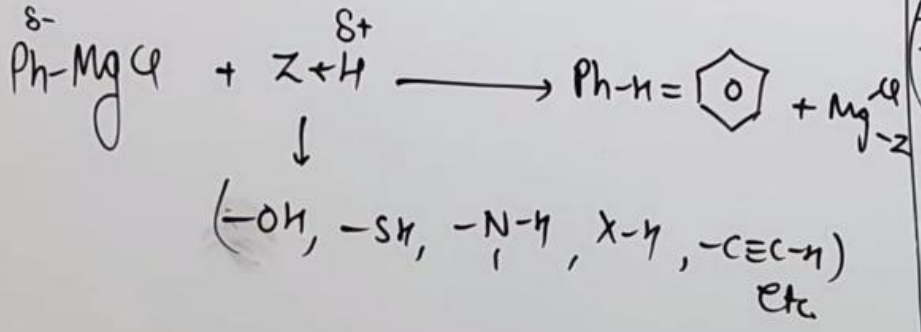
HYDROCARBON - (III) BENZENE

Methods of formation:-

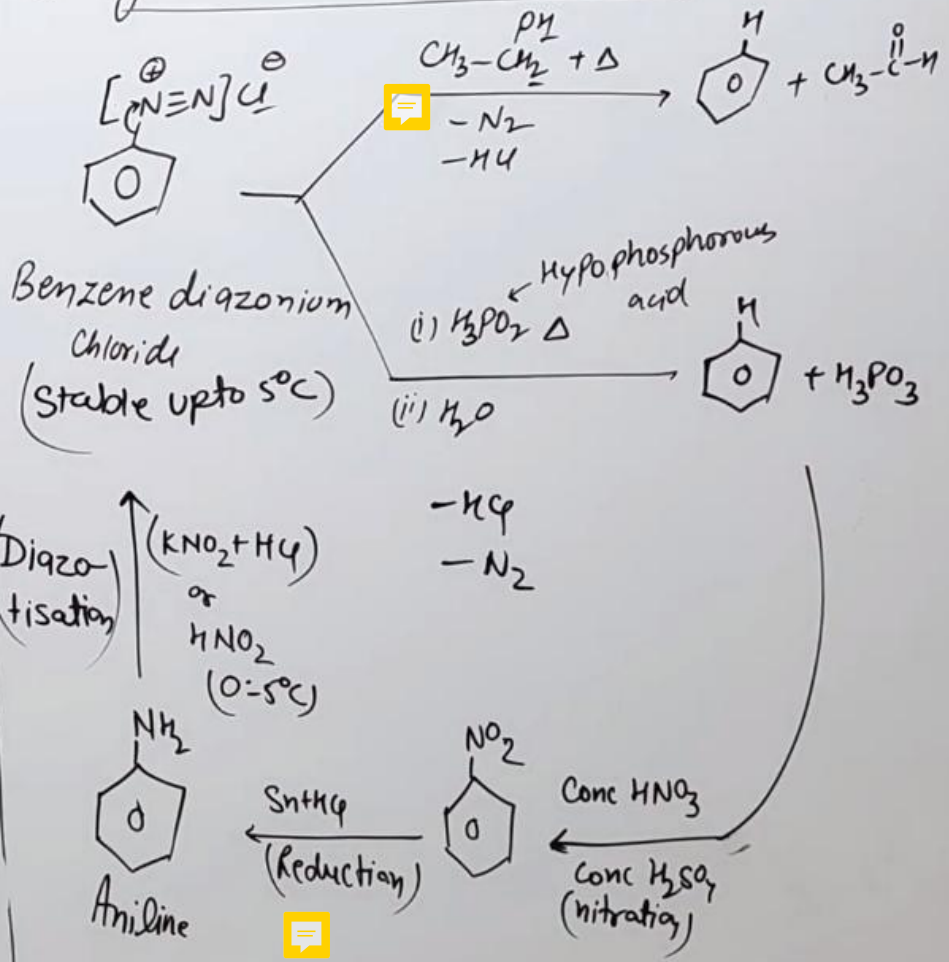
(7) By Reduction of chlorobenzene:-



(8) By Grignard Reagent:-



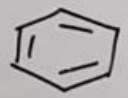
(9) By Benzenediazonium salt



HYDROCARBONS → (III) BENZENE

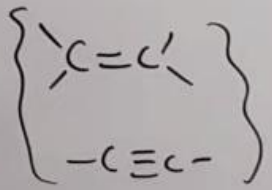
Aromatic compounds give electrophilic Substitution Reaction.

Electrophile → vacant orbital containing, e^- deficient, and attack on high e^- density containing position.



Electrophilic substitution

π e^- density



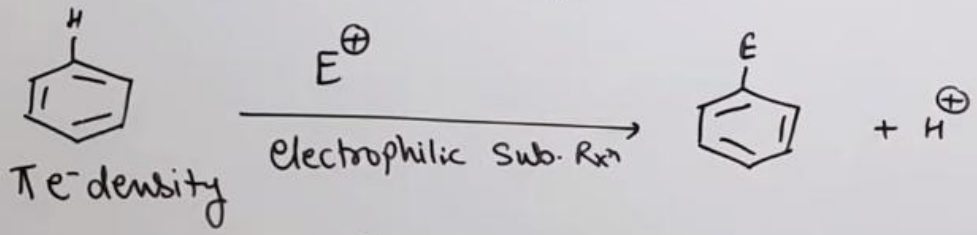
Electrophilic addition.

competishun

HYDROCARBONS → (III) BENZENE

Aromatic compounds give electrophilic Substitution Reaction.

Electrophile → vacant orbital containing, e^- deficient, and attack on high e^- density containing position.



$E^{\oplus} = \overset{\oplus}{N}O_2$ Nitration.

$E^{\oplus} = X^{\oplus}$ Halogenation.

$E^{\oplus} = SO_3$ Sulphonation

$E^{\oplus} = R^{\oplus}$ Friedel craft alkylation

$E^{\oplus} = R-\overset{O}{\overset{\|}{C}}{}^{\oplus}$ Friedel craft acylation etc

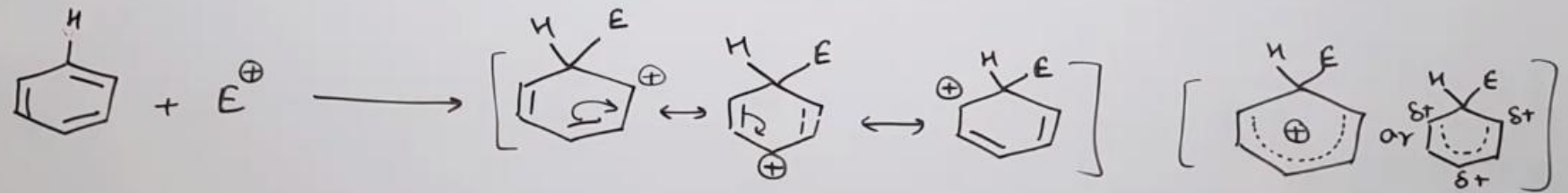
competishun

HYDROCARBONS → (III) BENZENE

Elelectrophilic Substitution Reactions of benzene:- ($Ar S_E$)

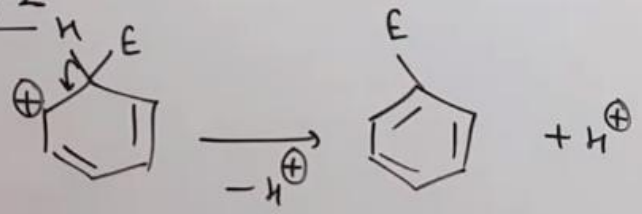
Mechanism of $Ar S_E$

Step-1



Hybrid = Arinium ion.

Step-2



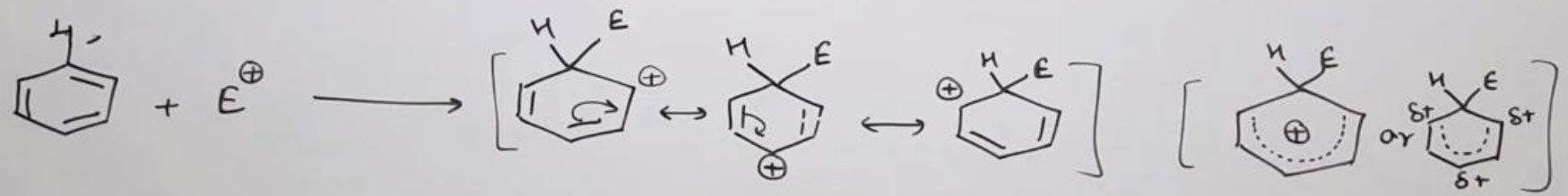
competishun

HYDROCARBONS → (III) BENZENE

Electrophilic Substitution Reactions of benzene:- ($Ar S_E$)

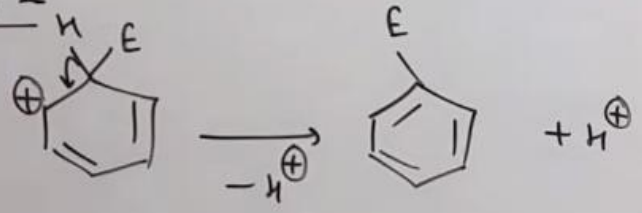
Mechanism of $Ar S_E$

Step-1



Hybrid = Arinium ion.

Step-2



⇒ Reactivity for $Ar S_E$ ∝ Stability of intermediate

⇒ " " " " ∝ Electron density of aro. ring

⇒ " " " ↑es in presence of EDG (+M, +H.C, +I)

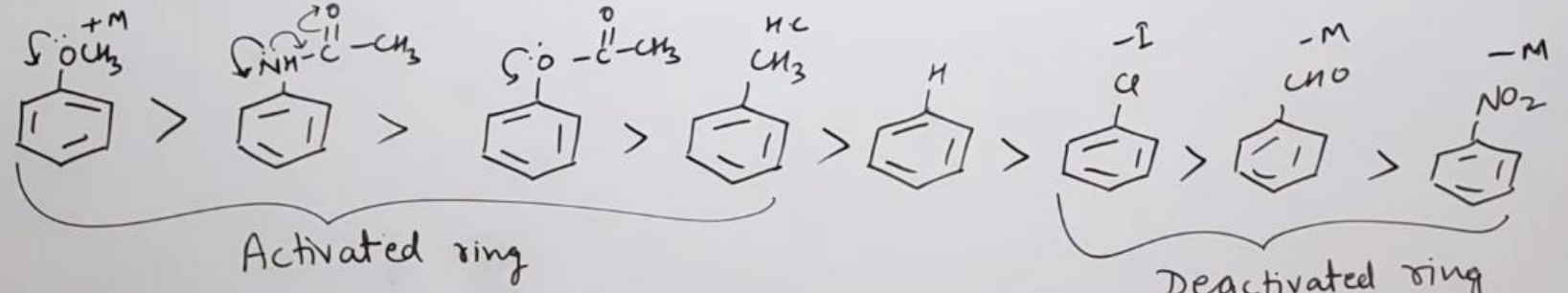
⇒ " " " ↓es " " " EWG (−M, −I)

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HYDROCARBONS ⇒ (III) BENZENE

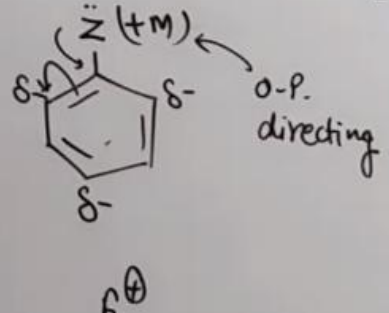
Elelectrophilic Substitution Reactions of benzene:- ($Ar S_E$)

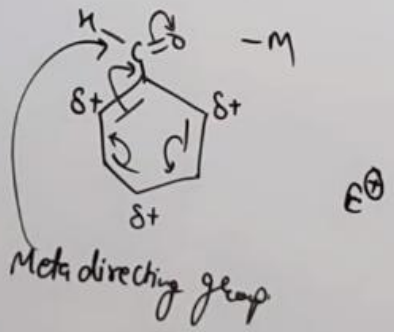
⇒ Reactivity for $Ar S_E$



⇒ Directing groups:-

- Ortho-Para directing group: (+M, H.C, +I) (-OH, -OR, $-O-\overset{O}{\overset{\|}{C}}-R$, $-NH_2$, -NHR, $-CH=CH_2$, alkyl, $-\ddot{X}:$)
- meta directing group: (-M, Strong -I) ($-NO_2$, -CHO, $-\overset{O}{\overset{\|}{C}}-R$, $-SO_3H$, $-COOH$, $-COCl$, $-CONH_2$, $-\overset{\oplus}{N}R_3$ etc)





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HYDROCARBONS → (III) BENZENE

Electrophilic Substitution Reactions of benzene:- ($Ar S_E$)

Name	Reagent	$E^{\oplus}$	Product
① Nitration	ⓐ Conc. HNO_3 + Conc. H_2SO_4 $HNO_3 + H_2SO_4 \rightleftharpoons \overset{\oplus}{NO_2} + H_2O + HSO_4^-$ ⓑ Fuming Nitric acid $2HNO_3 \rightleftharpoons \overset{\oplus}{NO_2} + \overset{\ominus}{NO_3} + H_2O$ ⓒ $N_2O_5 \rightleftharpoons \overset{\oplus}{NO_2} + \overset{\ominus}{NO_3}$ etc.	$\overset{\oplus}{NO_2}$ nitronium ion	$Ar-NO_2$
② Sulphonation	ⓐ Conc. H_2SO_4 $2H_2SO_4 \rightleftharpoons H_3O^{\oplus} + HSO_4^{\ominus} + SO_3$ ⓑ Oleum $H_2S_2O_7 \rightleftharpoons H_2SO_4 + SO_3$ fuming sulphuric acid	$O=S=O$	$Ar-SO_3H$

③ Halogenation

ⓐ $X-X$ + Lewis acid → $X^{\oplus} + X-L.A^{\ominus}$

(AlX_3, FeX_3, BX_3 etc)

$X^{\oplus}$ → $Ar-X$

- $Cl_2 + AlCl_3 \rightarrow Cl^{\oplus} + AlCl_4^{\ominus}$
- $Br_2 + Fe \xrightarrow{\Delta} FeBr_3$
  $Br_2 + FeBr_3 \rightarrow Br^{\oplus} + FeBr_4^{\ominus}$
- $I^{\delta+}-Cl^{\delta-} + AlCl_3 \rightarrow I^{\oplus} + AlCl_4^{\ominus}$

Reactivity

$F_2 > Cl_2 > Br_2 > I_2$

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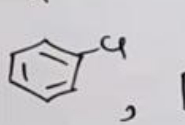
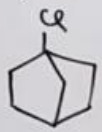
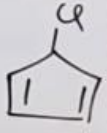
HYDROCARBONS → (III) BENZENE

Electrophilic Substitution Reactions of benzene:- ($Ar S_E$)

(4) Friedel craft alkylation

ⓐ $R-Cl \xrightarrow{AlCl_3} R^{\oplus} + AlCl_4^{\ominus}$

Note

$R-Cl \neq$ [$CH_2=CH-Cl$, $R-C\equiv C-Cl$,  ,  , ]

ⓑ $R-\ddot{O}H \xrightarrow[(H_2SO_4 / H_3PO_4 / HF / BF_3 \text{ etc})]{H^{\oplus}} R-\overset{H}{\underset{\oplus}{O}}-H \longrightarrow R^{\oplus} + H_2O$

ⓒ  $\xrightarrow[H_2SO_4 / H_3PO_4 / HF]{H^{\oplus}}$ 

⇒ Rearrangement possible in Carbocation.

$R^{\oplus}$

$Ar-R$

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HYDROCARBONS → (III) BENZENE

Electrophilic Substitution reactions of benzene:- ($Ar S_E$)

⑤ Friedel craft acylation

ⓐ $R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow{AlCl_3} R-\overset{\overset{O}{\|}}{C}\oplus + \overset{\ominus}{AlCl_4}$

ⓑ $R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R \xrightarrow{AlCl_3} R-\overset{\overset{O}{\|}}{C}\oplus \quad \left[R-\overset{\overset{O}{\|}}{C}-O-\overset{\ominus}{AlCl_3}\right]$

ⓒ $R-\overset{\overset{O}{\|}}{C}-\ddot{O}H \xrightarrow[H_2SO_4 / H_3PO_4]{H^{\oplus}\ (Strong)} R-\overset{\overset{O}{\|}}{C}\oplus + H_2O$

$R-\overset{\oplus}{C}=\ddot{O}: \leftrightarrow R-C\equiv\overset{\oplus}{O}$

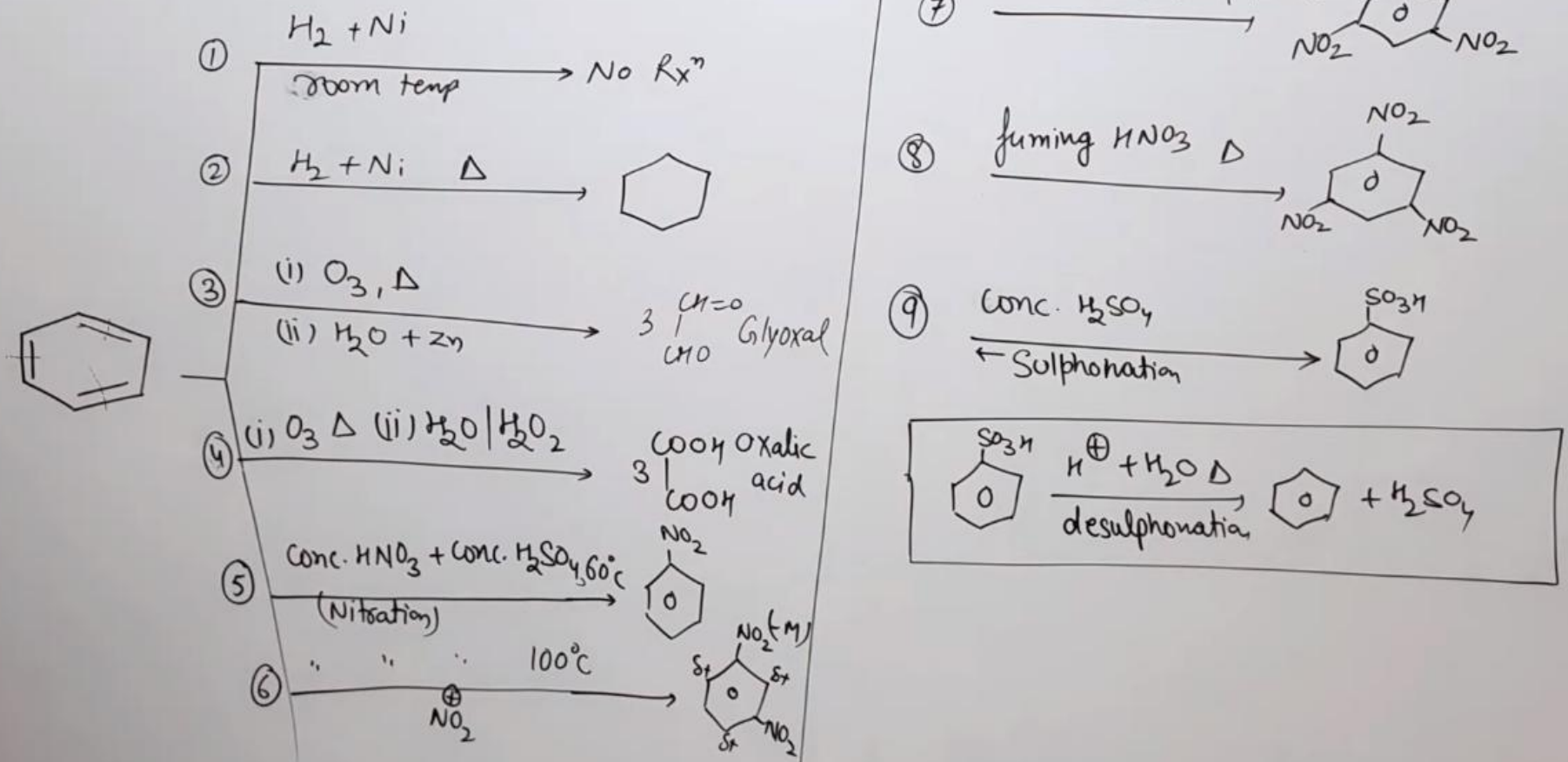
$Ar-\overset{\overset{O}{\|}}{C}-R$

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HYDROCARBONS → (III) BENZENE

1,3,5-trinitrobenzene or sym-trinitrobenzene

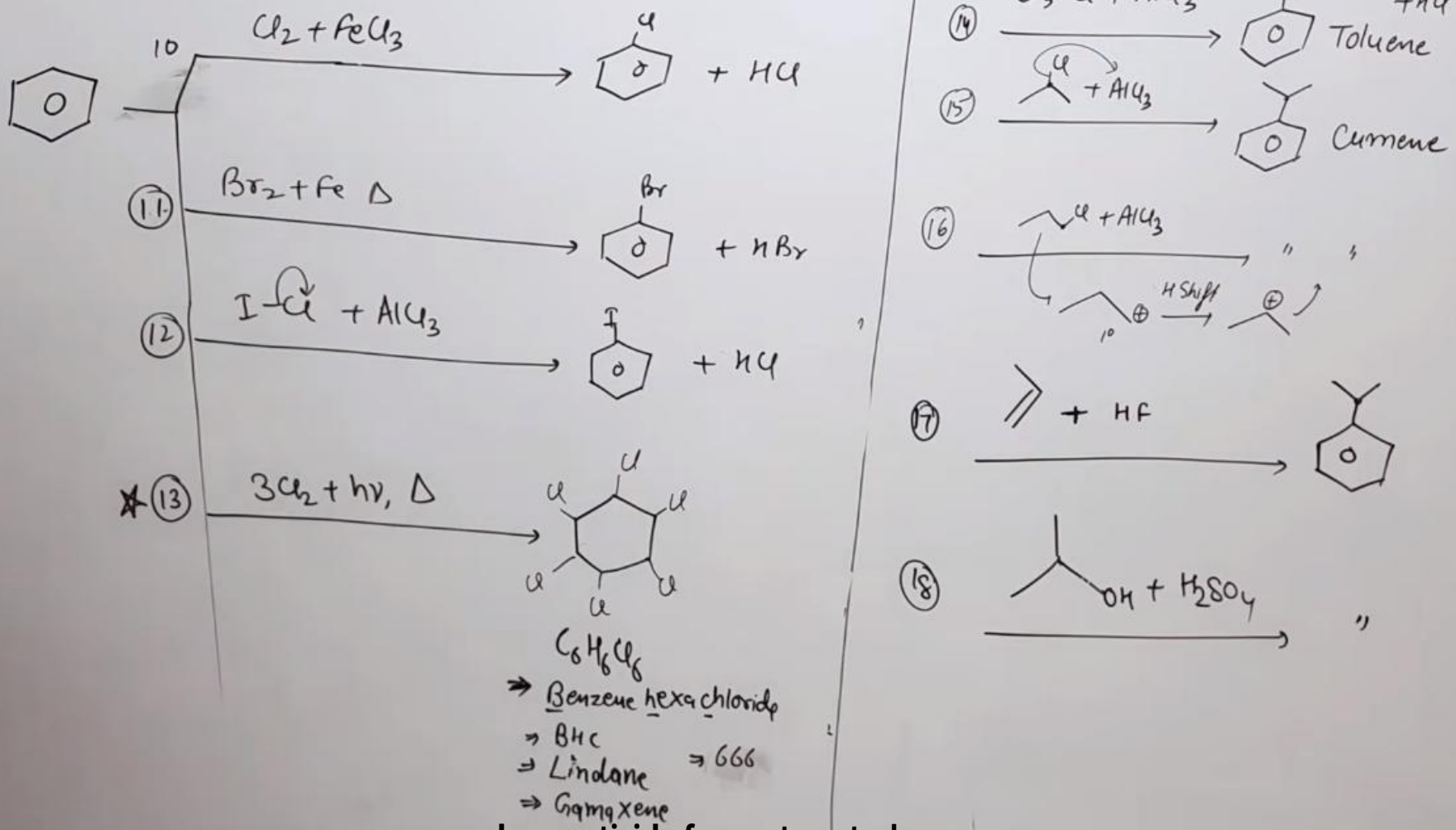
Chemical Reactions of Benzene:-



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HYDROCARBONS → (III) BENZENE

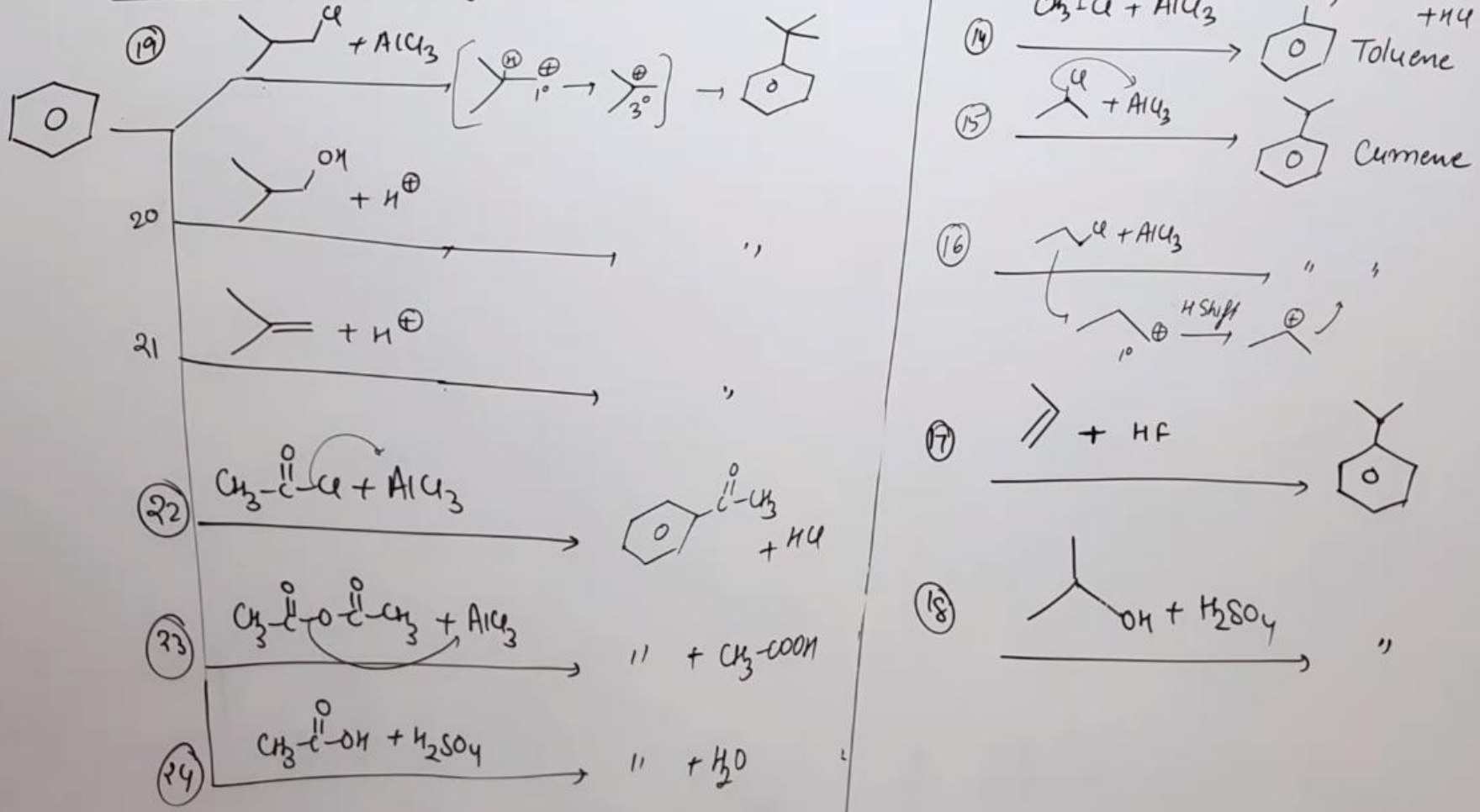
Chemical Reactions of Benzene:-



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HYDROCARBONS → (III) BENZENE

Chemical Reactions of Benzene:-



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